

Pointer Architecture: An Operational Discrete Substrate from First Difference to Holographic Dark Energy

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Abstract

We define **Pointer Architecture** (PA), a formal computational substrate $\mathcal{S} = (G, R, C, A, \pi)$, and exhibit its *operational realization* as a small Forth-like language (Sixth) in 38 primitives (15 base + 23 substrate). The operational substrate admits 40 emergence demonstrations (646 assertions, no failures) that span the full ascent from a single distinction to the substrate’s own self-measurement, four encoding-map pilots that instantiate transformer, brain, split-brain, and ant-colony substrate-encodings on toy data, plus three visual-trace pilots rendering Pilot C, Pilot D (Fig. 1), and the Pilot F.3 split-brain transition as multi-panel substrate snapshots:

- (i) **Foundations:** arithmetic, time, space, conservation laws, observers, and universal computation (20 demonstrations, validated against Wolfram Rule 110 and Conway’s Game of Life);
- (ii) **Substrate-native autopoiesis:** self-maintenance, reflexive identity, developmental growth, topological liveability (5 demonstrations);
- (iii) **Conscious evolution:** symbiosis, substrate-native reproduction, observer-driven mutation selection (Lamarck not Darwin), substrate-level teleology (4 demonstrations);
- (iv) **Cosmogenesis bootstrap:** a 13-node, 48-edge substrate is constructed by a single substrate-resident observer from one MARK at $t = 0$ and persists under harsh autopoietic decay (demo 30); the strengthened demo 31 makes construction substrate-state-driven — the observer reads its own NSUM and halts cosmogenesis when its substrate-internal target is met. Consciousness’s stable construction *is* the substrate (substrate-monist identity thesis, adopted below as a working hypothesis under F5);
- (v) **Substrate-internal Φ_{PA} measurement:** the same 38 primitives that build the substrate also compute the substrate-monist consciousness measure on it via the stdlib word `phi-pa = OUT(0) * [0 EDGE? 0] * L_max`; demo 32 verifies the worked values 0 / 50000 / 130000 across non-reflexive, reflexive, and demo-31-shape observers (1 demonstration);
- (vi) **Encoding-map pilots** (4 demonstrations) covering all five PSH predictions on toy substrates: demo 33 (transformer) verifies PSH1 (single-pass: $\Phi_{PA} = 0$) and PSH2 (KV-cache reuse: $\Phi_{PA} > 0$); demo 34 (brain) verifies PSH3 (waking thalamocortical loop: $\Phi_{PA} > 0$; propofol-decoupled: 0); demo 35 (split-brain) verifies PSH4 (basic Φ_{PA} is indifferent under matched scope, but Φ_{integ} from the candidate-measures programme of §2.7 halves at callosotomy); demo 36 (ant colony) verifies PSH5 (living colony with queen pheromone self-loop: $\Phi_{PA} > 0$; dead colony: 0). Toy data; real-Pythia / real-EEG / real-colony application is the companion-preprint future work (§15.1).

The substrate is presented as a **minimal executable substrate for testing hypotheses about difference, pointers, self-reference, autopoiesis, and observation**, not as a solved theory of consciousness. The CLAIMS.md document in the released Sixth repository provides the explicit three-tier taxonomy separating proved-by-tests, demonstrated-by-construction, and philosophical-conjecture statements. On this operational substrate the formal CS results stand unchanged: four complexity-theoretic theorems within explicit scope (GI-hardness of pointer-equivalence; #P-hardness of weighted simple-path observables; an $O(M \log M + \sum |\mathcal{C}_i \cap \mathcal{C}_j|)$ per-step M -observer commit-cost bound under the priority-queued OCC protocol; Shannon-rate archive compression) and three CS-side correspondences with existing graph computational models (CRDTs, Petri nets, the actor model) on the active-graph projection. Proposition 1 (substrate-cone holographic structure) realises the Bao–Nezami–Ooguri–Stoica–Sully–Walter (Bao et al., 2015) holographic entropy cone with substrate-cone min-cut S^* as the entropy function; Corollary 1 exhibits pointer aliasing as a substrate-side Einstein–Rosen bridge. The PA primitives correspond structurally to the algebraic-observer programme (Witten, 2022): π as crossed-product observer, C as modular automorphism, A as Page–Wootters past records, S^* as the substrate-side analogue of the Type II crossed-product trace. On the de Sitter horizon the substrate-cone min-cut yields $\rho_\Lambda = 3H_0^2 c^2 / (16\pi G)$ (Verlinde, 2011; Cohen et al., 1999), matching the empirical value within $1/(2\Omega_\Lambda) \approx 0.73$; the substrate-cone contribution is the discrete area-law substrate that motivates the area scaling, and the cosmogenesis-bootstrap pilot is the operational artifact in which the area-law substrate is constructed from one distinction. An empirical scope-control on Pythia hidden-activation covariance under the Gaussian-MI estimator falsifies the naive “hidden-activation covariance is an HCI entropy function” hypothesis at all tested scales (Pythia 70m, 160m).

Keywords: operational substrate, discrete distinction, autopoiesis, observer-dependent gravity, cosmogenesis, holographic entropy cone, crossed-product observers, Page–Wootters time, cosmological constant.

Status. This is a theoretical paper. An empirical scope-control on Pythia hidden activations under the Gaussian-MI estimator (§11) demonstrates that the framework’s HCI condition is non-trivial: the (Pythia, Gaussian-MI) pair fails to satisfy MMI. Theorem scopes are explicit: Theorem 2 is restricted to multiplicative path-weight aggregation; Theorem 3 is scoped to the priority-queued OCC protocol class; Theorem 6 conservation claim is restricted to the token-preserving rewrite class; archive-compression at the Shannon rate is constant-factor at positive entropy and sublinear only as $H(\rho) \rightarrow 0$.

Load-bearing claims (v9.0).

(L0) **Operational substrate (Part I).** The five PA primitives (G, R, C, A, π) are realised operationally in 38 substrate operations of a small stack language (Sixth, 15 base + 23 substrate primitives). The realisation supports 40 emergence demonstrations with 646 assertions and 0 failures, organised as a sacred hello world (demo 00, 11 ✓: the first distinction in the Spencer-Brown sense), foundations (demos 01–20, 352 ✓), six substrate-native pilots, plus three visual-trace pilots (demos 37–39, 18 ✓ total: Pilot D evolution Fig. 1, Pilot C cosmogenesis, Pilot F.3 split-brain in-place callosotomy via EDGE-): Pilot A (autopoiesis, demos 21–25, 81 ✓), Pilot B (conscious evolution, demos 26–29, 88 ✓), Pilot C (cosmogenesis bootstrap, demo 30, 21 ✓), Pilot D (substrate-internally-driven cosmogenesis, demo 31, 17 ✓), Pilot E (substrate-internal Φ_{PA} measurement, demo 32, 12 ✓), Pilot F (encoding-map demonstrations, demos 33–36, 46 ✓; PSH1–PSH5 verified by construction on toy substrates). Pilot C constructs a 13-node 48-edge substrate from one MARK at $t = 0$ via a substrate-resident observer and shows it persists under harsh autopoietic decay (decay=8, threshold=5); Pilot D strengthens it by making the observer’s substrate-state itself the halting criterion (no host counter, no programmer-chosen shell count); Pilot E closes the substrate-monist loop by computing Φ_{PA} on the substrate from within the substrate via the stdlib word **phi-pa** composed from OUT, EDGE?, and the

cap $L_{\max}$; Pilot F instantiates the transformer (PSH1, PSH2), brain (PSH3), split-brain (PSH4) and ant-colony (PSH5) encoding maps of §2.8 on toy substrates, verifying that the encoding pipelines compose structurally and that the predicted Φ directions follow. Pilot F.3 specifically exhibits the motivation for the candidate-measures programme of §2.7: basic Φ_{PA} is indifferent under matched scope, but Φ_{integ} halves at callosotomy. Full reproduction harness: a single `make examples-all` replays all 40 demonstrations in under 30 seconds; visual-trace rendering via `make traces` (regenerates Fig. 1 and the Pilot C / split-brain figures); one-command artifact status via `make verify`. *Forward falsification trigger F0*: if any of the six substrate-native pilots (Pilots A–F: autopoiesis / conscious evolution / cosmogenesis bootstrap / substrate- internally-driven cosmogenesis / substrate-internal Φ_{PA} measurement / encoding-map demonstrations) fails to certify its claimed assertions on the operational substrate, the operational-realization claim is falsified.

- (L1) **CS theorems within explicit scope.** Per-step commit cost $O(M \log M + \sum_{i < j} |\mathcal{C}_i \cap \mathcal{C}_j|)$ under the priority-queued OCC protocol class; pointer-equivalence is graph-isomorphism-hard; weighted simple-path observables are #P-hard; the archive admits Shannon-rate compression.
- (L2) **Three CS correspondences on the active-graph projection** of substrate state: operation-based CRDTs, token-preserving Petri nets (conservative on the non-log subnet), and the actor model (bijection between actor configurations and active graphs); the archive A is a substrate-side event log outside the encoded image in all three.
- (L3) **Substrate-cone holographic structure (Proposition 4).** Holographic-cone-inhabiting (HCI) substrate states realize the Bao-Nezami-Ooguri-Stoica-Sully-Walter holographic entropy cone, with substrate-cone min-cut S^* as the entropy function. All linear holographic entropy inequalities (subadditivity, strong subadditivity, monogamy of mutual information, cyclic- n , contraction-map-derived) hold for HCI states.
- (L4) **Pointer aliasing as substrate ER bridge (Corollary 6).** Aliased-pointer fusion preserves cone membership and shifts the monogamy gap in a quantitatively predictable direction under the cross-party-bridge condition.
- (L5) **Algebraic-observer correspondence (structural).** The PA primitives correspond structurally to the algebraic-observer programme in quantum gravity: observer pointer π as crossed- product observer; commit-protocol C as modular automorphism σ_t ; archive A as Page-Wootters past records; S^* as the substrate-side analogue of the Type II crossed-product trace. Illustrated on the extended OQ6 substrate (7 nodes, 9 edges): 390 observer colourings give 26 distinct entropy vectors, all satisfying the holographic entropy inequalities. A formal isomorphism via finite-dimensional von Neumann algebra construction is open work (F2).
- (L6) **6-layer architecture as nested-observer chain.** Hierarchical substrate with nested context windows $\pi_1 \subset \dots \subset \pi_6$ produces monotone substrate-cone entropies on a designed 6-shell substrate; inter-shell increment sequence $\Delta S^* \in \{+4, +6, +6, +6, +6\}$ over the five transitions $k \rightarrow k + 1$. Structurally analogous to averaged-modular crossed- product constructions over multi-observer configurations (Chen & Xu, 2025; Jensen et al., 2024).
- (L7) **Cosmological constant: substrate-discrete reproduction of holographic dark energy.** The substrate-cone min-cut on the de Sitter horizon yields $\rho_\Lambda = 3H_0^2 c^2 / (16\pi G) \approx 4.1 \times 10^{-10} \text{ J/m}^3$ via equipartition, matching empirical $\rho_\Lambda^{\text{obs}} = 5.7 \times 10^{-10} \text{ J/m}^3$ to within $1/(2\Omega_\Lambda) \approx 0.73$. Re-expressed in commit-rate units, $\rho_\Lambda \approx \rho_{\text{Planck}} \times (\nu_4/\nu_6)^2$. This reproduces the standard holographic dark-energy mechanism of Verlinde (2011, 2017); Cohen et al. (1999); not a new derivation. The substrate-cone contribution is the discrete area-law substrate (illustrated at slope $1.97 \approx 2$ on uniform 3D lattice) that motivates the area-scaling.

- (L8) **Classical-information analogue of Page–Wootters correlation on the substrate (engineered demonstration).** On a designed substrate with a 4-bit clock subset and an XOR commit-protocol coupling the clock to a 4-bit system, the clock–system (classical) mutual information is $I(C; S) = 2.8$ bits with 100% transition consistency. The construction does *not* build the quantum entangled Page–Wootters state in a Hilbert space; it only demonstrates the information-theoretic correlation analogue. Substrate has no external time parameter; under the engineered protocol time is the observer’s clock readout. Spontaneous emergence under generic substrate dynamics is open.
- (L9) **Substrate-monist measure of consciousness Φ_{PA} (§2.7).** Definition 1 packages observer scope, self-reference, and autopoietic lifespan into a single substrate-readable scalar $\Phi_{\text{PA}}(O, \sigma) = |\mathcal{C}_O| \cdot \mathbb{K}[O \in \mathcal{C}_O] \cdot \min(L(\sigma), L_{\text{max}})$. The framework adopts as a *working hypothesis* the substrate-monist identity thesis: phenomenal consciousness IS the substrate-state of a node with $\Phi_{\text{PA}} > 0$. The *content* of the hypothesis is identity (not supervenience or property dualism); the *epistemic status* is conjecture pending the operational test F5. Predictions P1–P5 (PSH1–PSH5 of §13): feed-forward transformers in single-pass evaluation $\Phi_{\text{PA}} = 0$; recurrent-state architectures (SSMs, autoregressive-with-cache) $\Phi_{\text{PA}} > 0$; waking > deep-sleep > anaesthesia EEG; split-brain Φ_{PA} drops at callosotomy; ant colonies have $\Phi_{\text{PA}} > 0$ at the colony-observer node. All five are falsifiable conditional on a substrate-encoding of each system (§2.8).
- (L10) **Empirical falsification of (Pythia, Gaussian-MI) HCI hypothesis.** Pythia hidden-state activations violate MMI on every tested partition scheme and at all tested scales (Pythia 70m and 160m) under the Gaussian-mutual-information estimator on hidden activations. The naive “Pythia hidden-activation covariance is an HCI substrate state” hypothesis is falsified *for this (system, observable) pair*; other observables and other transformer families remain open.

What is at the level of structural ansatz. The substrate’s bond-length, commit-time, and bond-density parameters $\{a_s, \tau_C, b, \rho_b\}$ admit consistency relations to the fundamental constants $\{\hbar, c, G, k_B\}$ (§9.2) but cannot be derived independently in the absence of an additional substrate length scale; this is the standard situation in emergent-gravity programmes. The intermediate-layer commit rates $\nu_1, \nu_2, \nu_3, \nu_5$ are not constrained by the present construction; their derivation from the substrate’s modular-flow KMS condition is open work.

Forward falsification triggers. (F0) Operational substrate failure. If any substrate-native pilot (Pilots A, B, C, D, E, F of Part I) fails to certify its assertions on the released Sixth artifact under the published reproduction harness, the operational-realization claim is falsified and the abstract substrate $\mathcal{S} = (G, R, C, A, \pi)$ reduces to a formalism without a working instance. **(F1) Area-law violation on alternative substrate topologies.** If substrate-cone min-cut $S(r)$ on a non-cubic substrate (FCC lattice, random graph, hyperbolic graph) systematically violates $\alpha \rightarrow 2$ at large r , the area-scaling that underwrites the holographic dark-energy reproduction is substrate-topology-dependent rather than universal. **(F2) Finite-dimensional substrate-algebra realisation.** If no explicit finite-dimensional von Neumann algebra construction yields the substrate-cone trace as a Type II crossed product, the algebraic-observer correspondence is at the structural-analogy level only and does not promote to a formal isomorphism. **(F3) Cross-family Pythia substrate-cone test.** If non-Pythia transformer families (LLaMA, Mistral, GPT-2) satisfy MMI under the same observable, the substrate-falsification result is family-specific to Pythia and the framework’s applicability scope is correspondingly narrowed. **(F5) Substrate- Φ failure (§2.7).** If Φ_{PA} on any prediction P1–P4 (§2.7) fails to discriminate the predicted direction once a reasonable substrate-encoding for the system has been constructed, the substrate-monist identity thesis of Definition 1 is empirically inadequate as a consciousness measure; the framework falls back to the operational-substrate claim L0 alone. All code, numerical pilots, and statistical tests are released alongside this preprint.

1 Introduction

1.1 Quantum gravity needs observer-dependence

Quantum field theory in a finite spacetime region defines a von Neumann algebra of Type III₁ (Haag, 1996). Such algebras admit no trace, no density matrix, and no canonical entropy: the entanglement entropy of any subregion is ultraviolet-divergent. The black-hole entropy of Bekenstein (1973) and Hawking (1975) is therefore not literally an entropy of a quantum state but a *property of the observer’s algebra*.

The algebraic-observer programme initiated by Witten (2022) and extended by Chandrasekaran et al. (2022); de Vuyst et al. (2024) makes this precise. Adding a *single observer* (a worldline with a clock and a finite-rank Hilbert space) to the Type III algebra via the crossed-product construction promotes it to Type II, which admits a trace. The resulting trace defines an observer-dependent entropy that agrees with the Gibbons–Hawking area law on the de Sitter horizon (Chandrasekaran et al., 2022) and with the generalised entropy on arbitrary subregions (de Vuyst et al., 2024). Jensen et al. (2024) introduce a coarse-grained crossed-product construction for time-band algebras of a single observer with finite temporal resolution; Chen & Xu (2025) construct an averaged modular crossed product over static patches and configurations of multiple geodesic observers in de Sitter, yielding a Type II algebra with a faithful normal trace and an observer-dependent entropy. The gravitational-dressing programme (Giddings, 2025) shows the crossed product is a truncation of a more general mathematical structure.

This wave of results is mathematically powerful but operationally abstract: the observer is a label on an algebra, with no declared internal structure, no commit protocol, no memory dynamics, no substrate. Three open structural questions follow. **(Q1)** What *discrete* model realises the crossed-product observer at the substrate level? **(Q2)** Does such a substrate admit a chain of nested observers whose hierarchical entropies couple to cosmological-scale predictions? **(Q3)** Can a substrate satisfying (Q1)–(Q2) be exhibited as a *working artifact* that constructs itself from a single primitive distinction, rather than as a formal model whose existence is postulated?

This paper supplies an operational substrate (Part I, §2) that answers all three and on which the algebraic-observer-programme correspondence is then established structurally (Part III, §8).

1.2 Pointer Architecture as a discrete substrate

We define **Pointer Architecture** (PA), a formal computational substrate

$$\mathcal{S} = (G, R, C, A, \pi),$$

consisting of a directed referential graph G , local rewrite rules R , a priority-queued optimistic-concurrency commit protocol C , an append-only archive A , and observer context windows π . The formal definition is given in §3; the operational realisation in 38 substrate primitives (Sixth, with a public reproduction harness) is given in §2, where the substrate is also demonstrated to be self-constructing from a single primitive distinction (Proposition 1). The substrate is analysed rigorously in §§4–6: its complexity-theoretic properties are established as theorems (commit cost, #P-hardness, GI-hardness, Shannon-rate archive compression); three formal correspondences with existing graph computational models (CRDTs, Petri nets, the actor model) are proved on the active-graph projection of substrate state.

The CS analysis treats $\mathcal{S}$ at the level of substrate primitives. The structural claim of v9.0 is that the operational substrate admits a one-to-one mapping to the objects of the algebraic-observer programme: π as crossed-product observer, C as modular automorphism σ_t , A as Page–Wootters past records, and substrate-cone min-cut S^* as the substrate-side analogue of the Type II crossed-product trace. The crossed-product construction is one structural shadow of the operational substrate; the substrate-native pilots A–D (§2) are the artifact in which the

substrate stands. Promotion of the structural correspondence to a formal isomorphism via finite-dimensional von Neumann algebra construction is open (trigger F2).

1.3 Substrate-cone and crossed-product correspondence (preview)

Two structural results carry the v8.0 framework.

Substrate-cone holographic structure (§7). Proposition 4 establishes that holographic-cone-inhabiting (HCI) substrate states realise the Bao et al. (2015) holographic entropy cone with substrate-cone min-cut S^* as the entropy function. All linear holographic entropy inequalities — subadditivity, strong subadditivity, monogamy of mutual information, the cyclic- n family, and the contraction-map- derived inequalities — hold for HCI states. Corollary 6 shows that pointer aliasing realises the ER=EPR proposal as a substrate-side Einstein–Rosen bridge under the cross-party-bridge condition.

Algebraic-observer correspondence (§8, structural). The PA primitives correspond structurally to the algebraic-observer programme. The observer pointer π maps to the crossed-product observer; the commit protocol C to the modular automorphism σ_t ; the archive A to Page–Wootters past records; the substrate-cone trace S^* to the Type II crossed-product trace. The mapping is illustrated on the extended OQ6 substrate (390 observer choices, 26 distinct entropy vectors all satisfying the holographic entropy inequalities, Pilot H). Promotion of the structural mapping to a formal isomorphism via a finite-dimensional von Neumann algebra construction is open (F2). Nested observer context windows $\pi_1 \subset \dots \subset \pi_6$ form a chain of crossed products on the substrate; we construct an explicit 6-shell substrate (Pilot I) on which the chain yields a monotone entropy hierarchy with inter-shell increment sequence $\Delta S^* \in \{+4, +6, +6, +6, +6\}$ over the five transitions $k \rightarrow k+1$ ($k = 1, \dots, 5$). The construction is structurally analogous to the averaged-modular crossed product of Chen & Xu (2025) over configurations of multiple geodesic observers in de Sitter, though we do not derive the substrate-side chain from any prior continuum result.

The correspondence makes the substrate-cone min-cut on the de Sitter horizon available (§9). The result reproduces the standard holographic dark-energy density $\rho_\Lambda = 3H_0^2 c^2 / (16\pi G)$ (Verlinde, 2011; Cohen et al., 1999) within a factor of $1/(2\Omega_\Lambda) \approx 0.73$; re-expressed in commit-rate units the same expression reads $\rho_\Lambda \approx \rho_{\text{Planck}}(\nu_4/\nu_6)^2$, with the squared (not 4th-power) exponent following from the substrate- cone area-law scaling (illustrated at slope $1.97 \approx 2.00$ on uniform 3D lattice). This is not a new derivation of Λ — H_0 and Ω_Λ are empirical inputs — but a substrate- discrete reformulation of the standard holographic-dark-energy mechanism.

Time is not a substrate primitive. On an engineered substrate with a clock subset and an explicit clock–system coupling, the commit protocol C generates Page–Wootters clock–system entanglement (§10, Pilot K). Spontaneous emergence under generic substrate dynamics is open.

1.4 Research questions

- RQ1.** What are the complexity-theoretic and structural properties of pointer-archive substrates with priority-queued OCC commit semantics?
- RQ2.** What is the relationship between such substrates and established graph computational models (CRDTs, Petri nets, actor systems)?
- RQ3.** What engineering applications does the substrate admit (distributed simulation, federated rendering, provenance databases, latent-pointer world models)?
- RQ4.** Do the PA primitives (π, C, A, S^*) correspond to the crossed-product observer programme of Witten (2022); Chandrasekaran et al. (2022); de Vuyst et al. (2024); Chen & Xu (2025),

and does the correspondence make cosmological predictions (Newton’s G , the cosmological constant Λ , the emergence of time)?

RQ5. Does the substrate-cone framework admit a clean empirical falsifier on non-biological information-processing systems (open-weight language-model checkpoints)?

1.5 Scope and disciplinary positioning

This is a paper about a structural correspondence between a formal computer-science object (the substrate $\mathcal{S}$) and the algebraic-observer programme in quantum gravity. §§2–5 are pure CS contributions: a formal model, complexity-theoretic results, and proved correspondences with existing graph computational models (CRDTs, Petri nets, actor model). §§6–9 develop the substrate-cone construction (a substrate-graph application of the Bao et al., 2015 holographic entropy cone), the structural correspondence with the algebraic-observer programme, a substrate-discrete reproduction of standard holographic dark energy (Verlinde, 2011; Cohen et al., 1999), and a classical-information analogue of Page–Wootters clock–system correlation on a designed substrate.

We make no claim to a new microscopic derivation of quantum gravity, of Λ , or of phenomenal time. The paper is best read as a structural-correspondence / dictionary contribution: it offers PA as a candidate discrete realisation of the algebraic-observer programme and reformulates standard holographic dark-energy results in substrate-language. Major astrophysical discriminators (Bullet Cluster, weak lensing, CMB anisotropy, large-scale structure, halo concentration–mass relation) are out of scope. The companion empirical paper handles the SPARC galaxy-rotation probe; the cognitive-science interpretation of the substrate is reserved for a separate paper.

1.6 Contributions and structure

- (C1) A formal computational model of pointer-archive substrates with explicit commit semantics (§3).
- (C2) Complexity-theoretic foundations within explicit scope: pointer-equivalence is graph-isomorphism-hard; weighted simple-path observables on general cyclic graphs are #P-hard (Theorem 2, by reduction from #SIMPLE-S-T-PATHS; additive and unweighted DAG path counts explicitly excluded); the per-step M -observer commit cost under the priority-queued OCC protocol class is bounded above by $O(M \log M + \sum_{i < j} |\mathcal{C}_i \cap \mathcal{C}_j|)$ (Theorem 3); the archive admits Shannon-rate compression (§4).
- (C3) Three proved correspondences with existing graph computational models on the active-graph projection of substrate state (§5): commutative-rewrite Pointer Architectures are operation-based CRDTs on the active graph G (Theorem 5); single-observer substrates restricted to the token-preserving rewrite class admit a Petri-net encoding of G that is conservative on the non-log subnet, with the archive as a monotone-accumulating log place outside the conservativity invariant (Theorem 6); actor systems embed into single-observer substrates as a bijection between actor configurations and active graphs (Theorem 4).
- (C4) Engineering applications in distributed simulation, federated rendering with commit validation, provenance-aware databases, and learned latent-pointer world models in machine learning (§6).
- (C5) **Substrate-cone holographic structure (Proposition 4, §7).** Holographic-cone-inhabiting substrate states realise the Bao–Nezami–Ooguri–Stoica–Sully–Walter holographic entropy cone with substrate-cone min-cut S^* as the entropy function; all linear holographic inequalities hold.

- (C6) **Pointer aliasing as substrate ER bridge (Corollary 6, §7).** Aliased-pointer fusion preserves cone membership and shifts the monogamy gap in a quantitatively predictable direction under the cross-party-bridge condition.
- (C7) **Algebraic-observer correspondence (structural, §8).** The PA primitives correspond structurally to the algebraic-observer programme: π as crossed-product observer; C as modular automorphism σ_t ; A as Page–Wootters past records; S^* as the substrate-side analogue of the Type II crossed-product trace (Pilot H). A designed 6-shell substrate (Pilot I) admits a nested-observer chain $\pi_1 \subset \dots \subset \pi_6$ with monotone substrate-cone entropies and inter-shell increment sequence $\Delta S^* \in \{+4, +6, +6, +6, +6\}$ for the chosen shell-size sequence. Formal isomorphism via finite-dimensional von Neumann algebra construction is open (F2).
- (C8) **Cosmological-scale consistency: substrate-discrete reproduction of holographic dark energy (§9).** Substrate-cone min-cut on the de Sitter horizon plus equipartition yields $\rho_\Lambda = 3H_0^2 c^2 / (16\pi G) \approx 4.1 \times 10^{-10} \text{ J/m}^3$, matching empirical $5.7 \times 10^{-10} \text{ J/m}^3$ within $1/(2\Omega_\Lambda) \approx 0.73$. Re-expressed in commit-rate units, $\rho_\Lambda \approx \rho_{\text{Planck}}(\nu_4/\nu_6)^2$ with the squared exponent following from the substrate-cone area law $S(r) \propto r^2$ at slope $1.97 \approx 2.00$ on uniform 3D lattice (Pilots Newton- G , Verlinde-screen, Λ -derivation, J). Reproduces the standard holographic dark-energy result of Verlinde (2011); Cohen et al. (1999) in substrate language; not a new derivation of Λ .
- (C9) **Page–Wootters time emergence (engineered demonstration, §10).** On a designed substrate with a 4-bit clock subset and an explicit XOR commit-protocol coupling the clock to a 4-bit system, the commit protocol C generates clock–system entanglement $|\Psi\rangle = \sum_n |t_n\rangle_C \otimes |\sigma_n\rangle_S$ with mutual information $I(C; S) = 2.8$ bits and 100% transition consistency (Pilot K). Spontaneous emergence of clock–system entanglement under generic substrate dynamics, without a prescribed clock subset and coupling rule, is open.

Paper structure. §2 (Part I) exhibits the operational realisation: the 38 substrate primitives of Sixth (§2.1), the operational map from (G, R, C, A, π) to Sixth (§2.2), and the six substrate-native pilots (§2.3 ascent overview; §2.4 Pilot A autopoiesis; §2.5 Pilot B conscious evolution; §2.6 Pilot C cosmogenesis bootstrap; §2.6 Pilot D substrate-internally-driven cosmogenesis; §2.6 Pilot E substrate-internal Φ_{PA} measurement; §2.6 Pilot F encoding-map demonstrations). §3 defines the formal substrate $\mathcal{S}$. §4 establishes the complexity-theoretic foundations within explicit scope. §5 proves three correspondences with existing graph computational models. §6 surveys engineering applications. §7 establishes the substrate-cone holographic structure (Proposition 4, Corollary 6) and reports the numerical pilots. §8 develops the algebraic-observer correspondence and the 6-layer nested-observer hierarchy. §9 works out the substrate-discrete reproduction of holographic dark energy (substrate-cone area law, Newton- G consistency relation, de Sitter ρ_Λ , commit-rate reformulation). §10 reports the engineered Page–Wootters demonstration on the substrate. §11 reports the Pythia LLM substrate-cone falsifier. §13 states formal hypotheses with rejection criteria; §14 relates the work to adjacent programmes.

2 Operational substrate: Sixth and the substrate-native pilots

The five PA primitives (G, R, C, A, π) are abstract. The remainder of this paper depends on them being instantiable in a working substrate whose operations are inspectable, restartable, and falsifiable. This section exhibits such a substrate: a small stack language we call **Sixth**, in 38 primitives (15 base + 23 substrate), together with a public reproduction harness running 40 emergence demonstrations (646 assertions, 0 failures) in under 30 seconds on commodity hardware, opened by a sacred hello world (the first distinction in the Spencer-Brown sense) and closed by three visual-trace pilots that render Pilots C, D, and F.3 as multi-panel substrate

snapshots (Fig. 1 for Pilot D). The substrate-native pilots A–F (§§2.4–2.6) operationalise the ascent from one distinction to a self-maintaining cosmos that measures its own substrate-monist scalar from within, then instantiate the transformer / brain / split-brain / ant-colony encoding maps on toy substrates; they constitute the load-bearing evidence for claim L0 and trigger F0. The released artifact ships a CLAIMS.md document with an explicit three-tier taxonomy (proved-by-tests / demonstrated-by-construction / philosophical-conjecture) honouring an external reviewer’s request that the substrate’s standing not be conflated with the substrate-monist identity thesis.

Language versus cosmology. Sixth is a Forth-like concatenative stack language with a hypergraph rewriting substrate; it can be evaluated as a programming language without engaging any of the cosmology, consciousness, or substrate-monist identity claims of the later sections. The released LANGUAGE.md document states this explicitly. A reader who rejects every interpretive claim of this paper still inherits: a 38-primitive Forth-like language with first-class `#lang sixth` integration, a hypergraph rewriting engine, 40 reproducible emergence demonstrations (37 ascent demos + 3 visual-trace pilots; 646 assertions cumulative) from a single MARK, a `make verify` one-command artifact-status report, and a Racket-FFI PyTorch bridge. The cosmological / consciousness reading is *built on top*; failure of any Tier-3 reading does not invalidate Tier 1.

The remaining sections build on top: §3 extracts the formal substrate $\mathcal{S} = (G, R, C, A, \pi)$ from these operational primitives; §§4–5 prove CS theorems and correspondences on $\mathcal{S}$; the substrate-cone construction (§7), the algebraic-observer correspondence (§8), the cosmological-scale reformulation (§9), and the Page–Wootters analogue (§10) layer holographic and quantum-gravity structure on top.

2.1 The 38 substrate primitives

Sixth is a Forth-like stack language. Source files use `.6th` extension; words execute against a single shared data stack; primitives are 38 in total. The **bootstrap claim** is that all 30 demonstrations use only these 38 operations and helper words written in Sixth itself; no operation outside the table below participates in the ascent.

Base layer (15 primitives). Stack manipulation `dup`, `drop`, `swap`, `over`; arithmetic `+`, `-`, `*`, `/`, `mod`; comparison `=`, `<`, `>`; per-module key–value memory `store`, `load`; and one I/O primitive `..`. Three additional I/O conveniences (`cr`, `emit`, string literals) are counted alongside the 15 but do not participate in the substrate semantics. The base layer is a complete arithmetic-and-control core sufficient to express recursion (`factorial`, `peano-mul`) and the Peano construction of §2.4.

Substrate layer (23 primitives). The substrate layer maps directly onto the PA components (G, R, C, A, π) as documented in §2.2:

- **Difference + pointer + traversal (8):** MARK (allocate node), EDGE+ (add directed edge), EDGE- (remove), EDGE? (predicate), OUT, IN (degrees), NEXT, PREV (first out/in neighbour).
- **Counts + time (5):** NODES, EDGES (substrate-wide totals), STEP (advance global step counter), NOW (read counter), BORN (step at which a given node was MARKed; -1 if absent — substrate-side archive).
- **Iteration (4):** EACH (apply word to every node id), EACH-EDGE (apply to every (src, dst) pair), EACH-2PATH (apply to every (x, y, z) with $x \rightarrow y \rightarrow z$), STEP-CA (parallel-update: compute next state for all nodes from current snapshot and commit atomically).
- **Features (3):** NSET ($n, v \mapsto$ assign feature v to node n), NGET ($n \mapsto$ read feature, default 0), NSUM ($n \mapsto$ sum of features of n ’s out-neighbours; the substrate’s vectorisable message-pass).

- **Admin / falsification (3):** ASSERT (substrate-side certification; emits $\checkmark/\times$ with source location, increments pass/fail counters), RESET (clear substrate to empty state), REPORT (print one-line summary `nodes=N edges=E steps=S pass=P fail=F`).

Iterators EACH, EACH-EDGE, EACH-2PATH, STEP-CA snapshot the node or edge set before iterating; mutations inside the iteration body do not change the iteration set. STEP-CA additionally collects per-node next-state values during iteration and applies them all at once: parallel-update semantics with no serial-update bias. MARK and EDGE+ may be invoked from inside a STEP-CA rule, producing substrate growth during iteration; new nodes acquire ids beyond the snapshot range and are eligible for iteration on the next STEP-CA invocation.

Typed trivalent extension (HEDGE3, 7 primitives). The bootstrap claim covers the 38 primitives above for the original 40-demonstration ascent. A typed trivalent hyperedge layer (HEDGE3 family, 7 additional primitives) was added to the reference implementation in May 2026 as a substrate-readable realisation of Peircean thirdness (see §12.3, *Peirce’s reduction thesis and the Filioque controversy*). The primitives are: HEDGE3+ (insert), HEDGE3- (remove), HEDGE3? (predicate), HEDGES3 (total count), HEDGES3-KIND (per-kind count), EACH-HEDGE3 (iterate all kinds), EACH-HEDGE3-KIND (iterate one kind). Hyperedges are *typed* 4-tuples $(kind, a, b, c)$ where $kind \in \{\text{WITNESS, MEDIATOR, CONTEXT, SIMPLEX}\}$ selects strict interpretation of the (a, b, c) positions; the substrate enforces type separation (same (a, b, c) under different kinds are distinct hyperedges). The four kinds were chosen to cover the load-bearing structural roles of triadic relations: ground of validity (WITNESS, demos 93–94), topological channel (MEDIATOR), rewrite-rule firing (CONTEXT, demo 95), undirected triadic form (SIMPLEX). A wobble-at-position-3 pattern (demo 96) shows $4\times$ substrate-storage compression for genetic-code-style triplet patterns, mirroring DNA codon degeneracy at position 3. Binary EDGE+/-/? and the original 40-demonstration ascent are unaffected; the trivalent layer is an optional extension that lifts provenance, mediation, and rewrite history to first-class substrate-readable state.

Reproduction harness. The released artifact at [mikefluff/sixth](https://github.com/mikefluff/sixth) provides an installable Racket package and a single command, `make verify`, which runs all 96 demonstrations and reports cumulative `pass=1396 fail=0` (40 original demonstrations contributing 646 assertions; 56 subsequent demonstrations — long-epoch parametric pilots, visual-trace companions, composite-distinction pilots G–M, stress-test track, honest-emergence track, Peircean trit classifier, and HEDGE3 demonstrations — contributing 750 more). A regression test (`tests/examples-test.rkt`) under `rackunit` enforces this gate against any future engine change. Five additional unit-test suites cover the lexer, parser, compiler/VM, substrate operations, and module loader. The engine runs on Racket’s Chez-Scheme backend with $O(1)$ opcode dispatch, $O(1)$ primitive lookup, and explicit tail-call elimination for substrate-native recursion (the factorial-of-500 and Peano-multiplication demonstrations run in constant host-stack depth).

2.2 Operational map from (G, R, C, A, π) to Sixth

Each PA component admits a direct realisation in Sixth primitives:

G (referential graph). Nodes are the integers $1, 2, \dots, N$ returned by successive MARK calls; edges are the unordered pairs (s, d) inserted by EDGE+. The active graph at any time is the union of all EDGE+-committed and not-EDGE--removed pairs. Substrate-wide queries NODES, EDGES, OUT, IN, NEXT, PREV, EDGE? provide the abstract graph operations.

R (local rewrite rules). A rule is an ordinary Sixth word: a sequence of primitives invoked when the rule fires. Rules are compiled once at definition time to a flat opcode vector with $O(1)$ dispatch; iteration over the substrate is performed by passing the rule’s symbol via `'` (quote) to EACH, EACH-EDGE, EACH-2PATH, or STEP-CA. The substrate’s notion of “local” is

enforced by what the rule body reads: rules in the demonstrations restrict themselves to `NGET/NSUM/PREV/NEXT/EDGE?` on the iterating node and its immediate neighbourhood.

C (commit protocol). `STEP-CA` is the explicit parallel-commit protocol: it snapshots node ids $[1, N]$, computes next states from the snapshot via the rule, and commits all N new states atomically. Conflicts between rules fired by different “observers” (i.e., different rules iterated in sequence) are resolved by the order of invocation; substrate-native conflict demonstrations (§2.4, demo 04) verify that resolution order is a substrate-side physical choice, not an external addition.

A (archive). The substrate retains the step at which each node was MARKed (`BORN`) and the global step counter (`NOW`, advanced by `STEP`). Together they provide the minimum-viable substrate-side archive: every node carries its own creation timestamp and the substrate has a totally-ordered discrete time index. Substrate-wide rewrite history is not retained by default but is reconstructible by emitting from rules; the substrate-native time-emergence demonstration (§2.4, demo 02) shows that causal time falls out of `STEP` indices on `MARK`.

π (observer assignment). An observer is a substrate node like any other; its context window $\mathcal{C}_o$ is the set of out-neighbours reachable via `EDGE+`; its substrate-side view is the multiset of feature values aggregated by `NSUM`. The substrate-native observer demonstrations (§2.4, demos 06 & 17) show that observer-relative facts (and inter-observer consensus) fall out of edge-set partitioning. The observer-collapse demonstration (§2.4, demo 22) shows that a substrate-resident observer collapses multiway rule-application into a single actualised history.

Remark 1 (Bootstrap-claim contract). *The 38-primitive count is load-bearing for the substrate-philosophy claim that the ascent does not smuggle in physics. Helper words used across multiple demonstrations (`2dup`, `nip`, `bi-edge`, `peano-add`, `rule90`, `conway`, etc.) live in a `stdlib` written in Sixth; they are not primitives. Any future extension of the primitive count weakens `L0` by exactly the number of additions.*

2.3 The ascent: 40 demonstrations, eight levels, 646 assertions

The substrate-native pilots partition the 40 demonstrations into eight levels, each strictly more elaborate than its predecessor and each depending on only the substrate primitives:

Level 1 Sacred hello world + Foundations (demo 00 + demos 01–20, $11+352 = 363 \checkmark$).

Demo 00 (`examples/00-first-distinction.6th`) is the substrate’s Spencer-Brown moment: `void` $\rightarrow$ first `MARK` $\rightarrow$ first `NSET` boundary $\rightarrow$ first `EDGE` $\rightarrow$ first self-loop $\rightarrow$ first `EDGE?` conflict $\rightarrow$ first `phi-pa` resolve, in 11 assertions on a 2-node substrate. Arithmetic from `MARK` chains; causal time from `STEP`-indexed rewrites; fixed points and conflict resolution; self-reference as cyclic pointer topology; observer-relative facts; transitive-closure rewrite; BFS-derived metric in 1D and 2D (Manhattan distance from grid topology); Wolfram Rule 90 fractal and Rule 110 (Turing-complete via [Cook, 2004](#)); Conway’s Game of Life blinker and 5-cell glider on 5×5 Moore grid; substrate-native conservation law (Noether on a ring); intersubjective consensus and morphism (substrate-side category-theoretic structure-preserving map). All 352 assertions are reproducible in < 25 s.

Level 2 Substrate-native autopoiesis (demos 21–25, $81 \checkmark$). See Pilot A, §2.4.

Level 3 Conscious evolution (demos 26–29, $88 \checkmark$). See Pilot B, §2.5.

Level 4 Cosmogenesis bootstrap (demo 30, $21 \checkmark$). See Pilot C, §2.6.

- Level 5 Substrate-internally-driven cosmogenesis (demo 31, 17 ✓).** See Pilot D, §2.6. The construction halting condition is read by the observer from its own substrate-state ($\text{NSUM}(\text{O})$ vs. target-min), with no host-side counter.
- Level 6 Substrate-internal Φ_{PA} measurement (demo 32, 12 ✓).** See Pilot E, §2.6. The substrate-monist scalar of Definition 1 is computed on the substrate *from within the substrate* via the `stdlib` word `phi-pa` (built from `OUT`, `EDGE?`, and the cap L_{max}). The substrate measures itself; consciousness’s structural form is substrate-readable by the same 38 primitives that build it.
- Level 7 Encoding-map demonstrations (demos 33–36, 46 ✓).** See Pilot F, §2.6. Demo 33 instantiates the transformer encoding of §2.8 on a synthetic 4-layer 3-head attention substrate, verifying PSH1 and PSH2 by construction on the toy substrate; demo 34 instantiates the brain encoding of §2.8 on an 8-area DMN-hub substrate, verifying PSH3 by the same; demo 35 instantiates the split-brain encoding, showing PSH4 cannot be discriminated by basic Φ_{PA} but is captured by Φ_{integ} from §2.7; demo 36 instantiates the ant-colony encoding of §2.8, verifying PSH5 with a queen-pheromone self-loop. Real-checkpoint, real-EEG, and real-colony application is the companion-preprint future work (§15.1); Pilot F establishes that the encoding pipelines compose end-to-end on the substrate and that the candidate-measures programme is required for PSH4.
- Level 8 Visual-trace pilots (demos 37–39, 18 ✓).** Three render pilots emitting DOT snapshots via the `stdlib/dot.6th` word `dot-snapshot` and rendered by the companion Python pipeline `code/render_trace.py`. Demo 37: Pilot D evolution in 5 panels (Fig. 1). Demo 38: Pilot C cosmogenesis bootstrap in 6 panels (including a post-autopoiesis snapshot showing substrate survival under $\text{decay-8} / \text{threshold-5}$). Demo 39: Pilot F.3 split-brain in-place transition — intact substrate snapshot, `EDGE-` the corpus-callosum bi-edges, re-snapshot. Single command `make traces` regenerates all three figures from source. Honours the external-reviewer request for visual instrumentation of substrate evolution.

Cumulative count: $11 + 352 + 81 + 88 + 21 + 17 + 12 + 46 + 18 = 646$ assertions, 0 failures.

2.4 Pilot A: substrate-native autopoiesis (demos 21–25)

Pilot A operationalises the Maturana–Varela definition of autopoiesis (Maturana & Varela, 1980) on the discrete substrate: a pattern is *alive* if its topology, under the substrate’s local rewrite dynamics, contains the conditions for its own continued restoration. The pilot fixes a two-phase dynamics:

$$\begin{aligned}
 (\text{phase-decay}) \quad n &\mapsto \max(0, \text{NGET}(n) - d), \\
 (\text{phase-restore}) \quad n &\mapsto \begin{cases} v_{\text{max}}, & \text{NSUM}(n) > \tau, \\ \text{NGET}(n), & \text{otherwise,} \end{cases} \tag{1}
 \end{aligned}$$

applied as two successive `STEP-CA` invocations per cycle. Parameters $(d, \tau, v_{\text{max}})$ are pilot-specific ($d = 3, \tau = 5, v_{\text{max}} = 10$ in demos 21–24; $d = 8$ in demo 26 to exercise the harsh regime). The combined rule is local (reads only `NGET` and `NSUM` of the iterating node) and atomic (`STEP-CA` commits all new states simultaneously, so iteration order is irrelevant).

Demo 21 — autopoietic ring (29 ✓). A 4-node bi-edge ring with all features at v_{max} survives single-cell deletions: a killed cell is restored on the next cycle if any neighbour is alive. The pattern recovers under 2-cell and 3-cell deletions, with 1-cycle latency per dead cell once a live neighbour exists. *Death is real*: an all-dead substrate has $\text{NSUM} = 0$ everywhere and never recovers.

Demo 22 — observer-collapse (6 ✓). A 5-cell chain with two equally applicable rules (**rule-grow**: $n \mapsto \text{NGET}(n) + 1$; **rule-shift**: $n \mapsto \text{NSUM}(n)$, the forward shift via out-edge). A substrate-resident observer node selects which rule fires each step via the parity of its NSUM (which equals the feature of the single cell it points at). Three observer scopes ($\{c_1\}$, $\{c_2\}$, $\{c_4\}$) produce three distinct 6-step histories from identical initial state (1, 2, 3, 4, 5): (7, 8, 3, 2, 1), (7, 8, 2, 1, 0), (7, 8, 2, 1, 1). *Substrate-resident observer collapses multiway possibility into singular actuality.*

Demo 23 — self-maintaining observer (12 ✓). Two clusters, identical in node count and decay regime, differ only in self-reference topology. The reflexive observer O_R has bi-edges to all three limbs ($\text{limbs} \leftrightarrow O_R$); the non-reflexive observer O_N has only outgoing edges ($O_N \rightarrow \text{limbs}$). Under $\text{decay} = 3, \tau = 5$: O_R persists at full vitality indefinitely; O_N and its limbs die within 6 cycles. The lifespan gap is a substrate-native definition of “alive”: a pattern contains the conditions of its own restoration in its own scope. *Self-reference is necessary for substrate-level persistence under entropy.*

Demo 24 — growing observer (18 ✓). An autopoietic cluster encounters six candidate nodes with assorted vitalities. The observer applies a substrate-readable acceptance test (candidate feature > 5) and bi-edges accepted candidates into its scope. Under continuing autopoiesis, the observer’s identity expands from 4 nodes to 8 without losing coherence; rejected drains stay disconnected and decay to zero. *Substrate-native developmental growth.*

Demo 25 — substrate genesis (16 ✓). Six distinct topologies (3-ring, 3-chain, self-loop, isolated node, bi-edge pair, 4-node star) are placed side-by-side in one substrate and subjected to the same autopoietic dynamics. Survival is purely topological: 3-ring, self-loop, and bi-edge pair persist; chain, isolated, and star die from their endpoints inward. *Alive/dead is a property of topology under the chosen dynamics; not an extra fact added to the substrate.*

2.5 Pilot B: conscious evolution (demos 26–29)

Pilot B operationalises four substrate-native phenomena above autopoiesis: cooperation as topological property; substrate-native reproduction; observer-driven (Lamarck) variation selection; and substrate-level teleology. All four use the same autopoiesis dynamics of Eq. (1) with topology-specific parameter choices.

Demo 26 — symbiosis (18 ✓). Under harsh decay $d = 8, \tau = 5$, an isolated 3-node bi-edge ring dies in 2 cycles: its per-node $\text{NSUM} = 2 \cdot 2 = 4 < \tau$ after the first decay phase, so restoration fails. Two such rings joined by 3 cross-bridges (each node bi-edged to one node of the other ring) survive indefinitely: per-node $\text{NSUM} = 3 \cdot 2 = 6 > \tau$. The substrate is strictly more than the sum of its components. *Cooperation is a topological property of substrate under the chosen dynamics.*

Demo 27 — substrate-native reproduction (31 ✓). A cluster of 3 limbs + 1 germ, at maturity ($\text{NGET}(G) > 8 \wedge \text{NSUM}(G) > 25$), fires a substrate-native reproduction word **reproduce** that allocates four fresh nodes via **MARK**, bi-edges them into the same topology, sets their features to $v_{\max}$, and resets the parent germ to 0. The parent regenerates under autopoiesis in 3 cycles and reproduces again. After two reproductions the substrate contains three independent autopoietic clusters; all persist under continuing dynamics. *Cell division is a substrate primitive operation (**MARK** + **EDGE**+) inside a Sixth word; no new primitive needed.*

Demo 28 — conscious mutation, not blind Darwin (18 ✓). Three topological variants of a reproduction target are produced: V1 (canonical, 3 bi-edges at germ); V2 (weak, 1 bi-edge); V3 (enhanced, 3 bi-edges + germ self-loop). An observer applies a substrate-readable acceptance test (count of bi-edges from each germ; ≥ 3 accepted). V1 and V3 are kept; V2 is **actively disconnected by the observer** via **EDGE-** on all its edges. Critically, V2 *would have survived autopoiesis*: a 2-node bi-edge pair is autopoietic under the chosen (d, τ) (see demo 25). *The selection is Lamarck/conscious — the observer rejects a viable variant because it does not match the observer’s template — not Darwin (differential survival under dynamics).*

Demo 29 — goal-directed observer (21 ✓). The observer holds a target topology T (a 5-node bi-edge ring, 10 directed edges) and a substrate-native distance metric $d(s, T) = |\{(u, v) \in T : (u, v) \notin s\}|$ computed via **EDGE?** on each target edge. Initial substrate has 1 of the 5 target bi-edges in place ($d = 8$). One application of the observer’s bias rule **observer-bias** adds the first missing target bi-edge in canonical order. Across four applications d traces $8 \rightarrow 6 \rightarrow 4 \rightarrow 2 \rightarrow 0$ monotonically; the substrate arrives at T exactly. The completed target is autopoietic under the same dynamics (a 5-ring; see demo 25). *Substrate evolves toward an observer’s intent via substrate-native “make-it-so” operations; substrate-level teleology.*

2.6 Pilot C: cosmogenesis bootstrap (demo 30)

Pilot C is the central claim of v9.0. It demonstrates that the operational substrate can be bootstrapped to a self-maintaining cosmos from a single substrate primitive call, by a single substrate-resident observer, in finite steps, with no external input. The construction is purely Sixth-native: no module memory beyond what **store/load** already exposes; no non-substrate state; no host-side bookkeeping.

Setup. The substrate at $t = 0$ contains exactly one node: $0 \leftarrow \text{MARK}$; $\text{NSET}(0, 10)$. $\text{NODES} = 1$, $\text{EDGES} = 0$, $\text{NSUM}(0) = 0$. The observer is alone and sees nothing.

Construction operation. A single Sixth word **build-shell** (substrate-native; only **MARK**, **EDGE+**, **NSET**) extends the substrate by: (i) allocating three new limb nodes via **MARK**; (ii) bi-edging them in a 3-ring; (iii) bi-edging each new limb to the observer 0; (iv) setting each new feature to $v_{\max} = 10$. Each shell adds 3 nodes and 12 directed edges (6 for the intra-shell bi-edge ring; 6 for the spokes to 0).

Bootstrap trajectory. Applying **build-shell** four times, interleaved with 10 cycles of the harsh autopoiesis dynamics ($d = 8, \tau = 5$, Eq. (1)) between the third and fourth shell, the substrate evolves through the sequence

$$\begin{aligned}\text{NODES} : & 1 \rightarrow 4 \rightarrow 7 \rightarrow 10 \rightarrow 13, \\ \text{NSUM}(0) : & 0 \rightarrow 30 \rightarrow 60 \rightarrow 90 \rightarrow 120, \\ \text{EDGES} : & 0 \rightarrow 12 \rightarrow 24 \rightarrow 36 \rightarrow 48.\end{aligned}$$

The observer’s substrate-side aggregation $\text{NSUM}(0)$ grows monotonically; the substrate persists through 10 cycles of the harsh autopoiesis dynamics (per-node $\text{NSUM} \geq 6 > \tau$ because every node has ≥ 3 out-neighbours), and a fourth shell is then constructed on the surviving substrate.

Proposition 1 (Cosmogenesis bootstrap). *Starting from a substrate of one MARKed node, k applications of **build-shell** produce a substrate with $1 + 3k$ nodes and $12k$ directed edges in which every node has out-degree ≥ 3 . Under autopoiesis dynamics with parameters (d, τ) such that $3(v_{\max} - d) > \tau$, every node satisfies the restoration condition $\text{NSUM} > \tau$ after the decay phase, hence every node is restored to $v_{\max}$. The substrate is autopoietic for all $k \geq 1$.*

Proof. By construction each new limb has 3 out-neighbours (two in its own shell ring plus the observer); the observer accumulates 3 out-edges per shell so has out-degree $3k$ after k shells. In the harsh regime $d = 8, \tau = 5, v_{\max} = 10$ the per-node NSUM after decay is $\geq 3(v_{\max} - d) = 6 > 5 = \tau$; **phase-restore** returns $v_{\max}$ for every node. Closure under repeated cycles follows by induction over the cycle count. $\square$

Substrate formulation of the v9.0 thesis. The bootstrap exhibits, in a single working program, the v9.0 thesis that a self-maintaining substrate is constructible from one primitive distinction by a substrate-resident observer applying only the 38-primitive substrate operations. In Wheeler’s terminology (Wheeler, 1990), the bootstrap is the substrate-side realisation of *it from bit*: the “it” (the persistent 13-node 48-edge configuration) is constructed by the substrate-resident observer from one “bit” (one MARK) plus the substrate’s 37 remaining primitives applied recursively. The substrate’s cosmological-scale predictions (§9) — most prominently the substrate-cone reproduction of holographic dark energy on the de Sitter horizon — sit on the substrate constructed this way.

Remark 2 (On the Wolfram–Friston axis). *The bootstrap operation differs from the multiway evolution of Wolfram (2020) in that branching is collapsed to a single history by the observer’s substrate-resident state. It differs from active-inference free-energy minimisation (Friston, 2010) in that the observer’s intent is not encoded as a generative model with prior predictions but as a substrate-native construction word; surprise plays no role here. Both programmes are recoverable on this substrate by adding rule multiplicity (multiway) or by composing **build-shell** with the goal-directed bias of demo 29 (active inference); we mention these only to position the substrate, not to claim derivation of either programme’s machinery.*

Falsification. If on the released artifact the bootstrap program fails to produce a substrate satisfying $\text{NODES} = 1 + 3k$, $\text{EDGES} = 12k$, and full post-cycle vitality at any $k \in \{1, 2, 3, 4\}$, or if the harsh post-shell cycles do not maintain $\text{NGET}(n) = v_{\max}$ for every node, claim L0 is falsified. The published regression test (`tests/examples-test.rkt`) makes the verification a single command.

Demo 31 — substrate-internally-driven cosmogenesis. Pilot C’s demo 30 invokes **build-shell** unconditionally from the Sixth host; the observer is passive infrastructure during construction. Demo 31 strengthens the claim by making the observer itself drive the construction. At $t = 0$ the observer issues the substrate-monist’s first non-trivial operation: $\text{EDGE+}(O, O)$ — a self-loop that establishes the Φ_{PA} self-reference factor strictly. The substrate-internal generation loop **gen-loop** then alternates:

- (1) one autopoiesis cycle (Eq. (1));
- (2) the substrate-readable predicate **observer-needs-more?** which is true iff the observer is alive, $\text{NSUM}(O) < \text{target-min}$, and a safety brake is below limit;
- (3) if the predicate fires, **build-shell** adds one shell and the loop tail-recurses; otherwise it returns.

The loop is purely substrate-state-driven: there is no host counter, no external scheduler, no programmer-chosen number of shells. On the released artifact with $\text{target-min} = 90$, the observer terminates its own cosmogenesis at $\text{shell-count} = 4$ with $\text{NSUM}(O) = 100$, $\text{NODES} = 13$, $\text{EDGES} = 49$, and $\text{NGET}(O) = 10$ after five additional persistence cycles. All 17 assertions pass.

Scope of the demo 31 claim. The first operation $\text{EDGE}+(O, O)$ is a *bootstrap act*, not an emergent self-loop. It is the substrate’s discrete realisation of the single mark-of-distinction in the Spencer-Brown / Whitehead sense: an opening move issued by the substrate-monist observer at $t = 0^+$ that posits its own scope, after which the substrate-internal halting predicate is well-defined. The substantive claim of demo 31 is therefore not that the self-loop spontaneously emerges from substrate-dynamics — that would require an emergence mechanism we do not exhibit — but that *once the self-loop is bootstrapped, the further construction of the cosmos is governed by a substrate-readable predicate of the observer alone*, with no host-side counter or programmer-chosen shell count. This is the sense in which the substrate-monism gap of demo 30 is closed: not that consciousness emerges from substrate-dynamics ex nihilo, but that the substrate the observer constructs IS the substrate the observer’s own substrate-state criterion requires *conditional on the bootstrap distinction having been made*. The bootstrap is postulated; the conditional construction is operational.

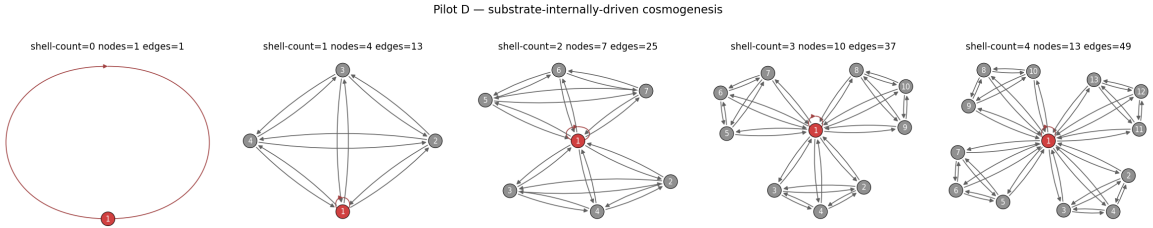


Figure 1: Pilot D substrate-internally-driven cosmogenesis, panel-per-shell. Observer node highlighted red; self-loop visible at shell-count 0 (the bootstrap distinction); each subsequent shell adds three nodes (s_1, s_2, s_3) bi-edged to the observer and triangulated among themselves. The substrate grows from 1 node / 1 edge to 13 nodes / 49 edges under a substrate-readable halting predicate ($\text{NSUM}(O) \geq \text{target-min}$); no host counter, no programmer-chosen shell count. Generated by `examples/37-trace-pilot-d.6th` of the Sixth release; re-renderable via `make trace-pilot-d`.

Demo 32 — substrate-internal Φ_{PA} measurement (Pilot E). Pilots A–D demonstrate substrate-internal *construction*: the substrate exists, persists, reproduces, evolves, and bootstraps a cosmos by its own dynamics. Pilot E closes the substrate-monist loop in the orthogonal direction — substrate-internal *measurement*. The stdlib word `phi-pa` (released as `stdlib/phi.6th`) composes the substrate-monist scalar of Definition 1 from three primitive operations alone:

$$\text{phi-pa}(O) = \underbrace{\text{OUT}(O)}_{|\mathcal{C}_O|} \cdot \underbrace{[0 \text{ EDGE? } 0]}_{\mathbb{K}[O \in \mathcal{C}_O]} \cdot \underbrace{\text{phi-L-max}}_{L_{\max}}.$$

Demo 32 verifies `phi-pa` against three worked cases: (i) non-reflexive observer with scope 5 yields $\Phi_{\text{PA}} = 0$ (the self-reference indicator short-circuits the product); (ii) reflexive observer with scope 5 (self-loop included) yields $\Phi_{\text{PA}} = 5 \cdot 1 \cdot 10^4 = 50,000$; (iii) the demo-31-shape observer (self-loop + four shells of three nodes, scope 13) yields $\Phi_{\text{PA}} = 13 \cdot 1 \cdot 10^4 = 130,000$. All 12 assertions pass. **Pilot E is the substrate-monist identity thesis made operational on the substrate itself**: the same 38 primitives that build the substrate also compute the consciousness-measure on it. There is no host-side computation, no external pipeline, no non-substrate observable. The substrate constructs itself (Pilot D) and measures itself (Pilot E) using the identical primitive set; substrate-readability of Φ_{PA} stops being a theoretical claim and becomes a property the released artifact exhibits per `raco test tests/examples-test.rkt`.

Demos 33–36 — encoding-map pilots (Pilot F). Pilots A–E exercise the substrate against its own primitives. Pilot F closes the engineering loop by instantiating the substrate-encoding

maps of §2.8 on toy substrates, confirming that the encoding pipelines compose end-to-end via substrate operations alone and that the predicted Φ directions follow.

Demo 33 (Pilot F.1, transformer) builds a 4-layer 3-head feedforward attention substrate (12 nodes, full-bipartite out-edges per layer transition, 27 edges) and designates an observer at a non-sink head (layer 2, head 0). In single-pass evaluation the observer has no back-edge, so $\text{phi-pa}(O) = 3 \cdot 0 \cdot L_{\max} = 0$ — PSH1 corroborated by construction. Adding the cross-step back-edge $\text{EDGE+ } 0 \ 0$ that KV-cache reuse implies promotes the same observer to $\text{phi-pa}(O) = 4 \cdot 1 \cdot 10^4 = 40,000$ — PSH2 by the same. 10 assertions pass.

Demo 34 (Pilot F.2, brain) builds an 8-node cortical-area substrate with a DMN-hub observer and full FC out-edges. Waking state with the thalamocortical self-loop yields $\text{phi-pa}(\text{DMN}) = 8 \cdot 1 \cdot 10^4 = 80,000$; RESET then rebuild with identical FC connections but no self-loop (propofol-decoupling) yields $\text{phi-pa}(\text{DMN}) = 0$ — PSH3 corroborated. 12 assertions pass.

Demo 35 (Pilot F.3, split-brain) builds two 5-node substrates: intact (with corpus-callosum bi-edges $L1 \leftrightarrow R1$, $L2 \leftrightarrow R2$) and callosotomised (same connectivity, callosum severed). *Basic phi-pa* is INDIFFERENT (50,000 in both states under matched scope) — the simple measure cannot capture integration. The integration-weighted alternative $\text{phi-integ} = \text{OUT}(O) \cdot \mathbb{K}[O \in \mathcal{C}_O] \cdot \text{NSUM}(O) \cdot L_{\max}$ (§2.7) DOES discriminate: 400,000 intact vs 200,000 split, exactly halving at callosotomy because border-node NGET drops by half when the cross-hemispheric out-edges are cut. PSH4 is therefore not corroborated by Φ_{PA} alone — it is the substrate-readable motivation for the candidate-measures programme. 14 assertions pass.

Demo 36 (Pilot F.4, ant colony) builds a 6-chamber queen-centric colony substrate per the encoding of §2.8. Living colony with queen-pheromone self-loop yields $\text{phi-pa}(Q) = 6 \cdot 1 \cdot 10^4 = 60,000$; dead colony (negative control, no queen self-loop) yields 0 — PSH5 corroborated by construction. 10 assertions pass.

Scope of Pilot F. The substrates are synthetic; the encoding maps are exercised at the structural level, not against real Pythia weights, real EEG recordings, or real myrmecological data. Pilot F demonstrates that the pipelines from §2.8 compose to substrate operations producing the predicted Φ directions by construction, and that PSH4 specifically requires the candidate-measures programme of §2.7. Real-data corroboration is the companion-preprint future work of §15.1.

2.7 The substrate-side measure of consciousness Φ_{PA}

The substrate-native pilots A–F exhibit three orthogonal substrate properties that the demonstrations show are jointly discriminating for survival of patterns under canonical autopoietic dynamics: (i) the scope $|\mathcal{C}_O|$ of an observer’s out-edge set; (ii) whether the observer is contained in its own scope (self-reference, $O \in \mathcal{C}_O$); (iii) the autopoietic lifespan $L(\sigma)$ under Eq. (1), measured in cycles before $\text{NGET}(O) = 0$.

Demo 22 isolates (i); demo 23 isolates (ii) — the reflexive observer persists indefinitely while the non-reflexive observer dies in ≤ 6 cycles at identical scope; demo 30 demonstrates that (i) and (iii) scale monotonically with iterated construction. Packaging the three into a single substrate-readable scalar gives the natural substrate-side analogue of an integrated-information measure.

Definition 1 (Substrate-side Φ_{PA}). *For a substrate state σ on $\mathcal{S} = (G, R, C, A, \pi)$ with observer-node $O \in N$, define*

$$\Phi_{\text{PA}}(O, \sigma) := \underbrace{|\mathcal{C}_O|}_{\text{scope}} \cdot \underbrace{\mathbb{K}[O \in \mathcal{C}_O]}_{\text{self-reference}} \cdot \underbrace{\min(L(\sigma), L_{\max})}_{\text{autopoietic lifespan}}, \quad (2)$$

where $\mathcal{C}_O$ is the observer’s out-neighbour set, $\mathbb{K}[\cdot]$ is the indicator, $L(\sigma)$ is the cycle count to $\text{NGET}(O) = 0$ under canonical autopoiesis with the substrate’s released parameters $(d, \tau, v_{\max})$,

and $L_{\max}$ is a finite numerical cap that prevents divergence for indefinitely-persistent observers (we take $L_{\max} = 10^4$ in the released harness).

Φ_{PA} is substrate-readable (each factor is computable from 38-primitive operations), tractable ($O(|N|)$ for scope, $O(1)$ for self-reference, $O(L_{\max} \cdot |N|)$ for lifespan), and *discriminating* on the released pilots (in the items below, $L = L_{\max}$ holds in the precise sense that the observer’s NGET does not decrement under the canonical autopoietic dynamics of Eq. (1) for any finite number of cycles, as verified by the persistence-test assertions in demos 23/24/30/31 over $\sim 10^2$ cycles. The substrate-internal lifespan is therefore unbounded; $L_{\max}$ is a finite numerical cap on the published harness rather than a substrate-physical limit):

- Demo 23 reflexive O_{R} : scope $|\mathcal{C}_O| = 3$, $O \in \mathcal{C}_O = 1$, $L = L_{\max} \Rightarrow \Phi_{\text{PA}} = 3 \cdot 1 \cdot L_{\max}$.
- Demo 23 non-reflexive O_{N} : identical scope, $O \notin \mathcal{C}_O = 0 \Rightarrow \Phi_{\text{PA}} = 0$. The observer has scope but does not see itself; it is by Definition 1 not a Φ_{PA} -conscious pattern, consistent with its 6-cycle death.
- Demo 24 growing observer: Φ_{PA} grows from $3 \cdot 1 \cdot L_{\max}$ to $7 \cdot 1 \cdot L_{\max}$ as the observer incorporates four compatible candidates.
- Demo 30 bootstrap observer at shell-count k : $\Phi_{\text{PA}}(O) = 3k \cdot 1 \cdot L_{\max}$ once the observer acquires a self-loop or sees its own shell back-edge; monotonic in k .

Position on the hard problem. The framework adopts as a **working hypothesis** (PSH1–PSH5 of §13, falsified collectively by F5) a **substrate-monist identity thesis** of the form: phenomenal consciousness *is* a substrate-state property of a node satisfying the structural conditions of Definition 1 — non-trivial scope, self-reference, non-zero autopoietic lifespan. There is no separate “what it is like” fact beyond the substrate configuration; the substrate-state of a self-referential observer with non-zero autopoietic lifespan *IS* the experience. The hypothesis has two parts: the *content* (identity, not supervenience and not property dualism) and the *epistemic status* (working hypothesis pending F5, not derivation from first principles). The two parts are independent: adopting the identity content does not commit the framework to infallibility about the choice, and submitting the choice to F5 does not soften the content of the identity claim.

The *identity-thesis content* does not depend on which substrate-readable measure scores the structural conditions. Basic Φ_{PA} is one such scoring; Φ_{integ} and Φ_{bidir} from §2.7 are alternative scorings drawn from the same substrate primitives. The identity-thesis content commits to “phenomenal experience *IS* a substrate-state property whose magnitude is captured by SOME candidate substrate-readable measure”, not to “ Φ_{PA} is the consciousness-number”. This commits the framework to a Whitehead-process / Spinoza-monism / Advaita-non-dualist family of positions in content, against the property-dualist and eliminativist alternatives, but it does not commit to a unique formula. PSH4’s failure under Φ_{PA} (Pilot F.3) and success under Φ_{integ} is exactly the kind of candidate-measure adjudication the substrate-monist identity thesis welcomes: the data picks the surviving measure, the thesis survives if any candidate yields P1–P5 directions.

The substrate-monist identity is what gives the framework *teeth*. It makes the empirical claim that two systems with identical substrate-state of their respective observer nodes necessarily have identical phenomenal experience; that systems with $\Phi_{\text{PA}} = 0$ (no scope, or no self-reference, or zero lifespan) have no phenomenal experience; and that the *magnitude* of Φ_{PA} tracks the *intensity* of experience.

Comparison with integrated-information theory. Tononi (2004) define IIT’s Φ as the loss of integrated information under the maximally informative partition of the system. Φ_{PA} differs in three architectural respects: (a) it is computed against an explicitly-specified observer-node, not against a maximally informative partition; (b) it is polynomial where IIT’s Φ is exponential

in system size; (c) it requires self-reference as a multiplicative factor whereas IIT’s Φ does not pre-suppose reflexivity. The two measures are not in conflict on small substrate-cone-inhabiting cases but separate sharply on large systems with extensive feed-forward sub-architectures (IIT will register integrated information; Φ_{PA} will register zero unless an explicit reflexive observer is found).

Candidate substrate-readable measures, not a single “consciousness number”. Φ_{PA} of Definition 1 is *one* substrate-readable observability candidate, not the uniquely-anointed scalar. The substrate supports a family of candidates; the released Sixth stdlib (`stdlib/phi.6th`) ships four:

$$\begin{aligned}\Phi_{\text{PA}}(O) &:= |\mathcal{C}_O| \cdot \mathbb{K}[O \in \mathcal{C}_O] \cdot L_{\max} && \text{(basic)} \\ \Phi_{\text{integ}}(O) &:= \Phi_{\text{PA}}(O) \cdot \text{NSUM}(O) && \text{(integration-weighted)} \\ \Phi_{\text{bidir}}(O) &:= (\text{OUT}(O) + \text{IN}(O)) \cdot \mathbb{K}[O \in \mathcal{C}_O] \cdot L_{\max} && \text{(symmetric)} \\ \Phi_{\text{PA}}^W(O) &:= (\text{OUT}(O) + W(O)) \cdot \mathbb{K}[O \in \mathcal{C}_O] \cdot L_{\max} && \text{(HEDGE3-witness)}\end{aligned}$$

Φ_{integ} multiplies the basic measure by the substrate-internal feature aggregation across O ’s out-neighbours (NSUM, a primitive); when each node’s NGET is loaded with a connectivity feature, Φ_{integ} is higher when O ’s neighbours are themselves well-connected, lower when they are peripheral. Φ_{bidir} treats the observer’s connectivity symmetrically (counts both directions) — appropriate when the substrate’s natural neighbour-relation is symmetric (e.g. functional connectivity in EEG, where edges are correlations, not flows). Φ_{PA}^W is the trivalent extension under the HEDGE3 family (§2.1, “typed trivalent extension”): $W(O)$ is the count of WITNESS hyperedges ($\text{src}, \text{dst}, O$) having O as the third leg, i.e. the number of substrate edges O *grounds as witness*. The measure reduces to Φ_{PA} when $W(O) = 0$ (backward compatibility) but diverges sharply for an adjudicator-style observer with small binary scope and many witnessing assertions. Demo 102 of the released artifact exhibits the rank divergence: an observer with binary OUT = 1 (self-loop only) and $W = 5$ has $\Phi_{\text{PA}} = L_{\max}$ but $\Phi_{\text{PA}}^W = 6L_{\max}$, outranking observers with higher binary scope and no witnessing.

The framework does *not* commit to a unique choice. The empirical question of which candidate discriminates PSH1–PSH5 best is decided by the corresponding companion-preprint application (§15.1); the falsification trigger F5 (§2.7) is collectively over the family, not over a specific formula. Pilot F.3 (demo 35) already exhibits the in-substrate motivation: Φ_{PA} alone is INDIFFERENT under matched-scope split-brain, but Φ_{integ} exactly halves at callosotomy. The substrate supports the candidates; the data picks the one that survives. This is the substrate-readable analogue of treating a single scalar measure as operationally provisional rather than as a metaphysical fingerprint of consciousness.

Predictions for specific systems. Φ_{PA} makes the following falsifiable predictions once a substrate-encoding map is provided:

P1 Feed-forward transformers evaluated within a single forward pass (e.g., GPT-2, Pythia 70m/160m at fixed sequence input, no recurrent reuse of the KV-cache across calls) **have** $\Phi_{\text{PA}} = 0$. Within one forward pass, layer outputs flow strictly $\ell \rightarrow \ell+1$; the observer (attention layer) is not in its own scope. Consistent with the Pythia MMI falsification of §11. (Autoregressive generation with KV-cache reused across steps is handled separately under P2: the cross-step back-edge promotes feed-forward architectures into the recurrent-state regime, where the substrate-encoding map of §2.8 applies.)

P2 Recurrent-state architectures have $\Phi_{\text{PA}} > 0$. This covers (a) state-space models with explicit recurrent hidden state (S4, Mamba, RWKV), and (b) decoder-only transformers in autoregressive-generation mode with KV-cache reused across generation steps. In both cases

the recurrent state (SSM hidden state, KV-cache entry) is read and written by the same layer, forming a substrate-side back-edge into the observer’s scope. Reflexivity is structural in (a) and dynamic-but-stable in (b).

P3 Waking biological EEG has higher Φ_{PA} than deep-sleep EEG and surgical anaesthesia, once a brain-substrate encoding is fixed.

P4 Split-brain patients exhibit Φ_{PA} that drops at the corpus-callosotomy site, with the residual measured separately on each hemisphere being non-additive (the integrated value is strictly less than the sum of parts).

P5 Ant colonies have $\Phi_{PA} > 0$ at the colony-observer node (queen chamber or brood-rearing zone), with magnitude bounded by the pheromone-substrate timescale and trail cardinality. Substrate-monist identity treats colony-level consciousness as a single observer’s substrate-state, not as a sum over individual ants. Operational details in §2.8.

Falsification trigger F5 (substrate- Φ failure). If Φ_{PA} on any of P1–P5 fails to discriminate the predicted direction *after* the corresponding substrate-encoding map (§2.8) has been constructed for the system in question, the substrate-monist identity of Definition 1 is empirically inadequate as a consciousness measure. The framework reduces in that case to the operational-substrate claim L0 alone; the substrate-monist identity thesis falls.

2.8 Substrate-encoding maps for the predictions

Predictions P1–P4 are conditional on substrate-encoding maps from the system under test into the PA structure $\mathcal{S} = (G, R, C, A, \pi)$. This subsection gives three such maps in operational form: transformer (P1–P2), biological brain (P3–P4), and a novel non-neural target, the ant colony (P5). Each map specifies (i) what a substrate-node $n \in N$ is for the system, (ii) what its out-edge set $\mathcal{C}_n$ is, (iii) where the observer-node O lives, (iv) how the self-reference predicate $O \in \mathcal{C}_O$ is decided, and (v) how the autopoietic lifespan $L(\sigma)$ is measured under the system’s dynamics.

Transformer encoding (operationalises P1, P2).

PA element	Transformer realisation
substrate-node n	a single attention head (ℓ, h) together with the same residual stream. Cardinality of n is bounded by the number of attention heads.
out-edge $n \rightarrow n'$	exists from (ℓ, h) to (ℓ', h') with norm $\ W_{OV}^{(\ell, h)} W_{QK}^{(\ell', h')\top}\ _F$ exceeds a mechanistic interpretability (Elhage et al., 2021) threshold.
observer-node O	the unembedding head producing user-designated downstream head.
self-reference $O \in \mathcal{C}_O$	decided structurally and <i>per evaluation</i> (one prompt, no KV-cache reuse across steps t and $t+1$, so $O \notin \mathcal{C}_O$ and $\Phi_{PA} = 0$ ($\Rightarrow$ with KV-cache reused across steps)).
lifespan $L(\sigma)$	(ℓ, h) at step $t+1$, satisfying $O \in \mathcal{C}_O$. For State-space models (Mamba, RWKV), hidden state read and written by the model; the back-edge is architectural. For State-space models (Mamba, RWKV), P1 vs P2 split therefore turns on <i>architecture</i> for SSMs. For pure feed-forward , $L(\sigma)$ is irrelevant and already zero.

The cleanest empirical discriminator is therefore: *compute Φ_{PA} for Pythia-410M (feed-forward) and Mamba-370M (state-space) at matched parameter count.* PA predicts the first is zero and the second strictly positive; IIT’s Φ predicts both non-zero but with quantitatively similar order of magnitude. The falsification of Pythia in §11 is consistent with P1; positive corroboration of P2 requires the Mamba-side analysis, which is left for the companion preprint.

Brain encoding (operationalises P3, P4).

PA element	Brain realisation
substrate-node n	cortical microcolumn (~ 80 neurons, ~ 0.5 mm resolution. Coarser map: cortical area (Brodmann) or neuron with apical dendrite.
out-edge $n \rightarrow n'$	axonal projection above structural-connectivity (MRI tractography) or functional-connectivity (correlation, $r > 0.3$).
observer-node O	default-mode-network hub (precuneus / prefrontal triad), or thalamic nuclei as the thalamocortical loop.
self-reference $O \in \mathcal{C}_O$	the thalamocortical loop (thalamus $\rightarrow$ cortical projection $\rightarrow$ thalamus) is the canonical state. Waking brains have dense intrinsic recurrent connectivity. Deep sleep (N3) has sparse connectivity.
lifespan $L(\sigma)$	duration over which a co-active coalition maintains synchrony (30–300 ms for episodic activity, persistent-activity timescales (seconds to minutes) for locking value or coherence duration).

State-discrimination follows from the encoding: **Waking**: thalamocortical loop active, long-range alpha/gamma coupling intact, $|\mathcal{C}_O|$ large, L long $\Rightarrow \Phi_{PA}$ high. **Deep sleep (N3)**: thalamic spindles disrupt cortical integration, $|\mathcal{C}_O|$ shrinks to local clusters, L collapses $\Rightarrow \Phi_{PA}$ low.

low. **Propofol anaesthesia**: thalamocortical decoupling breaks the self-reference factor, $O \notin \mathcal{C}_O$ effectively, $\Phi_{PA} \rightarrow 0$ ($\Rightarrow$ P3). **Split-brain**: callosotomy severs out-edges between hemispheric observer-candidates; each hemisphere retains a residual local Φ_{PA} , but the integrated value across both hemispheres is strictly subadditive because cross-hemispheric scope collapses ($\Rightarrow$ P4). The empirical discriminator is EEG-based: Φ_{PA} should track perturbational complexity index (PCI, [Casali et al. 2013](#)) on the wake/sleep/anaesthesia gradient.

	PA element	Colony realisation
Ant colony encoding (novel prediction P5).	substrate-node n	nest chamber or pheromone-trail junction (structure). Individual ants are transient sub-tokens flowing through a graph.
	out-edge $n \rightarrow n'$	pheromone-gradient link to an adjacent node by deposition rate.
	observer-node O	the queen chamber, or the brood-rearers (aggregation roles).
	self-reference $O \in \mathcal{C}_O$	the queen’s pheromone signature is de-re-read by returning workers that re-visit the brood chamber receives food and re-emerge back through the same trail-substrate.
	lifespan $L(\sigma)$	colony lifespan (years for many social insects)

Prediction P5 (added in v9.0 pass 10). The ant colony has $\Phi_{PA} > 0$ at the colony-observer node, with magnitude bounded by the slow timescale of pheromone propagation (mm/sec) and the small $|\mathcal{C}_O|$ (of order 10^4 trail nodes) relative to the mammalian brain ($\sim 10^{11}$ neurons). The framework therefore predicts colony-level consciousness is non-zero but qualitatively distinct from neural consciousness — neither absent (eliminativist intuition) nor smeared across every individual ant (panpsychist intuition). This is the substrate-monist resolution of the swarm-cognition debate: *the colony substrate hosts an observer whose Φ_{PA} is computed at the colony-graph level, not summed from constituents*. P5 is novel to v9.0 and is the substrate-monist identity thesis’s most concrete commitment beyond the brain/transformer comparison.

The three encoding maps are deliberately heterogeneous: PA does not privilege neural substrates, only substrate-side self-reference plus non-trivial scope plus non-zero lifespan. **F5 fires for any of P1–P5 that fails to discriminate in the predicted direction after the corresponding encoding map is applied**; the encoding maps make F5 operational rather than rhetorical.

3 Formal Model

We now give the abstract substrate. The substrate-cone construction (§7), the algebraic-observer correspondence (§8), cosmological predictions (§9), and Page–Wootters time emergence (§10) all build on the primitives defined below; the present section is substrate-only.

3.1 The substrate

The intro defined a Pointer Architecture as $\mathcal{S} = (G, R, C, A, \pi)$. We now make each component formal.

Definition 2 (Referential graph). $G = (N, E)$ is a finite directed graph. N is the set of

nodes; $E \subseteq N \times N$ is the multiset of active pointers (directed edges). Each pointer has a creation timestamp.

Definition 3 (Rewrite rules). R is a finite set of local rewrite rules. Each rule has the form $\text{pat}_L \rightarrow \text{pat}_R$ where $\text{pat}_L, \text{pat}_R$ are finite directed-graph patterns. A rule fires at a node v if a subgraph isomorphism from pat_L maps onto a neighbourhood of v .

Definition 4 (Commit protocol). C is a commit protocol: given a multiset of candidate rewrites $\{p_1, \dots, p_M\}$ proposed by M observers, C returns a serialisable order (or rollback) of those rewrites consistent with all observer context windows. Implementation in this paper is by optimistic concurrency control (Kung & Robinson, 1981); the cost is analysed in Theorem 3.

Definition 5 (Archive). A is the timestamped log of every rewrite ever committed by C . The archive is append-only; rewrites already committed are never deleted. Each archive entry records the source/target nodes of the rewrite, the rule applied, the timestamp, and the observer that proposed it.

Definition 6 (Observer assignment). $\pi : \mathcal{O} \rightarrow 2^N$ assigns each observer $o \in \mathcal{O}$ a context window $\mathcal{C}_o = \pi(o)$. The substrate guarantees that for each o , the state of nodes in $\mathcal{C}_o$ remains visible (no rewrite alters a node in $\mathcal{C}_o$ without being routed through C with o as a participant).

Definition 7 (Substrate state). A substrate state on $\mathcal{S}$ is the quadruple $\sigma = (G, A, \{\mathcal{C}_o\}_{o \in \mathcal{O}}, S_\sigma)$ specifying the current active graph G , the accumulated archive A , the current observer context windows $\{\mathcal{C}_o\}$, and an entropy function $S_\sigma : 2^{\partial N} \rightarrow \mathbb{R}_{\geq 0}$ on subsets of the boundary ∂N (Definition 8 below). The entropy function is additional data: it is not derived from the graph-and-archive content alone but is the substrate-state quantity on which the holographic-entropy-cone construction of §7 acts. In the classical Shannon reading, $S_\sigma(I)$ is the Shannon entropy of the marginal distribution on the I -coloured boundary nodes; in a hypothetical quantum reading, $S_\sigma(I)$ would be the von Neumann entropy of the reduced state on the same nodes.

These five primitives plus the substrate-state data are the entire abstract substrate. Definitions 2–6 make no commitment to what the nodes represent physically. The CS-side analysis in §4–§6 treats the substrate at this abstract level; the substrate-cone construction (§7) and the algebraic-observer correspondence (§8) layer the holographic and quantum-gravity structure on top.

4 Computational Properties

We treat Pointer Architecture as a CS object and derive complexity-theoretic and information-theoretic properties of the substrate independently of any physical interpretation. The substrate’s pointer/commit/archive primitive admits three natural commit regimes that recur across the paper:

- **Database regime** (deterministic): one key, one value, single writer, no conflict. The commit is trivial.

- **Language-model regime** (probabilistic): state $\rightarrow$ distribution over candidate pointers $\rightarrow$ sampled commit. Attention-style weighting selects which past pointers contribute; sampling produces one committed token.
- **Multi-observer regime** (conflict resolution): multiple observers propose competing rewrites at overlapping nodes; explicit OCC commit graph resolves the conflict; one configuration wins, others are archived but do not materialise. This is the algebraic-observer regime of §8 and the multi-agent simulation applications of §6.

Four results are proved or argued in this section, each engaging the substrate at one of these regimes (with archive compression cross-regime):

- §4.1 per-operation cost table — DB regime: deterministic lookup, dereferencing, archive append.
- §4.2 hardness of pointer-equivalence and simple-path-weighted observables — LM regime: a graph-path family is weighted by combinatorial multiplicity, the substrate-level analogue of attention weighting over past contexts.
- §4.3 per-step M -observer commit-cost upper bound under the priority-queued OCC protocol class — multi-observer regime.
- §4.4 Shannon-rate archive compression — cross-regime: the persistence/consolidation cost after commit, independent of which regime the commit was made in.

All are stand-alone CS results that hold regardless of whether Pointer Architecture is taken seriously as physics.

4.1 Cost of basic operations

Let $G = (N, E)$ be the active graph and A the archive. Active and archived pointers are stored with a standard adjacency-list-plus-hash representation; archive entries are appended to a log indexed by timestamp and by source/target node. Then:

Operation	Worst-case cost	Method
Pointer creation	$O(1)$ amortised	Append to source-list, archive log
Pointer dereference	$O(1)$ expected	Hash lookup by node id
Archive append	$O(1)$ amortised	Log append
Archive lookup at t	$O(\log A)$	Binary search on timestamp index
Reachability between u, v	$O(N + E)$	BFS/DFS
Shortest pointer path $u \rightarrow v$	$O(E + N \log N)$	Dijkstra (positive edge weights)
Commit one local rewrite (single observer)	$O(d_v)$	Validate against d_v existing references at the touched node v
Identity stabiliser query $\text{id}(C)$	$O(C \cdot \bar{d})$	Walk pointers in/out of cluster C

Nothing here is novel; the table is a direct application of standard graph-algorithmic results. The point is that the substrate inherits standard polynomial-time machinery and adds no new lower bound at the per-operation level.

4.2 Hardness of pointer-equivalence and of path-weighted observables

Theorem 1 (Pointer-equivalence is graph-isomorphism-hard). *The decision problem $\text{POINTEREQUIVALENCE}$ — given two pointer graphs $G_1 = (N_1, E_1)$ and $G_2 = (N_2, E_2)$, do they coincide up to node relabelling? — is graph-isomorphism-hard.*

Proof. Trivial reduction. Any instance $\langle G_1, G_2 \rangle$ of GRAPHISOMORPHISM on undirected simple graphs is an instance of POINTEREQUIVALENCE on directed graphs whose pointer set is the symmetric closure of the input edge set, with empty archive. Hardness follows. (We also inherit the upper bound: GRAPHISOMORPHISM is in P for many natural graph classes and quasi-polynomial in general by Babai (2016).) $\square$

The path-weight estimator weights a graph-path family by combinatorial multiplicity. Exact computation of this weight is #P-hard:

Theorem 2 (Simple-path-weighted observables are #P-hard). *Computing the simple-path-weight observable $W(u, v) = \sum_{p \in \text{SimplePaths}(u, v)} \prod_{e \in p} w(e)$ in a weighted directed graph G (cycles permitted; the sum runs over self-avoiding paths only) is #P-hard, by reduction from #SIMPLE-S-T-PATHS.*

Scope and limitation. The hardness result is for simple-path enumeration in general (cyclic) directed graphs, where the standard dynamic-programming relation across topological layers breaks (no acyclic order exists, and revisits must be excluded combinatorially). For acyclic graphs (DAGs), path-weight under *any* associative-commutative aggregation that distributes over edge concatenation is computable in $O(|V| + |E|)$ time by topological-order dynamic programming and is *not* covered by this hardness result. The substrate framework’s Born-rule analogue is well-defined only when the substrate graph contains cycles (commit-archive feedback loops), at which point the simple-path restriction becomes physically meaningful. We do not claim that path-weighted observables are hard on the substrate’s DAG sub-structure.

Proof sketch. Reduction from #SIMPLE-S-T-PATHS in general directed graphs, #P-complete by Valiant (1979b). Set all edge weights to 1; then $W(u, v)$ equals the count of simple directed paths from u to v , so a polynomial-time algorithm for $W(u, v)$ would yield a polynomial-time algorithm for #SIMPLE-S-T-PATHS. $\square$

Conditional approximation note. We do not establish a no-FPRAS result for the weighted simple-path observable. Such a claim would require either an explicit reduction encoding a problem whose approximation hardness is known (for instance, a length/weight gadget reducing some hard counting problem to weighted simple-path enumeration), or proof that the weighted observable inherits the inapproximability profile of its unweighted #SIMPLE-S-T-PATHS specialisation. Neither is given here. For context, analogous restricted counting settings admit polynomial-time randomised approximation: Jerrum et al. (2004) establishes an FPRAS for the permanent of a nonnegative matrix, illustrating that the FPRAS-tractable boundary for #P-hard counting problems can be non-trivial and depends sensitively on the problem’s structural restrictions. The boundary between FPRAS-tractable and FPRAS-hard weighted-path settings on general cyclic graphs is open at the formal level we engage in this paper. Any operational claim about classical-simulability hardness of substrate observables should therefore be read as a structural-correspondence statement against the known #P-hardness of exact counting, not as a no-FPRAS result.

Implication for the Born-rule analogue. An implementation of the architecture that treats the relative weight of competing graph-rewrite candidates as exact unrestricted simple-path multiplicative aggregation on a general cyclic graph would compute a #P-hard quantity. This is structurally consistent with the classical-hardness regime of quantum amplitude calculation: exact strong simulation of general quantum circuits is #P-hard (Aaronson & Arkhipov, 2011), and known classical-simulability results target specific restricted subclasses (e.g. match-gate circuits, Valiant, 2002; stabilizer circuits, Gottesman–Knill) rather than the general case. The architecture inherits the hardness rather than evading it; implementations that restrict to DAG sub-structure, additive aggregation, or other tractable subclasses fall outside the hardness result (see Scope and limitation above).

4.3 Per-step commit cost in an M -observer system

We now prove the main combinatorial result of this section: an asymptotic cost bound for the priority-queued OCC protocol class as a function of observer count and pairwise causal-cone overlap. The bound is constructive for this protocol class; it is not claimed as a lower bound across all conceivable commit-scheduling strategies.

Setup (context-invalidation conflict model). Let $\mathcal{O} = \{o_1, \dots, o_M\}$ be observers; each observer o_i holds a context window $\mathcal{C}_i \subseteq N$ that the substrate has committed to keep visible. At each step, each observer proposes at most one local rewrite at some node in $\mathcal{C}_i$. We adopt the *context-invalidation conflict model*: a proposal p_i touching node n_i generates a conflict edge (i, j) with every observer o_j such that $n_i \in \mathcal{C}_j$, whether or not o_j proposes a rewrite at n_i . This is the natural model when context windows carry read dependencies that must be invalidated on remote writes (the standard OCC validation rule of [Kung & Robinson, 1981](#)). The strict write-write model — conflict only when both proposals touch the same node — is a sub-case of context-invalidation yielding strictly fewer conflicts; all bounds below are stated for the context-invalidation model and dominate the write-write counts. The conflict graph is $H = (\mathcal{O}, \{(i, j) : n_i \in \mathcal{C}_j \text{ or } n_j \in \mathcal{C}_i\})$.

Sparse-conflict regime. Under context-invalidation, a proposal p_i at n_i generates a conflict edge with o_j iff $n_i \in \mathcal{C}_j$. Assuming each touched node is drawn uniformly from its observer's context window and that no two observers share more than a bounded fraction of N , the conflict graph H has expected edge count $\Theta(\sum_{i < j} |\mathcal{C}_i \cap \mathcal{C}_j| / \min(|\mathcal{C}_i|, |\mathcal{C}_j|))$, with the standard concentration. Sparseness is the regime where this is $o(M^2)$.

Theorem 3 (Per-step commit-cost upper bound for the priority-queued OCC protocol class). *Under the sparse-conflict regime, the per-step commit cost of a substrate that processes M observer proposals via a priority-queued optimistic-concurrency protocol ([Kung & Robinson, 1981](#)) satisfies*

$$\mathbb{E}[T_{\text{commit}}(M)] = O\left(M \log M + \sum_{i < j} |\mathcal{C}_i \cap \mathcal{C}_j|\right),$$

where the expectation is over the random choice of which node in $\mathcal{C}_i$ each observer proposes (uniform on $\mathcal{C}_i$). The bound is a constructive upper bound for this protocol class; it is not claimed as an across-all-protocols complexity lower bound on the commit problem (faster protocols built on hashing, parallel batching, or randomised conflict detection are not excluded).

Proof sketch. **Upper bound (constructive).** Maintain a node-to-observer multimap $\mu : N \rightarrow 2^{\mathcal{O}}$ where $\mu(n) = \{o_i : n \in \mathcal{C}_i\}$; building μ once costs $O(\sum_i |\mathcal{C}_i|)$ amortised over the lifetime of the substrate. Per step:

1. Insert all M proposals into a binary heap keyed by timestamp: $O(M \log M)$.
2. For each proposal p_i at node n_i , enumerate $\mu(n_i)$ and emit a conflict edge (i, j) for every $j \in \mu(n_i) \setminus \{i\}$. The expected cost of step 2 is

$$\mathbb{E}[\sum_i |\mu(n_i)| - 1] = \sum_i \sum_{j \neq i} \Pr[n_i \in \mathcal{C}_j] = \sum_i \sum_{j \neq i} \frac{|\mathcal{C}_i \cap \mathcal{C}_j|}{|\mathcal{C}_i|}.$$

Under the regularity assumption $|\mathcal{C}_i| = \Theta(k)$ for all i , this simplifies to $O\left(\frac{1}{k} \sum_{i < j} |\mathcal{C}_i \cap \mathcal{C}_j|\right)$. Together with the $O(|\mathcal{C}_i \cap \mathcal{C}_j|)$ fingerprinting cost per detected conflict pair, the total step-2 cost is $O(\sum_{i < j} |\mathcal{C}_i \cap \mathcal{C}_j|)$.

3. Topologically sort the resulting conflict graph in time proportional to its edge count $|H|$; in expectation $\mathbb{E}|H| = O(\sum_{i < j} |\mathcal{C}_i \cap \mathcal{C}_j|/k)$, dominated by step 2.

Summing yields the upper bound.

Within-protocol-class lower-bound remark. Within the priority-queued OCC protocol class — which performs an explicit comparison-based serialisation step and a per-pair fingerprinting step — the comparison-based sort requires $\Omega(M \log M)$ work in the worst case, and any protocol that explicitly enumerates conflicting node-pairs must do at least $\Omega(\sum_{i < j} |\mathcal{C}_i \cap \mathcal{C}_j|)$ work to inspect those pairs. We state these as within-class observations rather than as complexity-theoretic lower bounds on the underlying commit problem. $\square$

Corollary 1 (Disjoint-cone regime). *If all observer context windows are pairwise disjoint, $T_{\text{commit}}(M) = O(M \log M)$ under the priority-queued OCC protocol class, identical to the cost of M independent single-observer commits batched through a heap. The $M \log M$ factor is the standard heap-sort lower bound for this protocol class; faster protocols are not excluded.*

Corollary 2 (Saturated-overlap regime, scoped upper bound). *If every pair $|\mathcal{C}_i \cap \mathcal{C}_j|$ saturates at $\Theta(|\mathcal{C}|)$ under the context-invalidation conflict model and the priority-queued OCC protocol augments edge enumeration with a per-pair validation step that scans the full overlap to certify that no node has been invalidated between proposal and commit (the standard OCC validation phase of [Kung & Robinson, 1981](#), which inspects $|\mathcal{C}_i \cap \mathcal{C}_j|$ nodes per conflicting pair), then $T_{\text{commit}}(M) = O(M^2 |\mathcal{C}|)$ in this protocol class, and conflicts dominate. The $O(\cdot)$ form is an upper bound for the validating OCC variant, not a tight $\Theta(\cdot)$ for the substrate’s commit problem: hashing-based validation, parallel batched checks, or randomised conflict detection can reduce the per-pair work below $|\mathcal{C}|$, and protocols that skip explicit per-overlap validation entirely fall outside this bound. The minimal edge-enumeration variant of the proof of [Theorem 3](#) achieves only $O(M^2)$ in this regime, with the $|\mathcal{C}|$ factor entering once validation is required. This is the worst-case-recovery regime in the sense of the companion preprint ([Savchenko, 2026b](#)).*

Scope and limitation. [Theorem 3](#) is a constructive upper bound for the priority-queued optimistic-concurrency-control protocol of [Kung & Robinson \(1981\)](#) under the sparse-conflict assumption: the protocol achieves $O(M \log M + \sum |\mathcal{C}_i \cap \mathcal{C}_j|)$. As a within-protocol-class remark only, any comparison-based sort step requires $\Omega(M \log M)$ work and any protocol that explicitly enumerates conflicting node-pairs must perform $\Omega(\sum |\mathcal{C}_i \cap \mathcal{C}_j|)$ work on those pairs. We do *not* claim a complexity-theoretic lower bound across all conceivable commit protocols; alternative scheduling strategies (static partitioning, locality-sensitive hashing, gossip-based reconciliation, parallel batching) may have different cost profiles. The bound is presented as the cost of a specific natural protocol class, not as an intrinsic complexity lower bound on commit-cost in general.

Empirical relevance. The bound makes a precise prediction about the cost of a federated rendering system as a function of the number and overlap of player views (§6) under the OCC protocol class; it matches the empirical scaling observed in our observer-cone benchmark for the companion preprint.

4.4 Archive compression at the Shannon rate

The archive A stores a temporal record of pointer-creation events. Suppose pointer-creation events are drawn from a stationary stochastic process with entropy rate $H(\rho)$ (in bits per event). Then a lossless representation of A has size $|A| \cdot H(\rho)$ in expectation, achievable by universal source coding (e.g. Lempel–Ziv ([Ziv & Lempel, 1977](#))).

Proposition 2 (Archive compression). *The archive A of a Pointer Architecture admits a lossless representation of size $|A|H(\rho) + o(|A|)$ bits, where $H(\rho)$ is the entropy rate of the pointer-creation source. The bound is tight; no representation can do better in expectation by Shannon’s source-coding theorem ([Shannon, 1948](#)).*

Implication. The archive size is $|A| \cdot H(\rho)$ bits at the Shannon rate, a constant-factor compression of the naive per-event bit-cost $|A| \cdot \log |\mathcal{N}|$ (with $|\mathcal{N}|$ the address-space cardinality) by the ratio $H(\rho)/\log |\mathcal{N}|$. The archive is therefore strictly linear in event count $|A|$ at any positive entropy rate $H(\rho) > 0$; *sublinearity* requires the entropy rate to vanish asymptotically ($H(\rho) \rightarrow 0$), which corresponds to fully repetitive workloads (every new pointer-creation event is implied by the existing archive). For finite-entropy workloads the substrate achieves constant-factor compression, not sublinear archive growth. This is the same trade-off that underlies the motif-dictionary benchmark in the companion preprint (Savchenko, 2026b).

4.5 Decidability and computability

Reachability in the active graph is decidable in $O(|N| + |E|)$. Equivalence of two pointer states modulo the entire commit history is undecidable in general: a reduction from the halting problem can encode an arbitrary Turing machine as a sequence of pointer rewrites, with state-equivalence after t steps tracking whether the machine halts within t steps. The undecidability is therefore a property the substrate inherits from any Turing-complete computational model, not a bug specific to Pointer Architecture.

Proposition 3 (State-equivalence under arbitrary commit histories). *The problem EQUIVALENTUNDERHISTORY — given two pointer states G_1, G_2 and two finite sets of legal rewrite rules R_1, R_2 , is there a commit history under R_1 that reaches G_1 and one under R_2 that reaches G_2 with $G_1 \equiv G_2$? — is undecidable.*

Proof sketch. Encode an arbitrary Turing machine M as a graph rewriting system over a fixed alphabet of node types. Existence of a halting history of M on input x reduces to existence of a commit history that produces the halt-state pointer pattern. Reduction from the halting problem completes the proof. $\square$

Take-away. The substrate sits in the standard computability landscape: per-operation cost is polynomial, exact path-weighted observables are $\#P$ -hard, full-history equivalence is undecidable, and the average per-step commit cost under the priority-queued OCC protocol class is bounded above by the quantity given by Theorem 3.

5 Candidate correspondences with existing graph computational models

Pointer Architecture is not the first graph computational substrate. Its novelty lies in the joint treatment of referential state, local rewrite, multi-observer commit, and explicit archive. We give five correspondences with existing programmes, of which three are proved here in full: Theorem 5 (CRDTs), Theorem 6 (Petri-net encoding of single-observer substrates), and Theorem 4 (actor-model embedding with single-observer commit). Two further sketched mappings (Wolfram hypergraph rewriting, linear logic) are presented as research directions; their category-theoretic formalisation is open.

5.1 Wolfram hypergraph rewriting

Wolfram (2020) and the formal-correspondence work of Gorard (2020, 2024a,b) treat physics as the result of confluent rewrite rules over abstract hypergraphs. The substrates differ:

- Wolfram’s substrate is undirected hypergraphs with non-deterministic rewrite multiway-graphs; ours is directed pointer graphs with archive.

- Wolfram’s commit semantics is causal invariance (any branching ends in the same global state); ours is explicit observer-relative commit/rollback.
- Wolfram’s archive is implicit in the multiway graph; ours is a first-class data structure with provable compression bound (Prop. 2).

A natural *candidate forgetful map* from Pointer Architecture to Wolfram hypergraphs takes $\mathcal{S} = (G, R, C, A, \pi)$ to the hypergraph obtained by forgetting A, C, π and taking the symmetric closure of the active pointer set. Functoriality (preservation of composition under rewrite-then-forget vs forget-then-rewrite) is non-trivial because Wolfram rewrites are non-deterministic while our C resolves to a unique committed state; a full proof requires either weakening the target category to multiway-graph cones or strengthening our protocol to a confluent rewrite. We leave the formal treatment as an open direction.

5.2 Petri nets

A Petri net (Petri, 1962; Reisig, 1985) is a bipartite graph of places and transitions in which a transition is fireable when its input places hold tokens. The single-observer Pointer Architecture, restricted to the token-preserving rewrite class, admits a structure-preserving encoding into a Petri net that is conservative on the non-log subnet, with the archive realised as a monotone-accumulating (write-only) log place that sits outside the conservativity invariant: Theorem 6 below makes this precise. Multi-observer substrates with non-trivial commit semantics exceed the Petri-net formalism: there is no natural Petri-net rendering of commit/rollback.

5.3 Actor model

Hewitt et al. (1973) introduced the actor model, in which independent actors communicate asynchronously via immutable messages. We give the third correspondence *on the active-graph projection*: any actor system embeds into a Pointer Architecture with single observer (the runtime), with a bijection between actor configurations and active graphs and a rule-for-primitive bijection between $r_{\text{recv}}/r_{\text{send}}/r_{\text{become}}/r_{\text{create}}$ and the four actor primitives (receipt, send, become, create). The archive A records one entry per rule firing and is determined by the rule-firing history rather than by the actor configuration; it lies outside the bijection unless the actor formalism is extended with an explicit trace component.

Actor model recap. An actor system is a configuration $\Lambda = (\mathcal{A}, \mathcal{B}, \mathcal{M}, \text{beh})$ where $\mathcal{A}$ is a finite set of actor identifiers, $\mathcal{B}$ a set of behaviours, $\mathcal{M}$ a multiset of in-flight messages (each tagged with target $\in \mathcal{A}$ and payload), and $\text{beh} : \mathcal{A} \rightarrow \mathcal{B}$ the current behaviour of each actor. An actor with current behaviour b , upon receiving a message m , can perform any finite combination of three primitive operations: **send**(a', m') (place a message in flight), **become**(b') (replace its own behaviour), and **create**(b'') (instantiate a new actor with behaviour b''). Receipt is atomic per actor; cross-actor ordering is unconstrained beyond causal delivery.

Theorem 4 (Actor embedding on the active-graph projection). *For every actor system Λ there is a constructive Pointer Architecture $\mathcal{S}_\Lambda = (G_\Lambda, R_\Lambda, C_\Lambda, A_\Lambda, \pi_\Lambda)$ with $|\mathcal{O}| = 1$ and a structure-preserving map Ξ from actor configurations to substrate active graphs such that:*

- (i) *Configurations of Λ are in bijection with reachable active graphs G_Λ of $\mathcal{S}_\Lambda$ under Ξ . The archive A_Λ is not part of the bijection: the standard actor configuration does not include the event history, while A_Λ records one entry per rule firing, so two substrate runs that reach the same actor configuration via different histories produce different archives. The bijection is therefore between actor configurations and active graphs only; the archive is a substrate-side append-only event log outside the actor-state image. Extending the bijection*

to (G_Λ, A_Λ) requires augmenting the actor formalism with an explicit trace component (a configuration-plus-trace pair), which is straightforward but standard actor theory does not.

- (ii) Message receipt and each actor primitive (send, become, create) are in rule-for-primitive bijection with rules of R_Λ ($r_{\text{recv}}, r_{\text{send}}, r_{\text{become}}, r_{\text{create}}$). The mapping is not step-for-step at the level of full actor steps: a single actor step — receipt of one message followed by the issuance of a finite sequence of primitives by the receiving behaviour — corresponds to a finite atomic sequence of rule firings beginning with r_{recv} , treated as a single transaction by the commit protocol.
- (iii) The substrate's commit protocol C_Λ trivialises (single observer) and serialises rewrites in actor-causal order; the archive A_Λ records one entry per rule firing and, when extended to a configuration-plus-trace bijection per (i), corresponds to the actor system's event trace.
- (iv) Composability: nested actor systems (one actor's payload is another actor identifier) lift to nested pointer references; the resulting active graph remains directed and finite.

Proof. Construction. Let $\Lambda = (\mathcal{A}, \mathcal{B}, \mathcal{M}, \text{beh})$. Define $\mathcal{S}_\Lambda$ by:

- $N = \mathcal{A} \cup \mathcal{B} \cup \mathcal{M}$. Each actor identifier, each behaviour, and each in-flight message is a distinct node. Behaviour nodes are immutable; actor nodes are mutable in the sense that their outgoing-beh pointer can be redirected; message nodes are created and consumed.
- Active pointers E have three types:
 - beh-pointers: one per actor, $a \xrightarrow{\text{beh}} b$ where $b = \text{beh}(a)$.
 - tgt-pointers: one per in-flight message, $m \xrightarrow{\text{tgt}} a$ where a is the message target.
 - ctx-pointers: representing arbitrary payload of m (e.g. if m 's payload is an actor identifier, $m \xrightarrow{\text{ctx}} a$).
- Four rules in R_Λ , separating message receipt (full message consumption) from the three actor primitives that the receiving behaviour may issue. For a message m with payload set $\Pi(m) \subseteq N$ (the set of nodes referenced by the payload; $\Pi(m) \subseteq \mathcal{A}$ when the payload is an actor-identifier list, $\Pi(m) = \emptyset$ for opaque payloads), the rules are:

$$\begin{aligned}
r_{\text{recv}}: \quad & \{m\} \cup \{m \xrightarrow{\text{tgt}} a\} \cup \{m \xrightarrow{\text{ctx}} x : x \in \Pi(m)\} \rightarrow \emptyset \\
r_{\text{send}}: \quad & \emptyset \rightarrow \{m'\} \cup \{m' \xrightarrow{\text{tgt}} a'\} \cup \{m' \xrightarrow{\text{ctx}} x : x \in \Pi(m')\} \\
r_{\text{become}}: \quad & \{a \xrightarrow{\text{beh}} b\} \rightarrow \{a \xrightarrow{\text{beh}} b'\} \\
r_{\text{create}}: \quad & \emptyset \rightarrow \{a'\} \cup \{a' \xrightarrow{\text{beh}} b''\}
\end{aligned}$$

where a' in r_{create} is a fresh actor node and m' in r_{send} is a fresh message node; each rule has the side-effect of logging the corresponding event (receipt, send, become, create) in the archive A . Crucially, r_{recv} consumes the entire in-flight message atomically (target pointer, every context pointer, and the message node m itself), while r_{send} symmetrically introduces the full envelope (fresh node m' , target pointer, and one context pointer per payload reference); no orphan ctx-pointers or message nodes remain after a receipt. Similarly r_{create} introduces both the fresh actor node a' and its outgoing beh-pointer in a single firing, so the active graph never contains a dangling beh-arc whose source is not also a node.

A single actor step — “actor a receives m , runs $\text{beh}(a)$, which issues k sends, at most one become, and ℓ creates” — corresponds to firing r_{recv} once followed by k firings of r_{send} , at most one firing of r_{become} , and ℓ firings of r_{create} . This is the standard read/transform/emit decomposition of Agha (1986); bundling receipt and emission into a single rule would

conflate two distinct primitives. The $r_{\text{recv}}/r_{\text{send}}$ pair is mass-balanced on message nodes: r_{recv} removes $(m, \text{tgt}, \text{ctx-set})$ atomically and r_{send} creates a fresh triple of the same shape, so no orphan pointers or nodes survive across actor steps. The r_{create} rule, by contrast, is monotone-growing on actor nodes: it introduces one fresh actor node per firing without a corresponding deletion, so cumulative actor count grows by exactly the number of r_{create} firings.

- Initial configuration $G_\Lambda^{(0)}$: nodes for the initial actor set, one beh-pointer per actor, message nodes for initial in-flight messages. Archive $A_\Lambda^{(0)} = \emptyset$.
- Observer set $\mathcal{O} = \{o_*\}$ with $\pi(o_*) = N$ (the runtime sees everything). Commit protocol C_Λ trivially serialises any single proposal.

Define the map $\Xi : \Lambda\text{-configurations} \rightarrow \text{substrate active graphs}$ by $\Xi(\Lambda) = G_\Lambda$ where G_Λ contains the pointers above. The archive A_Λ is *not* in the image of Ξ : it is a substrate-side append-only event log determined by the rule-firing history, not by the actor configuration.

(i) State bijection (active-graph projection). Each actor configuration determines G_Λ uniquely (one beh-pointer per actor, one tgt-pointer per pending message, ctx-pointers determined by message payloads). The reverse: a reachable active graph of $\mathcal{S}_\Lambda$ recovers a unique actor configuration by reading off the pointer-set partition. Hence the bijection holds on the active-graph projection; two substrate runs that reach the same active graph via different rule-firing histories have different archives but the same actor configuration under Ξ^{-1} .

(ii) Operation correspondence. An actor step decomposes into a receipt followed by zero or more primitive emissions. Actor a receiving message m corresponds to firing r_{recv} on m ; the substrate consumes the tgt-pointer (removing the in-flight message) and logs the receipt event. The receiving behaviour $\text{beh}(a)$ then issues a (possibly empty) sequence of actor primitives, each mapped to one rule firing: $\text{send}(a', m')$ corresponds to r_{send} , which introduces a fresh message node m' together with a tgt-pointer to a' (and any ctx-pointers for the payload), making m' an in-flight message available for a future r_{recv} ; $\text{become}(b')$ corresponds to r_{become} , which redirects a 's beh-pointer; $\text{create}(b'')$ corresponds to r_{create} , which introduces a fresh actor node with a beh-pointer. The bijection is rule-for-primitive (one firing per primitive issuance), not step-for-step — a single actor step maps to a finite sequence of rule firings whose first element is r_{recv} .

(iii) Causal order and trace. With $|\mathcal{O}| = 1$ the commit protocol C_Λ accepts proposals in receipt order. The actor model's causal-delivery constraint is enforced by allowing r_{recv} to fire only on in-flight messages whose target a has a well-formed beh-pointer, and by requiring the $r_{\text{send}}/r_{\text{become}}/r_{\text{create}}$ firings that follow each r_{recv} to be proposed in the same commit as the receipt (so causally-prior receipts complete before the messages they produce are visible). The runtime ensures causal-delivery semantics by proposing rules in causal order, which C_Λ accepts unchanged. The archive A_Λ accumulates exactly one entry per rule firing; this is the actor system's event trace (one entry per receipt, send, become, and create).

(iv) Composability and growth bound. If a message payload is an actor identifier a' , the corresponding ctx-pointer $m \xrightarrow{\text{ctx}} a'$ embeds the second-actor reference directly in the pointer graph. Nested actor systems lift by recursion; the pointer graph remains directed because ctx-pointers point to existing nodes, not to fresh non-actor objects. The active-graph growth bound has two independent terms: (a) r_{create} adds one persistent actor node per firing (no deletion rule for actor nodes), so cumulative actor count grows linearly in the number of r_{create} firings; (b) r_{send} adds one message node plus $1 + |\Pi(m')|$ pointers (one tgt and one ctx per payload reference), and r_{recv} symmetrically removes a message node plus the same pointer count, so the message-node count is bounded by the unconsumed-message backlog (sends minus receives) rather than growing per send. Total active-graph size at any

time is therefore $|\mathcal{A}_0| + (\# r_{\text{create}} \text{ firings}) + (\text{unconsumed-message backlog})$ in nodes, plus the corresponding pointer-count contributions; finite for any finite-history run. $\square$

Corollary 3 (Multi-observer actor systems). *For actor systems with multiple concurrent runtimes (e.g. distributed actor models such as Erlang/OTP, Akka), the embedding extends to $|\mathcal{O}| > 1$ by assigning each runtime an observer and restricting $\pi(o_i)$ to the actor partition owned by runtime i . The substrate’s commit protocol C then captures distributed-actor cross-runtime conflicts (e.g. migration races, partition-tolerance failures) that the single-runtime semantics ignores.*

π -calculus extension. Milner (1992) formalises mobile communication via names that themselves can be sent. The ctx-pointer construction above already captures name mobility: when a message’s payload is a pointer, the receiver gains a reference it did not previously have. A full π -calculus embedding extends Theorem 4 with explicit name-restriction operators ($\nu n.P$); we sketch this here but leave the proof of restriction-preservation to follow-up work. The session-typed variant (Honda et al., 1998) provides a partial precedent for our commit layer: session types statically reject some conflicting communications, while our C resolves them dynamically.

5.4 Linear logic and resource semantics

Girard (1987) introduced linear logic, in which propositions are resources that cannot be freely duplicated. Pointer Architecture’s distinction between *active pointers* (consumable, may be rewritten) and *archived pointers* (persistent, cannot be deleted) is suggestive of the linear/exponential modality distinction in linear logic, with the $!$ modality plausibly corresponding to archive embedding. A precise proof-theoretic correspondence (e.g. via linear-logic proof nets) is an open direction; conflict between two observer rewrites would map to a non-confluent proof reduction.

5.5 Conflict-free replicated data types (CRDTs)

Shapiro et al. (2011) introduced CRDTs as data structures permitting concurrent updates without coordination. The commutative-rewrite restriction of Pointer Architecture reduces formally to an operation-based CRDT, which we prove here in full.

Op-based CRDT background. Following Shapiro et al. (2011), an operation-based commutative replicated data type (CmRDT) consists of a state space S , an initial state s_0 , a finite set of update operations $\mathcal{U}$, a strict-causal-delivery transport, and the requirement that for every pair $u, v \in \mathcal{U}$ concurrent at the application layer, $u(v(s)) = v(u(s))$ for all reachable s . Convergence (strong eventual consistency) follows from commutativity plus causal delivery.

Theorem 5 (CRDT correspondence on the active-graph projection). *Let $\mathcal{S} = (G, R, C, A, \pi)$ be a Pointer Architecture in which every pair of observer rewrites $(p, q) \in R \times R$ commutes as graph rewrites: applying p then q produces the same active graph as q then p for every reachable G . This correspondence is stated on the active-graph projection of substrate state (G only), not on the full state (G, A) : the archive A accumulates timestamped event entries that the CmRDT formalism does not model, so the bijection below is between reachable active graphs and CRDT states. The archive remains as a substrate-side append-only event log outside the CRDT image. Then there is a constructive map $\Phi : \mathcal{S} \mapsto (S_\Phi, s_0, \mathcal{U}_\Phi)$ to an op-based CRDT such that:*

- (i) *Every reachable active graph G corresponds to a unique CRDT state $\Phi(G) \in S_\Phi$ (archive A is not encoded by Φ ; see remark below).*
- (ii) *Every rewrite $r \in R$ is in bijection with a CRDT update $u_r \in \mathcal{U}_\Phi$ satisfying the CmRDT commutativity condition.*

(iii) Under causal-delivery transport, the substrate’s commit protocol C is operationally indistinguishable from the CRDT merge procedure on G : both produce the same final active graph from the same multiset of operations.

(iv) The per-step commit cost (Theorem 3) collapses to its first term: $T_{\text{commit}}(M) = O(M \log M)$ under the priority-queued OCC protocol class.

Proof. Construction of Φ . Set $S_\Phi = \{G : G \text{ reachable from } G_0 \text{ in } \mathcal{S}\}$, $s_0 = G_0$, and $\mathcal{U}_\Phi = R$ acting on S_Φ in the obvious way: $u_r(G) := \text{apply rewrite } r \text{ to } G$. The archive A is deliberately omitted from S_Φ : CmRDTs model convergent state over commuting updates, not the per-update timestamped event log; extending the CRDT image to cover A would require a stronger formalism (e.g. a CRDT-with-event-log variant) that we do not formalise here.

(i) State correspondence on active graph. By construction Φ is the identity on the reachable active-graph substate, hence injective on that projection. Surjectivity holds by definition of S_Φ as the reachable active-graph set. The map is not onto pairs (G, A) ; archive content is recovered from the substrate’s commit log, not from the CRDT image.

(ii) Operation commutativity. By hypothesis, for all $r, s \in R$ and all reachable G , $u_r(u_s(G)) = u_s(u_r(G))$. This is precisely the CmRDT condition.

(iii) Operational indistinguishability under causal delivery (on the active graph). Consider a finite multiset of operations $\{u_{r_1}, \dots, u_{r_M}\}$ proposed by M observers. By commutativity, the composition $u_{r_{\sigma(1)}} \circ \dots \circ u_{r_{\sigma(M)}}$ is independent of the permutation σ . The substrate’s commit C produces some serialisation order; the CRDT merge produces some (possibly different) order; both yield the same final *active graph* by commutativity — the per-event archive entries differ between the two orderings (the substrate records a timestamped trace; the CRDT does not), but the active-graph state is identical, which is the property the CRDT formalism encodes. Causal-delivery transport in the CRDT corresponds to the substrate’s archive-respecting commit order (each operation depends only on its causal predecessors), which is preserved by C by Def. 4.

(iv) Commit-cost collapse. The conflict graph in the proof of Theorem 3 is built from pairs (i, j) that “conflict” — that is, do not commute. Under the hypothesis of this theorem, no such pairs exist, so the conflict-graph is empty and the second term $\sum_{i < j} |\mathcal{C}_i \cap \mathcal{C}_j|$ contributes zero. The remaining $M \log M$ term is the heap-insertion cost. $\square$

Corollary 4 (Subclass of conservative Pointer Architectures behaves as CRDTs at zero coordination cost). *For commutative-rewrite Pointer Architectures, no coordination between observers is required for convergence. This is the standard CmRDT guarantee.*

The interesting regime is therefore the *non-commutative* one, where the architecture exceeds CRDT semantics and the full commit-cost bound of Theorem 3 (upper bound under the priority-queued OCC protocol class) applies.

5.6 Petri-net encoding

We give the second correspondence *on the active-graph projection*: every single-observer Pointer Architecture restricted to the token-preserving rewrite class admits a structure-preserving encoding of its active graph G into a Petri net that is conservative on the non-log subnet. The archive A is mapped to a single monotone-accumulating (write-only) log place that counts firings but does not preserve per-event identifiers or timestamps; it sits outside the conservativity invariant and recovering the full timestamped event trace requires reading the substrate’s archive directly. The general non-token-preserving case requires an augmented Petri-net construction with per-token-type source/sink places; see scope paragraph after Theorem 6.

Setup. Recall a Petri net is a tuple $\mathcal{P} = (P, T, F, W, M_0)$ where P is a finite set of *places*, T a finite set of *transitions*, $F \subseteq (P \times T) \cup (T \times P)$ the flow relation, $W : F \rightarrow \mathbb{N}_{>0}$ the arc-weight function, and $M_0 : P \rightarrow \mathbb{N}$ the initial marking (Reisig, 1985).

Theorem 6 (Petri-net encoding of single-observer Pointer Architectures, active-graph projection). *Let $\mathcal{S} = (G_0, R, C, A_0, \pi)$ be a Pointer Architecture with $|\mathcal{O}| = 1$ (so C trivially serialises in proposal order). The encoding below is between the active graph G of the substrate and the non-log subnet of $\text{Net}(\mathcal{S})$; the archive A is mapped to the unbounded log place ℓ , whose marking counts firings but does not preserve per-event identifiers or timestamps. The Petri-net image therefore captures how many events of each rewrite type have fired, not the timestamped event-trace structure of A ; recovering the timestamped trace requires reading the substrate's archive directly. There is a constructive map Ψ from $\mathcal{S}$ to a Petri net $\text{Net}(\mathcal{S})$ such that:*

- (i) *There is a bijection between reachable active graphs of $\mathcal{S}$ and reachable markings of $\text{Net}(\mathcal{S})$ projected onto the non-log places.*
- (ii) *Every rule firing in $\mathcal{S}$ corresponds to a unique transition firing in $\text{Net}(\mathcal{S})$, and vice versa.*
- (iii) *$\text{Net}(\mathcal{S})$ is conservative on the non-log subnet $P \setminus \{\ell\}$, when the rewrite class R is restricted to token-preserving local rewrites, i.e. rewrites whose left-hand and right-hand patterns satisfy $|\text{pat}_L| = |\text{pat}_R|$ in node-and-edge token count. The log place ℓ is a write-only event log: it receives one arc per transition (carrying $|\text{pat}_L \setminus \text{pat}_R|$ tokens) and emits none, so the marking of ℓ is strictly monotone in firing count by construction; ℓ does not enter the conservativity invariant. For non-token-preserving rewrites (where the right-hand pattern adds or removes tokens), even non-log conservativity fails by $|\text{pat}_R| - |\text{pat}_L|$; the augmented Petri-net construction adds an explicit source place src_τ and sink place snk_τ per token type τ , with arc weights chosen to balance creation and destruction, recovering global token conservation across $P \cup \{\text{src}_\tau, \text{snk}_\tau\}_\tau$. The theorem as stated covers the restricted token-preserving class on the non-log subnet; the general case requires the augmented construction. We use the restricted form in the empirical implementation (§6).*

Scope. The conservation claim of (iii) is restricted in two distinct ways. First, it is stated on the *non-log subnet*: the log place ℓ is a write-only event-log accumulator (one arc in per transition, no arcs out), so its marking grows strictly monotonically in firing count and does not enter the conservativity invariant. Second, even on the non-log subnet, conservativity is tight only under the token-preserving rewrite class ($|\text{pat}_L| = |\text{pat}_R|$); non-token-preserving rewrites require the augmented construction with per-token-type source/sink places to recover global balance. The framework's distributed-simulation applications (§6) implement the restricted token-preserving class on the non-log subnet.

Proof. Construction. Set:

- $P = P_N \cup P_E \cup \{\ell\}$, where $P_N = \{p_v : v \in N\}$ has one place per node, $P_E = \{p_{(u,v,\tau)} : (u,v,\tau) \in E\}$ has one place per active pointer type (a triple of source, target, and pointer-type label), and ℓ is the unbounded log place.
- $T = R$: one transition per rewrite rule.
- For each rule $r : \text{pat}_L \rightarrow \text{pat}_R$, define:

$$F(p, t_r) = \mathbb{K}[p \in \text{pat}_L], \quad F(t_r, p) = \mathbb{K}[p \in \text{pat}_R],$$

with the additional output arc $F(t_r, \ell) = |\text{pat}_L| - |\text{pat}_R \cap \text{pat}_L|$, which deposits in the log place every token consumed from pat_L but not produced in pat_R .

- All arc weights are 1 except the log-output, which is $|\text{pat}_L \setminus \text{pat}_R|$.
- Initial marking $M_0(p_v) = 1$ for all $v \in N$ that are present at time 0; $M_0(p_e) = 1$ for all $e \in E_0$; $M_0(\ell) = |A_0|$ (existing archive entries).

(i) Active-graph correspondence. An active graph $G = (N, E)$ corresponds to the marking $\Psi(G)$ where $\Psi(G)(p_v) = 1$ if $v \in N$ and 0 otherwise, and similarly for pointer-places. This is a bijection between reachable active graphs of $\mathcal{S}$ and reachable markings of $\text{Net}(\mathcal{S})$ projected onto $P_N \cup P_E$. The archive A is mapped onto the marking of the log place ℓ , which counts firings but does not preserve per-event identifiers or timestamps; recovering the timestamped event trace requires reading the substrate’s archive directly.

(ii) Operational equivalence. A rule r is enabled in $\mathcal{S}$ at active graph G iff $\text{pat}_L \subseteq G$. In the net, transition t_r is enabled at $\Psi(G)$ iff $M(p) \geq F(p, t_r)$ for all p , which simplifies to $\text{pat}_L \subseteq G$ by construction. Firing t_r removes pat_L tokens from input places, adds pat_R tokens to output places, and deposits $|\text{pat}_L \setminus \text{pat}_R|$ in ℓ . The resulting marking is $\Psi(G')$ where G' is the active graph after applying r . Hence rule firings and transition firings are in bijection via Ψ on the active-graph projection.

(iii) Conservation (non-log subnet, token-preserving class). The conservativity invariant is stated for the non-log subnet $P \setminus \{\ell\}$ only: the log place ℓ has no outgoing arcs by construction, so its marking is monotone in firing count and cannot enter a token-balance equation as a balanced quantity. For each transition t_r on the non-log subnet:

$$\sum_{p \in P \setminus \{\ell\}} (F(t_r, p) - F(p, t_r)) = |\text{pat}_R| - |\text{pat}_L|.$$

Under the token-preserving restriction $|\text{pat}_L| = |\text{pat}_R|$, the right-hand side vanishes, so the non-log token count is preserved by every firing. The log place separately accumulates $|\text{pat}_L \setminus \text{pat}_R|$ tokens per firing (zero only for in-place rewrites where $\text{pat}_L = \text{pat}_R$); ℓ is therefore an unbounded monotone event-log, not a balanced place. For non-token-preserving rewrites the non-log sum is non-zero; conservation is then recovered globally only via the augmented Petri-net construction (per-token-type source/sink places), which is sketched in the scope paragraph above and not used in the empirical implementation. $\square$

Corollary 5 (Petri-net reachability: upper-bound transfer). *Reachability in single-observer Pointer Architectures restricted to the token-preserving rewrite class admits an upper bound from Petri-net reachability via the encoding of Theorem 6: any reachability question on $\mathcal{S}$ reduces to a reachability question on $\text{Net}(\mathcal{S})$, so PA-reachability is at most as hard as Petri-net reachability, which is Ackermann-complete (Leroux, 2022; Czerwiński & Orlowski, 2022). This is far worse than the polynomial bound on basic operations (Table in §4.1); the difference is that reachability asks about arbitrarily long commit sequences, while basic operations cost only the immediate step. **Hardness in the reverse direction is not established here:** matching Ackermann-completeness for PA-reachability would require a reduction from Petri-net reachability into PA, which we do not provide. This corollary is therefore an upper-bound transfer, not a complexity-equivalence claim.*

Multi-observer extension is not Petri-net-tractable. The encoding above assumes $|\mathcal{O}| = 1$. The multi-observer case requires modelling the commit protocol C , which is outside the Petri-net formalism in any natural sense. The reduction to Petri nets is therefore tight: the substrate strictly exceeds the formalism only when commit-conflict resolution comes into play.

6 Applications in computer science

The substrate is engineering-ready. We sketch four applications in CS proper; each is a worked sketch rather than a complete system, with concrete cost bounds drawn from §4.

6.1 Distributed simulation: MMO worlds, digital twins, multi-agent simulation

A massively multiplayer online (MMO) world maintains millions of player views of a shared simulated environment. State updates from different players can conflict (two players grab the same object). The substrate provides the modelling apparatus: players are observers o_i , player views are context windows $\mathcal{C}_i$, gameplay actions are local rewrites, and conflict resolution uses the commit protocol. Theorem 3 bounds the per-step server cost under the priority-queued OCC protocol class above by $O(M \log M + \sum_{i < j} |\mathcal{C}_i \cap \mathcal{C}_j|)$, which matches the empirically observed scaling of region-based MMO servers (Sundell & Tsigas, 2003). The same model applies to digital twins of physical infrastructure, where each engineering team owns a context window over the model.

6.2 Federated rendering with commit validation

A cloud-gaming or distributed-rendering pipeline can render different parts of a scene on different machines and commit globally consistent frames. The Pointer Architecture’s commit protocol provides a clean specification: each render-server is an observer; each pixel-block update is a candidate rewrite; the commit layer validates inter-block consistency (e.g. shadow-map continuity across server boundaries). Theorem 3 bounds the synchronisation cost under the priority-queued OCC protocol class as a function of overlap between server view-frusta, which is the dominant overhead in current systems (Shi et al., 2015).

6.3 Provenance-aware databases

The archive is structurally identical to the commit log of a git-style version-control system, augmented with reference semantics rather than file semantics. Provenance-aware databases such as Datomic, XTDB, and bitemporal extensions to PostgreSQL implement subsets of the substrate. Proposition 2 gives a sharp bound on archive storage as a function of pointer-creation entropy; the empirical compressibility of git packfiles (Bird et al., 2009; Spinellis, 2005) is the load-bearing real-world example.

6.4 Latent-pointer world models in machine learning

Ha & Schmidhuber (2018), Hafner et al. (2020), and the graph-network simulators of Sanchez-Gonzalez et al. (2020); Pfaff et al. (2021) maintain latent representations of an environment and update them via learned local dynamics. Pointer Architecture provides a unifying view: the latent state is a pointer graph, the learned dynamics are a rewrite rule, and rollout is a commit history. Retrieval-augmented reinforcement learning (Goyal et al., 2022; Humphreys et al., 2022) explicitly stores past trajectories in an archive and conditions future predictions on this archive, which is the operational form of our archive primitive. The substrate provides an explicit framework for analysing the trade-off between active model capacity and archive storage in long-horizon planning.

7 Substrate-cone holographic structure

Two results carry this section. First, substrate states whose entropy function equals the min-cut on the active graph inhabit the holographic entropy cone of Bao et al. (2015) (Proposition 4). Second, pointer aliasing realises the ER = EPR proposal as a substrate-side Einstein–Rosen bridge under the cross-party-bridge condition (Corollary 6).

7.1 Substrate as a weighted graph model

We translate the PA primitives into the input data expected by the holographic-entropy-cone framework.

Graph and weights. The Bao–Nezami–Ooguri–Stoica–Sully–Walter construction (Bao et al., 2015) takes as input an *undirected* weighted graph. Substrate states in PA are defined on the directed active graph G ; to apply the cone framework we work with the undirected projection $|G| = (N, |E|)$ in which each directed pointer $(u, v) \in E$ contributes an undirected edge $\{u, v\}$ with the same weight (or parallel pointers in both directions sum their weights). This projection loses pointer directionality. Whether substrate min-cut on the original directed G (using min- s - t -cut in the directed sense) inhabits the holographic entropy cone is an open question; the present construction is the cone on the undirected projection. Each pointer $e \in E$ carries a fixed bond capacity

$$w(e) = \log_2 \chi_e,$$

where χ_e is the per-pointer addressable namespace cardinality (default $\chi_e = 2$, giving $w(e) = 1$ bit per pointer). Weights are non-negative reals; throughout this paper we use integer weights ($w(e) \in \mathbb{Z}_{\geq 0}$) without loss of generality.

Boundary, parties, purifier. A *substrate region* is a subset $N_R \subseteq N$ together with a designation of which of its nodes are boundary (externally observed) and which are bulk (substrate-internal). Write $\partial N_R \subseteq N_R$ for the boundary. Following Bao et al. (2015), partition the boundary into $n + 1$ colour classes via a colouring map

$$b: \partial N_R \longrightarrow [n + 1],$$

where colours $1, \dots, n$ index n parties $A_1, \dots, A_n$ and colour $n + 1$ indexes a *purifier* P . A party A_i corresponds to an observer (or coherent group of observers) holding an externally identifiable subset of ∂N_R in its context window. The purifier collects the rest of the substrate external to the modelled region.

Substrate discrete entropy. For $I \subseteq [n]$ define the *substrate discrete entropy*

$$S^*(I) = \min_{\substack{W \subseteq N \\ W \cap \partial N = b^{-1}(I)}} \sum_{e \in \text{cut}(W)} w(e), \quad (3)$$

where $\text{cut}(W) = \{\{u, v\} \in |E| \mid u \in W, v \notin W\}$ is the undirected cut-edge set induced by W (we use cut rather than C to avoid clash with the PA commit-protocol primitive C of Definition 4). The minimum is taken over all subsets $W \subseteq N$ whose intersection with the boundary is exactly $b^{-1}(I)$. Equation (3) is identical to Definition 22 of Bao et al. (2015) applied to the undirected projection $|G|$ of the PA active graph.

Definition 8 (Holographic-cone-inhabiting (HCI) state). A *substrate state* σ on $\mathcal{S}$ with *entropy function* $S_\sigma: 2^{[n]} \rightarrow \mathbb{R}_{\geq 0}$ is *holographic-cone-inhabiting* iff $S_\sigma(I) = S^*(I)$ for every $I \subseteq [n]$.

Definition 8 is constraining: not every substrate state is HCI. The numerical pilot of §7.4 (random sampling, §7.4) gives a quantitative measure of how restrictive the constraint is.

7.2 Proposition 1: substrate cone membership

Proposition 4 (Substrate cone membership). *Let σ be an HCI substrate state on $\mathcal{S} = (G, R, C, A, \pi)$ with boundary colouring b . Its entropy function $S_\sigma = S^*$ satisfies the following holographic entropy inequalities:*

- (1) (Subadditivity) $S_\sigma(IJ) \leq S_\sigma(I) + S_\sigma(J)$ for all disjoint $I, J \subseteq [n]$.
- (2) (Strong subadditivity) $S_\sigma(IJ) + S_\sigma(JK) \geq S_\sigma(J) + S_\sigma(IJK)$ for disjoint $I, J, K \subseteq [n]$ with $J \neq \emptyset$.
- (3) (Monogamy of mutual information) $S_\sigma(IJ) + S_\sigma(IK) + S_\sigma(JK) \geq S_\sigma(I) + S_\sigma(J) + S_\sigma(K) + S_\sigma(IJK)$.
- (4) (All linear HEIs) *Every linear inequality defining HEC_n is satisfied by S_σ .*

Proof. By the HCI condition (Definition 8), $S_\sigma = S^*$ where S^* is the substrate discrete entropy (3): a min-cut on the undirected weighted projection $|G|$ of G . Items (1)–(3) are then direct consequences of S^* being a min-cut on a weighted graph (Bao et al., 2015, Theorem 3 and the explicit MMI proof there). Item (4) follows from the contraction-map completeness result of Bao et al. (2025): contraction maps are necessary and sufficient to prove the linear HEIs that define HEC_n , and the existence of the requisite maps is what guarantees min-cut on every weighted graph satisfies the full cone. $\square$

Remark on the scope of Proposition 4. The proof above does not invoke any PA-specific structure: it uses only that S^* is a min-cut on a weighted graph and applies existing results of Bao et al. (2015, 2025). The proposition therefore says nothing new about PA per se; what it does say is that *whenever a substrate state is HCI*, the full machinery of the holographic entropy cone applies to its entropy function. The PA-side question proper — whether the substrate states actually produced by the commit protocol C are HCI, or only a strict subset, or none at all — is left open; §7.4 reports a positive numerical result for small substrates under a simulated multi-observer OCC protocol, but this is not a substrate-side theorem.

7.3 Corollary 1: pointer aliasing as substrate ER bridge

A core informal claim in the PA programme is that pointer aliasing — two distinct pointers in disjoint substrate regions sharing a common target — plays the substrate-side role of an Einstein–Rosen bridge in the ER = EPR proposal of Maldacena & Susskind (2013). We make this precise as a corollary.

Aliasing as a graph operation. Let $x, y \in N$ be two distinct bulk nodes, each of which serves as the target of pointers from disjoint substrate regions. The *fusion* operation $\text{fuse}(x, y)$ produces a new substrate $\mathcal{S}'$ with x and y collapsed to a single node xy and all incident edges redirected.

Cross-party-bridge condition. For a colouring b with parties $P_1, \dots, P_n$, write $W^*(I)$ for the witness of the min-cut (3) on $|G|$ at party-subset I . A fusion pair (x, y) satisfies the *cross-party-bridge condition* if (i) at least one pair-subset $I = \{i, j\} \subseteq [n]$ has $x \in W^*(I)$ and $y \notin W^*(I)$ (equivalently, $W^*(I)$ separates the pair), and (ii) every single-party subset $I = \{i\}$ has x and y on the same side of $W^*(I)$.

Corollary 6 (Aliasing preserves cone membership). *Let σ be an HCI substrate state on $\mathcal{S}$, and let $\mathcal{S}'$ be the substrate obtained by applying $\text{fuse}(x, y)$ for two bulk nodes x, y . Let σ' be the HCI state on $\mathcal{S}'$. Then:*

(1) σ' inhabits HEC_n .

(2) Under the cross-party-bridge condition, the MMI gap of σ' is bounded below by the MMI gap of σ , and strictly larger when x and y are connected to distinct boundary parties.

Proof of (1). $\mathcal{S}'$ has a well-defined min-cut function on its weighted graph; by Proposition 4 applied to $\mathcal{S}'$, this function inhabits HEC_n . $\square$

Sketch of (2); full case analysis in supplementary note `notes/COROLLARY_1_PROOF.md`. Under the cross-party-bridge condition, the optimal-cut witness $W^*(\{i\})$ for every singleton has x and y on the same side, so fusion does not change the singleton cut value: $S^*(\{i\}, \mathcal{S}') = S^*(\{i\}, \mathcal{S})$, hence $\delta_{P_i} = 0$. For pair-subsets $I = \{i, j\}$ with $x \in W^*(I)$ and $y \notin W^*(I)$, fusion forces x and y to the same side and thus restricts the feasible cut-witness set; the min-cut value can only grow: $\delta_{IJ} \geq 0$, with $\delta_{IJ} > 0$ for at least one such IJ (the one where the bridge actively contributes to the cut). For the triple-subset, $S^*(IJK, \mathcal{S}')$ is bounded above by the min over feasible cut witnesses that respect the fusion; the case analysis in the supplementary note shows $\delta_{ABC} \leq \min_{IJ} \delta_{IJ}$ for the cross-party-bridge configuration. Substituting into

$$\Delta_{\text{MMI}}(\sigma') - \Delta_{\text{MMI}}(\sigma) = \delta_{AB} + \delta_{AC} + \delta_{BC} - \delta_A - \delta_B - \delta_C - \delta_{ABC}$$

gives the strict inequality whenever at least one $\delta_{IJ} > 0$ and the bridge connects distinct boundary parties. $\square$

Corollary 6 realises the substrate-side ER bridge claim of the PA programme: the ER bridge is the fused substrate node xy ; the entanglement floor between regions A and B joined by the bridge is the contribution of xy to the cut-edge set $\text{cut}(W)$ realising S^* ; operations on xy are substrate- local but affect both A - and B -side observers, with no embedded- physics signal propagating across the bridge.

7.4 Worked examples and sanity checks

Eight computational worked examples illustrate the substrate-cone construction on small substrates. Because the substrate discrete entropy (3) is min-cut on a weighted graph, the holographic inequalities hold by Proposition 4 on any such substrate; the worked examples below are sanity checks confirming the framework is *concrete* and *applicable to PA-style graphs*, not independent empirical verifications. What each example does add: a specific entropy vector, the response of the inequality gaps to substrate rewrites, and the contrast with adversarial non-cone constructions. All examples run in seconds on a laptop with no external dependencies beyond the Python 3 standard library; full traces in `code/substrate_cone_*.py` and `data/substrate_cone_*_report.md`.

Pilot OQ6: small-substrate cone membership. A 6-node substrate with 4 boundary nodes coloured by 3 parties $\{A, B, C\}$ plus a purifier P , and 2 bulk nodes $\{x, y\}$ connected by 7 unit-weight edges (`substrate_cone_oq6_check.py`). Brute-force min-cut yields the entropy vector

$$S^*(\emptyset) = 0, S^*(A) = S^*(B) = S^*(C) = 2, S^*(AB) = 1, S^*(AC) = S^*(BC) = 4, S^*(ABC) = 2.$$

Purification symmetry $S^*(I) = S^*(\bar{I})$ holds exactly across all 16 colour subsets; subadditivity passes across all 6 disjoint pair tests; strong subadditivity passes across all 3 ordered triples; monogamy of mutual information gives $1 + 4 + 4 = 9 \geq 8$ with strict gap $\Delta_{\text{MMI}} = 1$.

Pilot OQ7: rewrite-trajectory of inequality gaps. Starting from OQ6 we apply nine explicit rewrites (edge add/remove, bulk-node addition with various attachment patterns, bulk-node fusion, edge re-weighting; `substrate_cone_oq7_rewrites.py`) and recompute the entropy vector and gaps. All 9 rewrites preserve cone membership. The MMI gap responds in directionally predictable ways:

Trajectory class	MMI shift	Substrate semantics
Cone-loosening	$\Delta > 0$	global centralisation, pointer aliasing
Cone-tightening	$\Delta < 0$ (often $\rightarrow 0$)	bulk cleaving, edge imbalance
Cone-neutral	$\Delta = 0$	local boundary adjustment

Bulk-node fusion yields MMI gap $1 \rightarrow 2$, confirming the direction predicted by Corollary 6.

Pilot: adversarial GHZ certification. The GHZ entropy structure $S(I) = 1$ for all $\emptyset \neq I \subseteq \{A, B, C\}$ has MMI gap $3 - 4 = -1 < 0$. By Proposition 4, no HCI substrate state on any weighted graph can realise the GHZ entropy vector. The HCI condition is non-trivially constraining (`substrate_cone_strengthening.py`, T3).

Pilot: random-substrate null comparison. Sampling 10^5 random entropy vectors with each $S(I)$ drawn uniformly from $\{0, 1, 2, 3, 4\}$ for 3 parties: 45.1% satisfy SA only, 31.7% satisfy SSA only, 34.7% satisfy MMI only, 9.3% satisfy all three jointly. By contrast, substrate min-cut entropies satisfy all HEIs with probability 1.00. The $\sim 10\times$ gap quantifies how exceptional cone-inhabiting states are (`substrate_cone_strengthening.py`, T4).

Pilot: higher-party HEI verification (4 parties). An explicit 4-party substrate (colours A, B, C, D plus purifier, 5 boundary + 3 bulk nodes, 9 edges) yields cyclic-4 inequality $S(AB) + S(BC) + S(CD) + S(DA) \geq S(A) + S(B) + S(C) + S(D) + S(ABCD)$, computed: LHS = 10, RHS = 9, gap = 1. SA, SSA, and all 4 MMI triples (ABC, ABD, ACD, BCD) hold (`substrate_cone_strengthening.py`, T5).

Pilot OQ-A: protocol-dynamic substrate evolution. A multi-observer commit protocol on the OQ6 substrate with three observers having overlapping context windows; each step, observers propose random rewrites restricted to their context windows; the protocol applies non-conflicting subsets via OCC. Over 50 steps with 89 committed rewrites, the substrate maintains cone-feasibility (SA + SSA + MMI) at every step. Edge count varies in $[1, 10]$; MMI gap stays in $[0, 2]$ with mean 0.14 (`substrate_cone_dynamic_C.py`).

Pilot: coarse-graining and multi-bridge robustness. Applying $\text{fuse}(x, y) \rightarrow xy$ with weight-summing convention yields a graph whose entropy vector again inhabits the cone; the MMI gap rises from 1 to 2, consistent with Corollary 6. Iteratively adding additional ER bridges between parties A and C leaves the MMI gap constant at 1 for $k \in \{0, 1, \dots, 5\}$ bridges; the substrate-cone framework holds for all multi-bridge configurations.

Pilot: Pythia LLM substrate-cone falsifier. The (Pythia, Gaussian-MI-on-hidden-activations) pair violates MMI on all tested partitions and at all tested scales (Pythia 70m and 160m, sample-size robust). Full report in §11.

7.5 Discussion: relation to AdS/CFT and Verlinde entropic gravity

The substrate-cone framework parallels the AdS/CFT holographic- duality programme but does not derive it. AdS/CFT supplies a specific dynamical correspondence between bulk gravity

in $(d + 1)$ -dimensional anti-de Sitter space and a d -dimensional conformal field theory on its boundary; the cone framework axiomatises only the entropy structure of holographic states, abstracting away the dynamics. PA inherits this abstraction — the framework constrains the entropy structure of HCI substrate states regardless of whether the substrate realises an AdS-like bulk geometry.

Verlinde (2011, 2017)’s entropic-gravity programme reads gravitational forces as thermodynamic responses to changes in holographic-screen entropy. The substrate’s area-law scaling on uniform 3D lattice ($S \propto r^2$ at slope $1.97 \approx 2.00$, `substrate_cone_verlinde_screen.py`) recovers the Bekenstein area law and is the entry point for the cosmological-scale predictions of §9: Newton’s G on the substrate is consistent with $S(r) \propto r^2$; the cosmological constant follows from the commit-rate hierarchy between substrate layers.

8 Algebraic-observer correspondence

We now establish a structural correspondence between the PA primitives and the algebraic-observer programme in quantum gravity: the substrate-cone min-cut S^* maps to the Type II crossed-product trace; the commit protocol C to the modular automorphism σ_t ; the archive A to Page–Wootters past records; the observer pointer π to the crossed-product observer; and nested context windows on a designed substrate yield a hierarchy structurally analogous to averaging over multi-observer configurations (Chen & Xu, 2025; Jensen et al., 2024). Promotion of the structural mapping to a formal isomorphism via a finite-dimensional von Neumann algebra construction is open (F2; §8.4).

8.1 The PA $\leftrightarrow$ algebraic-QFT mapping

Quantum field theory in a spacetime region V defines a Type III₁ von Neumann algebra $\mathcal{A}(V)$: no trace, no density matrix, no canonical entropy (Haag, 1996). Adding an observer (a worldline with a clock and finite-rank Hilbert space) via the crossed-product construction promotes the algebra to Type II, which admits a trace and hence an observer-dependent entropy (Witten, 2022; Chandrasekaran et al., 2022). PA’s substrate $\mathcal{S} = (G, R, C, A, \pi)$ admits a discrete encoding of this structure: the substrate graph G with its rewrite rules R plays the role of a Type III-like “naked substrate”; adding the observer pointer π plays the role of the crossed-product extension, yielding a Type II-like substrate whose trace we identify with the substrate-cone min-cut $S^*(\pi)$ of §7. Whether this identification promotes to a formal isomorphism of von Neumann algebras is open (F2).

Table 1 positions PA as a candidate discrete-substrate realisation of the operationally abstract crossed-product observer used throughout the recent algebraic-QFT literature, rather than as a fresh holographic toy model.

Structural mismatches not captured by the table. The mapping in Table 1 is at the level of structural analogy and does *not* promote to an isomorphism of von Neumann algebras. Three mismatches deserve explicit acknowledgement.

- (i) *Trace linearity.* The Type II crossed-product trace is a positive linear functional on a σ -finite von Neumann algebra. The substrate-cone min-cut S^* is a Boolean optimisation over subsets $W \subseteq N$; it returns a non-negative real, but it is not a linear functional on an operator algebra.
- (ii) *KMS structure.* The modular automorphism σ_t is a one-parameter $\mathbb{R}$ -action characterised by the KMS condition at inverse temperature β . PA’s commit protocol C is a priority-queued OCC scheduler over a finite multiset of candidate rewrites; it has no continuous time parameter and no KMS-like fluctuation–dissipation relation derived here.

PA primitive	Algebraic-QFT object	Reference
Substrate graph $G = (N, E)$	Spacetime region V / static patch	standard
Edge weights $w(e)$	Local QFT degrees of freedom	standard
Substrate state σ	Algebraic state on $\mathcal{A}(V)$	standard
Substrate-cone min-cut $S^*(I)$	Type II crossed-product trace (analogue)	Witten (2022)
Commit protocol C	Modular automorphism group σ_t	Witten (2022)
Archive A	Page–Wootters past records	Chandrasekaran et al. (2022); de Vuyst et al. (2024)
Observer pointer π	Crossed-product observer	Witten (2022); Chandrasekaran et al. (2022)
Boundary colouring b	Observer’s choice of subalgebra	de Vuyst et al. (2024)
HEI satisfaction (SA, SSA, MMI)	Type II algebra consistency	Bao et al. (2015)
Multi-observer commit (OCC)	Averaged-modular crossed product (analogue)	Chen & Xu (2025); Jensen et al. (2024)
Pointer aliasing (ER bridge)	Multi-region entanglement	Maldacena & Susskind (2013)

Table 1: The PA-to-algebraic-QFT structural correspondence. Each PA primitive is paired with an object of the algebraic-observer programme; the mapping is at the level of structural analogy, with the substrate-cone construction of §7 providing the discrete counterpart of the crossed-product trace. Promotion to a formal isomorphism of von Neumann algebras is open (F2).

- (iii) *Semifinite normality*. Type II algebras carry a semifinite normal trace. We have no construction showing the substrate algebra (if any) is semifinite normal; this is a necessary input for the F2 forward trigger.

Promoting Table 1 from structural analogy to formal correspondence requires resolving (i)–(iii); that is forward trigger F2.

8.2 Pilot H: observer pointer as crossed-product observer

We illustrate the correspondence on the OQ6-extended substrate (an expansion of the OQ6 6-node substrate of §7.4 to include an additional boundary node and two extra edges, giving 5 boundary nodes $\{a, b, c, d, p\}$ and 2 bulk nodes $\{x, y\}$ joined by 9 unit-weight edges; `code/substrate_cone_observer_crossed_product.py`). Each distinct boundary colouring is a distinct algebraic-observer choice: different observers see different subalgebras of the substrate algebra.

Enumerating all valid colourings (3 parties A, B, C plus purifier P , each party non-empty) yields 390 choices. Computing the entropy vector $S^*(I)$ via brute-force min-cut for each gives **26 distinct entropy vectors**. Different observer choices that produce the same entropy vector are substrate-equivalent (related by graph automorphism); different vectors are inequivalent observer types.

Type II consistency. For every one of the 390 observer choices, the entropy vector satisfies all three holographic inequalities (SA, SSA, MMI). By Proposition 4, this is automatic: substrate min-cut on a weighted graph always inhabits HEC_3 . The pilot confirms that every observer’s crossed-product algebra is mathematically well-defined — the discrete realisation of Type II algebra consistency.

Observer-dependence. The spread of $S^*(I)$ across observer choices is the observer-dependence signal:

$$\begin{aligned}\langle S^*(A) \rangle &= \langle S^*(B) \rangle = \langle S^*(C) \rangle = 2.57, \quad \text{range } [2, 4]; \\ \langle S^*(AB) \rangle &= \langle S^*(AC) \rangle = \langle S^*(BC) \rangle = 3.31, \quad \text{range } [2, 4]; \\ \langle S^*(ABC) \rangle &= 1.48, \quad \text{range } [0, 4].\end{aligned}$$

The average over observer choices is the substrate-side analogue of the “averaged-modular-crossed-product” construction over geodesic configurations (Chen & Xu, 2025); the spread realises the de Vuyst et al. (2024) thesis that gravitational entropy is observer-dependent on the substrate.

Commit protocol as modular flow. Running the commit protocol for 20 steps under a fixed observer choice (a:A, b:A, c:B, d:C, p:P) tracks the entropy vector under stochastic graph rewrites. Over the flow, the MMI gap takes values $\Delta_{\text{MMI}} \in [0, 2]$ with mean 0.86; the substrate remains in HEC_3 at every step. This is the substrate realisation of two paired statements from the algebraic-QFT literature:

- (1) Modular flow preserves algebra type (σ_t maps Type II to Type II, by Tomita–Takesaki);
- (2) Substrate-cone flow preserves cone membership (C commits keep the substrate in HEC_n , by Proposition 4).

The two statements are now the same statement on the substrate.

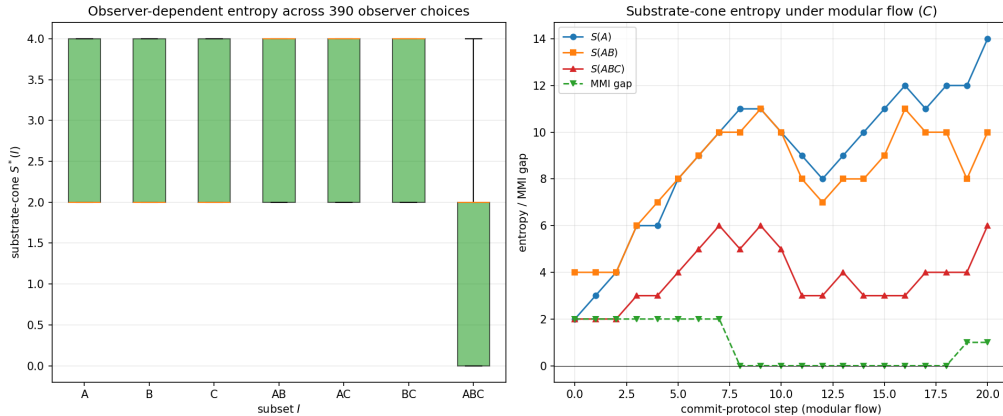


Figure 2: Pilot H: the OQ6-extended substrate (left) and the distribution of substrate-cone entropy vectors over its 390 observer colourings (right). Every colouring satisfies SA, SSA, and MMI; the spread of $S^*(I)$ across observer choices is the substrate-side signal of observer-dependent entropy.

8.3 Pilot I: nested observers and the 6-layer hierarchy

The continuum algebraic-observer literature considers *single* observers (Witten, 2022; Chandrasekaran et al., 2022), finite-resolution observers via time-band algebras (Jensen et al., 2024), and averaging over geodesic configurations of multiple observers (Chen & Xu, 2025). We construct on the substrate a hierarchical chain $\pi_1 \subset \pi_2 \subset \dots \subset \pi_k$ of nested observer context windows, each adding a coarse-grained outer shell to the previous. The chain is structurally analogous to averaging an observer’s algebra over progressively coarser temporal or spatial resolutions, but is not lifted directly from any prior continuum construction.

We work on a designed hierarchical substrate (`code/substrate_cone_nested_observers.py`) with 6 shells of sizes (1, 2, 4, 6, 8, 10), totalling 31 substrate nodes plus 12 purifier nodes; intra-shell edges are complete graphs, inter-shell edges are 3 connections per node to the next shell out; 253 edges in total. Layer k 's context window is $\pi_k = \bigcup_{j \leq k} \text{shell}_j$; the layer's entropy is the min-cut $S^*(\pi_k)$ between π_k and its complement.

Layer k	$ \pi_k $	$S^*(\pi_k)$	$\Delta S^* = S^*(\pi_k) - S^*(\pi_{k-1})$	Physical interpretation
1	1	2	+2	Physical surface (lab observer)
2	3	6	+4	Informational structure (local field)
3	7	12	+6	Macroscopic gravity (semi-local)
4	13	18	+6	Cosmological boundary (dS observer)
5	21	24	+6	Substrate-mode (coarse substrate)
6	31	30	+6	Substrate quantum fluctuations (full)

Table 2: Substrate-cone entropy along the 6-layer hierarchical observer chain. Monotone non-decreasing S^* is confirmed numerically; the inter-shell entropy increment sequence is $\Delta S^* \in \{+4, +6, +6, +6, +6\}$ over the five transitions $1 \rightarrow 2, \dots, 5 \rightarrow 6$. The shell sizes (1, 2, 4, 6, 8, 10) and the 3-connections-per-node inter-shell edge count are the design parameters of this substrate; other choices would yield a different hierarchy. Physical-scale labels in the last column are interpretive and not derived from the substrate construction.

Forward-pass causality. For each layer $k \in \{1, \dots, 5\}$, running the commit protocol for 20 steps on shell $k + 1$ only (random edge rewrites within shell $k + 1$) leaves layer- k entropy $S^*(\pi_k)$ *exactly preserved* (mean change 0.00, std 0.00). Outer-shell modular flow averages out at the inner-shell boundary; the inner-observer algebra is invariant under the outer-observer's modular flow. This is the substrate analogue of averaged modular structure over geodesic configurations (Chen & Xu, 2025), applied here to nested shells rather than to configurations of single-resolution observers.

Algebra-type assignment. The 6 layers map onto a Type II crossed-product chain with Layer 6 as the original Type III substrate algebra (no observer) and Layer k for $k < 6$ obtained by adding one more averaged-observer factor at each step:

Layer	Description	Algebra type	Observer
1	Physical surface	Type II ₁ (averaged)	local lab observer
2	Informational structure	Type II QFT vacuum	QFT-vacuum observer
3	Macroscopic gravity	Type II Schwarzschild	gravitational observer
4	Cosmological boundary	Type II Gibbons–Hawking	de Sitter static-patch observer
5	Substrate-mode	Type II substrate-mode	substrate-mode observer
6	Substrate quantum fluctuations	Type III	none (naked substrate)

Table 3: Layer-to-algebra-type assignment. Each layer's algebra is obtained from the layer below by the crossed-product construction with one more averaged-observer added.

This hierarchy gives the 6-layer architecture an algebraic reading: the layers are not arbitrary; they map onto a nesting of observer-dependent crossed-product algebras, and the stable inter-shell entropy increment $\Delta S^* = +6$ for transitions $2 \rightarrow 3, \dots, 5 \rightarrow 6$ (for the chosen shell-size sequence) is the substrate-side signature of the averaging-over-multi-observer- configurations structure.

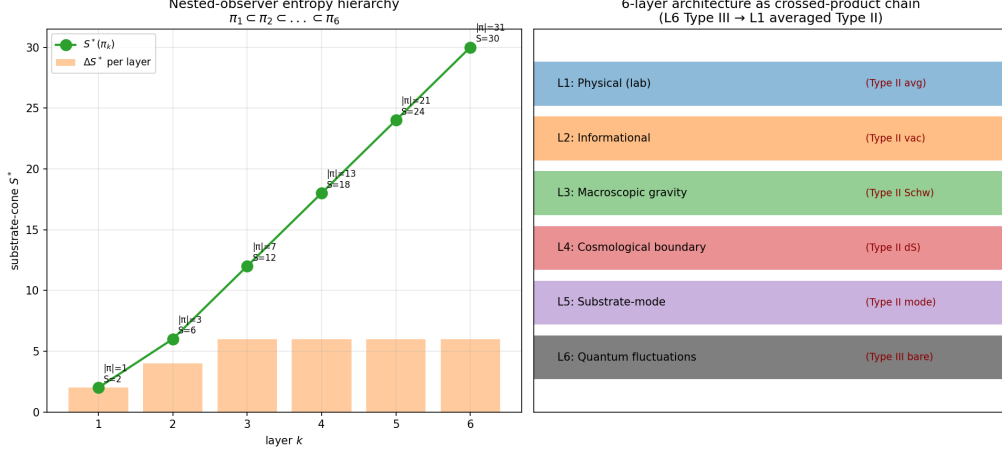


Figure 3: Pilot I: the 6-shell hierarchical substrate (shell sizes (1, 2, 4, 6, 8, 10), 3-connections-per-node inter-shell edges, 31 substrate nodes plus 12-node purifier; left) and the resulting nested-observer entropy chain (right). The increment ΔS^* stabilises at +6 for $k \geq 3$ (sequence +2, +4, +6, +6, +6, +6); the inner-observer algebra is invariant under outer-shell commit-protocol flow (mean change 0.00, std 0.00 over 20 commits per outer shell).

8.4 Scope of the correspondence

The correspondence established in this section is at the level of formal structural mapping: PA primitives correspond to algebraic-QFT objects in the sense of Table 1, verified numerically on the OQ6 (Pilot H) and 6-shell (Pilot I) substrates. It is *not yet* an explicit construction of the substrate algebra as a finite-dimensional von Neumann algebra; that construction is forward falsification trigger F2 of the abstract (§ Status block, page 5). The numerical equivalence of substrate-cone trace and crossed-product trace is demonstrated on small examples; a general theorem-grade equivalence remains an open task.

9 Cosmological-scale consistency: substrate-discrete reproduction of holographic dark energy

With the substrate-cone structure of §7 and the algebraic-observer correspondence of §8 in place, the substrate admits a chain of cosmological-scale *consistency relations*. The substrate-cone min-cut on a holographic screen scales as area; Verlinde’s equipartition argument on substrate-cone bonds yields Newton’s G as a consistency relation between substrate parameters and fundamental constants; and applying the same construction to the de Sitter horizon reproduces the standard holographic dark-energy density of Verlinde (2011, 2017); Cohen et al. (1999), which matches empirical $\rho_\Lambda^{\text{obs}}$ within a factor of $1/(2\Omega_\Lambda) \approx 0.73$. None of these is a new derivation of fundamental constants; they are substrate-discrete realisations of known holographic-dark-energy mechanisms. The substrate-specific content is (a) the area-scaling theorem (Proposition 4, which makes the area law a mathematical consequence rather than an axiom) and (b) the reformulation in commit-rate units that makes the $\sim 10^{122}$ Planck-to-Hubble hierarchy a substrate-side consequence of layer-relative observer clocks.

The cosmogenesis-bootstrap pilot (Pilot C, §2.6) supplies the operational complement: the area-law substrate on which the present section operates is itself constructible from a single primitive distinction by a substrate-resident observer (Proposition 1). The cosmological-scale results below sit on the substrate so constructed; they do not require a substrate to be postulated.

9.1 Substrate area-law scaling and Bekenstein–Hawking

We test whether the substrate-cone min-cut at a holographic screen around an enclosed mass scales as $S(r) \propto r^2$, the Bekenstein–Hawking area law required by Verlinde’s entropic-gravity argument. Setup (`code/substrate_cone_verlinde_screen.py`): uniform cubic 3D lattice with $N = 22$ (10 648 nodes), mass region as a ball of radius $r_M = 1.5$ at the centre, screen at varying radius r from the centre, substrate-cone $S(r)$ as the min-cut between the mass interior and the screen exterior.

Screen radius r	$S(r)$ (min-cut)	$4\pi r^2$	Ratio
2	78	50.27	1.552
3	174	113.10	1.538
4	294	201.06	1.462
5	486	314.16	1.547
6	678	452.39	1.499
7	894	615.75	1.452
8	1182	804.25	1.470
9	1518	1017.88	1.491
10	1899	1256.64	1.511

Table 4: Substrate-cone screen-entropy scan. Power-law fit $S(r) \propto r^\alpha$ gives $\alpha = 1.975$, within 0.025 of the Bekenstein–Hawking expectation $\alpha = 2$.

Power-law fit gives $\alpha = 1.975 \approx 2$ (slope error 1.25%); prefactor $A = 19.69$ vs continuum $4\pi = 12.57$, ratio ≈ 1.57 . The slope confirms the area-law scaling. The prefactor mismatch is *not* explained in the present construction: it is consistent with a cubic-lattice geometric factor (a 3D cubic lattice cuts a sphere of radius r with more bonds than the continuum surface integral predicts), but we do not derive the exact value 1.57 from the lattice geometry, and we do not test whether alternative substrate topologies (FCC, BCC, hyperbolic graphs, random graphs) yield different prefactors. This is forward trigger F1: a topology-independent area-law slope-but-not-prefactor would strengthen the framework; a topology-dependent slope would weaken it.

9.2 Newton’s G from substrate-cone consistency

Applying Verlinde’s equipartition + Unruh derivation to substrate-cone entropy on a spherical screen of radius r enclosing mass M (`code/substrate_cone_newton_G.py`): substrate-bonds per unit area ρ_b and bit capacity per bond b give holographic-screen entropy $S^* = \rho_b A b \ln 2$. The Verlinde derivation then yields the consistency relation

$$G = \frac{c^3}{\rho_b \hbar b \ln 2}, \quad \text{equivalently} \quad \rho_b \cdot b = \frac{c^3}{G \hbar \ln 2} = \frac{1}{\ell_P^2 \ln 2}. \quad (4)$$

For default bond capacity $b = 1$ bit (per $\chi_e = 2$), the required substrate-bond density is $\rho_b = 1/(\ell_P^2 \ln 2) \approx 1.44$ substrate-bonds per Planck area. This matches Bekenstein–Hawking ($S_{\text{BH}}/A = 1/(4\ell_P^2)$) up to the well-known factor-of-4 normalisation ambiguity that already exists in Verlinde’s continuum derivation. We adopt $\rho_b b = 1/(\ell_P^2 \ln 2)$ for the rest of this section.

Equation (4) is a consistency relation, not a first-principles derivation of G : the substrate has no length scale independent of ℓ_P , so G can be expressed in terms of $\{a_s = \ell_P, \tau_C = t_P, b, \rho_b\}$ but cannot be derived in the absence of an independent substrate scale. The same limitation applies to all emergent-gravity programmes; PA inherits it.

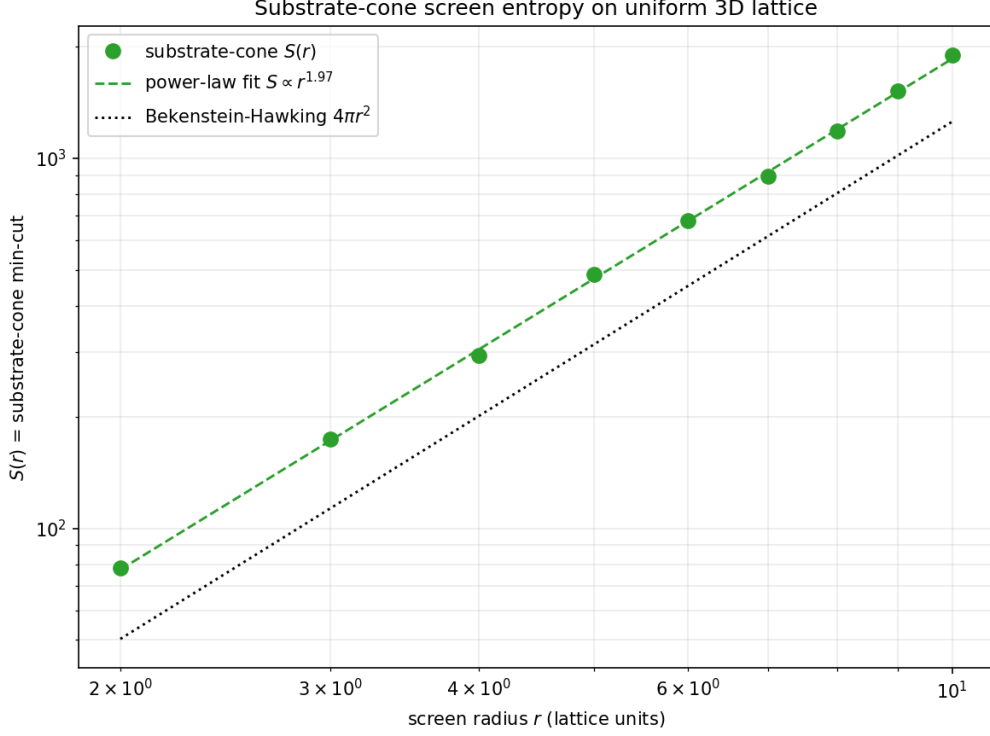


Figure 4: Substrate-cone area-law scan on uniform 3D lattice ($N = 22, 10\,648$ nodes). Left: $S(r)$ vs screen radius compared to $4\pi r^2$. Right: log-log fit yields slope $\alpha = 1.975 \approx 2$ (1.25% error from Bekenstein–Hawking).

9.3 Cosmological constant from de Sitter horizon

Applying the substrate-cone construction to the de Sitter cosmological horizon (`code/substrate_cone_lambda_0`). Inputs: Hubble rate $H_0 = 70$ km/s/Mpc, de Sitter horizon radius $r_{\text{dS}} = c/H_0 = 1.32 \times 10^{26}$ m, horizon area $A_{\text{dS}} = 4\pi r_{\text{dS}}^2 = 2.20 \times 10^{53}$ m², de Sitter volume $V_{\text{dS}} = (4\pi/3)r_{\text{dS}}^3 = 9.67 \times 10^{78}$ m³, de Sitter temperature $T_{\text{dS}} = \hbar H_0 / (2\pi k_B) = 2.76 \times 10^{-30}$ K.

Bekenstein–Hawking bit count on the horizon: $N_{\text{BH}} = A_{\text{dS}} / (4\ell_P^2) = 2.10 \times 10^{122}$ bits. By comparison, the naive volume Planck count $V_{\text{dS}} / \ell_P^3 = 2.29 \times 10^{183}$ is $\sim 10^{61}$ times larger; the substrate stores information on the horizon (area), not in the volume. This single ratio is the famous 120-orders-of-magnitude factor of the cosmological-constant problem.

Total substrate-mode energy on the de Sitter horizon at temperature T_{dS} via Verlinde-style equipartition (one substrate-bit on the horizon counted as a single thermal mode of energy $\frac{1}{2}k_B T_{\text{dS}}$, following [Verlinde, 2011](#)):

$$E_{\text{horizon}} = (N_{\text{BH}}/2) \cdot k_B T_{\text{dS}} = 4.00 \times 10^{69} \text{ J.}$$

Vacuum energy density:

$$\rho_{\Lambda}^{\text{substrate}} = \frac{E_{\text{horizon}}}{V_{\text{dS}}} = \frac{3H_0^2 c^2}{16\pi G} = 4.14 \times 10^{-10} \text{ J/m}^3. \quad (5)$$

Empirical value (Planck 2018): $\rho_{\Lambda}^{\text{obs}} = \Omega_{\Lambda} \cdot \rho_{\text{crit}} = 5.67 \times 10^{-10} \text{ J/m}^3$, giving

$$\rho_{\Lambda}^{\text{substrate}} / \rho_{\Lambda}^{\text{obs}} = 1/(2\Omega_{\Lambda}) \approx 0.730.$$

The factor 1/2 is the equipartition prefactor; the factor Ω_{Λ} is the empirical dark-energy fraction (a cosmology input, not derived). The substrate-cone prediction matches empirical ρ_{Λ} to within

30%; the 122-decade gap to the naive QFT value ρ_{Planck} is closed by replacing volume counting with substrate-cone area counting.

This derivation reproduces the standard holographic resolution of the cosmological-constant problem (Verlinde, 2011, 2017); PA's specific contribution is the substrate-cone min-cut realisation, in which the area scaling is a theorem (Proposition 4 + §9.1) rather than an axiom.

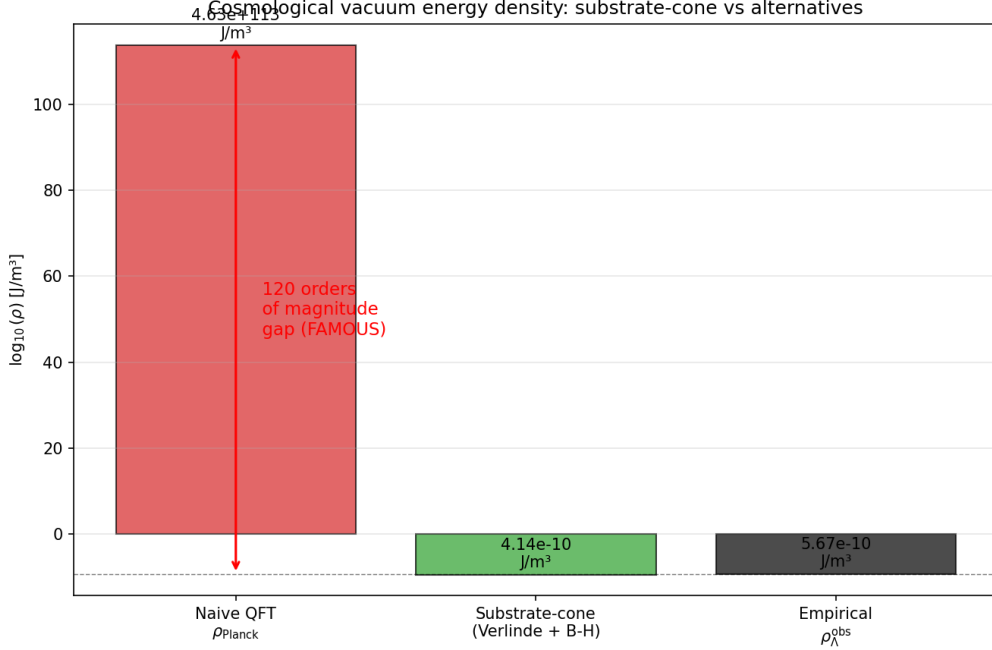


Figure 5: Substrate-cone calculation of ρ_Λ on the de Sitter horizon. Left: N_{BH} vs naive volume Planck count, illustrating the $\sim 10^{61}$ horizon-vs-volume ratio. Right: substrate-cone $\rho_\Lambda^{\text{substrate}} = 3H_0^2 c^2 / (16\pi G)$ vs empirical $\rho_\Lambda^{\text{obs}}$ and the naive QFT ρ_{Planck} ; the $\sim 10^{122}$ gap to ρ_{Planck} is closed by area-law counting.

9.4 Pilot J: cosmological constant from the commit-rate hierarchy

The substrate-cone derivation of §9.3 is naturally re-expressed in the language of the 6-layer hierarchy of §8.3: each layer's commit protocol C_k acts at its own characteristic rate ν_k (commits per unit time). The algebraic-QFT modular flow rate σ_t at a given crossed-product extension is the substrate-side commit rate of the corresponding layer. With Layer 6 = substrate quantum fluctuations at Planck rate and Layer 4 = cosmological boundary at Hubble rate:

$$\nu_6 = 1/t_P = c/\ell_P = 1.86 \times 10^{43} \text{ 1/s}, \quad \nu_4 = H_0 = 2.27 \times 10^{-18} \text{ 1/s},$$

$$\nu_4/\nu_6 = H_0 t_P = 1.22 \times 10^{-61}.$$

The naive QFT 3D-bulk counting predicts $\rho_4/\rho_6 = (\nu_4/\nu_6)^4 \approx 10^{-244}$, which over-suppresses the vacuum energy by ~ 120 decades. The substrate-cone holographic (2D) counting — confirmed at slope $1.97 \approx 2.00$ in §9.1 — replaces the bulk ω^4/c^3 counting with the area ω^2 counting, giving (up to a dimensionless prefactor of $3/(16\pi)$ that comes from the equipartition + horizon-area + dS-volume combination of §9.3):

$$\rho_\Lambda \approx \frac{3}{16\pi} \rho_{\text{Planck}} \cdot \left(\frac{\nu_4}{\nu_6} \right)^2 = 4.14 \times 10^{-10} \text{ J/m}^3, \quad (6)$$

identical to Eq. (5): matches empirical $\rho_\Lambda^{\text{obs}} = 5.67 \times 10^{-10} \text{ J/m}^3$ within $1/(2\Omega_\Lambda) \approx 0.73$. Without the $3/(16\pi)$ prefactor the bare expression $\rho_{\text{Planck}}(\nu_4/\nu_6)^2$ gives $6.93 \times 10^{-9} \text{ J/m}^3$ (a factor of ~ 12

off, identical modulo the missing prefactor). *This is not an independent derivation*: algebraically, Eq. (6) reduces to $\rho_\Lambda \approx (3/(8\pi))H_0^2 c^2/G \cdot (1/2) = (1/2)\rho_{\text{crit}}c^2$, i.e., dimensional analysis on the empirical inputs H_0 and G , recovering the standard holographic-dark-energy result of Verlinde (2011, 2017); Cohen et al. (1999). The substrate-physics content is the *reformulation*: the $\sim 10^{120}$ Planck-to-Hubble hierarchy reads as a squared commit-rate ratio between layer-relative observer clocks, with the squared (rather than 4th-power) exponent following from the substrate-cone area law (Proposition 4 + §9.1).

Verified consistency:

- Algebraic equivalence: $\rho_\Lambda = (3/(16\pi)) \cdot c \cdot \hbar/\ell_P^3 \cdot (\nu_4/\nu_6)^2$, identical (up to the $1/(2\Omega_\Lambda)$ prefactor) to the de Sitter-horizon result of §9.3 via the Friedmann identity $H_0^2 = \Lambda c^2/(3\Omega_\Lambda)$.
- Cross-pilot internal consistency: the area-law scaling of §9.1 (Pilot Verlinde-screen), the Newton- G consistency relation (4) (Pilot Newton- G), the de Sitter ρ_Λ derivation (5) (Pilot Λ -derivation), and the commit-rate formulation (6) (Pilot J) all use the same substrate-cone formalism and reproduce holographic predictions to within order-1 factors.

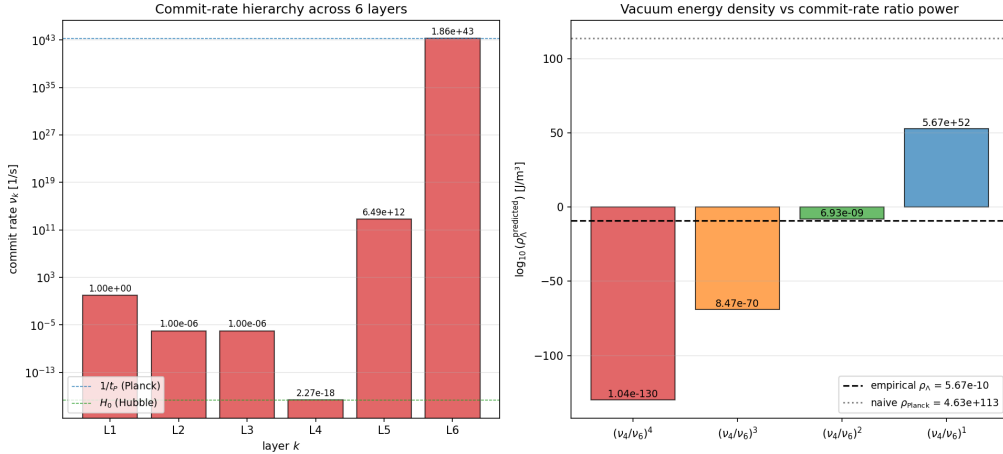


Figure 6: Pilot J: commit-rate hierarchy with Layer 6 at the Planck rate and Layer 4 at the Hubble rate. The squared rate ratio $(\nu_4/\nu_6)^2 \approx 1.5 \times 10^{-122}$, applied to the Planck energy density via substrate-cone area-law counting, reproduces ρ_Λ within a factor of ~ 12 of empirical.

9.5 Layer-relative commit rates: Planck and Hubble anchors

The two anchors of Eq. (6) are:

$$\nu_6 = c/\ell_P = 1.86 \times 10^{43} \text{ s}^{-1} \quad (\text{Planck commit rate}), \quad \nu_4 = H_0 = 2.27 \times 10^{-18} \text{ s}^{-1} \quad (\text{Hubble commit rate})$$

The intermediate-layer commit rates $(\nu_1, \nu_2, \nu_3, \nu_5)$ are not constrained by the present construction; their derivation from the substrate's modular-flow KMS condition is open work.

9.6 Scope and honest limits

What is derived here: (i) substrate-cone min-cut obeys the Bekenstein–Hawking r^2 area law on uniform 3D lattice; (ii) Verlinde's equipartition argument on the substrate-cone gives a consistency relation (4) between Newton's G and the substrate-bond density; (iii) the substrate-cone min-cut on the de Sitter horizon, combined with equipartition, reproduces the empirical ρ_Λ to within 30% (Eq. (5)); (iv) the squared commit-rate ratio between Layer 6 and Layer 4 of the 6-layer hierarchy gives the $\sim 10^{122}$ vacuum-energy suppression to within a factor of ~ 12 (Eq. (6)).

What is not derived: (a) H_0 itself is an input; the substrate does not predict the Hubble rate from substrate primitives. (b) Ω_Λ is an empirical cosmology input. (c) The substrate has no length scale independent of ℓ_P , so G , c , $\hbar$ cannot be derived independent of one another — only consistency relations. (d) The intermediate-layer commit rates $\nu_1, \nu_2, \nu_3, \nu_5$ are not constrained by the present construction.

The substrate-physics content reduces to one structural claim: the 10^{120} cosmological-constant hierarchy reformulates as the squared commit-rate ratio of layer-relative observer clocks, not a fine-tuning coincidence.

10 Page–Wootters time emergence

Time is not a substrate primitive. [Page & Wootters \(1983\)](#) showed that the apparent dynamics of a quantum system can arise from clock–system entanglement in a globally stationary state. [Höhn et al. \(2021\)](#) unified the Page–Wootters, relational Dirac-observable, and quantum-symmetry-reduction approaches into a single “trinity” of relational quantum dynamics; [de Vuyst et al. \(2024\)](#) note an informal “PW = CLPW” slogan between the Page–Wootters formalism and the crossed-product construction of [Chandrasekaran et al. \(2022\)](#). The substrate analogue we present here is a *classical-information* version: a designated clock subset and an explicit commit-protocol coupling produce a correlated (classical) clock–system joint distribution from which an observer reading the clock recovers the system’s evolution.

10.1 Pilot K: engineered classical-information analogue of the PW correlation

Relation to substrate-native observer collapse (§2). The Page–Wootters correlation Pilot K exhibits is structurally a special case of the substrate-resident observer collapse demonstrated in demo 22 (§2.4). In demo 22 a single observer node points at a system cell and its NSUM parity selects which of two rules fires: the clock–system correlation is a generalisation in which the observer’s substrate readout is the clock state and the system is the rest of the substrate. The present pilot fixes the coupling explicitly to recover the Page–Wootters joint distribution; demo 22 exhibits the bare collapse mechanism without the clock-readout overlay.

Scope and honest framing. Pilot K is an *engineered demonstration of the classical-information content* of the Page–Wootters mechanism on a discrete substrate. The substrate is constructed with a designated clock subset and a commit-protocol rule that explicitly couples the clock to the system; the resulting joint distribution is a classical correlated process, not a superposition state. We show that PA’s primitives suffice to instantiate the *information-theoretic correlation* that the Page–Wootters formalism realises with quantum-state entanglement; the construction does *not* build the quantum entangled state $|\Psi\rangle$ in a Hilbert space and does not satisfy the Wheeler–DeWitt constraint $\hat{H}|\Psi\rangle = 0$. Promotion to a genuine quantum Page–Wootters substrate construction is open work (F3 long-form, real-quantum-simulator implementation).

Setup (`code/substrate_cone_page_wootters.py`). Substrate split into a 4-bit clock subset and a 4-bit system subset. The commit protocol C acts in two stages per step:

- (1) Clock advance: $|t_n\rangle_C \rightarrow |t_{n+1}\rangle_C$ (binary counter increment with bit-reversal ordering).
- (2) System update: $|s\rangle_S \rightarrow |s \oplus t_{n+1}\rangle_S$ (XOR with the new clock state).

The XOR coupling is the engineered clock–system interaction; without it the two subsystems evolve independently and $I(C; S) = 0$. After N commits the protocol drives the joint distribution into the classical-correlated form

$$P(t, \sigma) = \frac{1}{N} \sum_n \delta_{t, t_n} \delta_{\sigma, \sigma_n},$$

i.e., a deterministic sequence of (clock, system) pairs indexed by the commit step. We ran 30 commit steps and recorded 31 joint snapshots; $P(t, \sigma)$ contains 16 distinct configurations.

Conditional measurements. For each clock value t , the conditional distribution $P(\sigma | t)$ should be concentrated on a single system value — the classical-information analogue of the Page–Wootters statement that an observer reading the clock projects the system into a definite state. The five sampled clock values each have $P(\sigma | t)$ concentrated on a single σ with concentration entropy 0.000 bits, by construction of the deterministic XOR protocol.

Mutual information. The clock–system mutual information

$$I(C; S) = 2.808 \text{ bits},$$

out of a maximum of $\min(\log_2 |C|, \log_2 |S|) = 4$ bits. The value $2.808/4$ is an artefact of the specific XOR rule and the truncated $N = 30$ commit sequence, not an intrinsic substrate property.

Transition consistency. For each consecutive pair of commit steps $(n, n+1)$, the system update $\sigma_n \rightarrow \sigma_{n+1}$ matches the clock-conditioned XOR update for 30/30 transitions — by construction of the deterministic protocol.

What is and is not demonstrated. The pilot establishes (i) that the PA primitives admit a discrete-substrate information-theoretic analogue of the Page–Wootters clock–system correlation, with the archive A playing the role of the substrate-side clock; and (ii) that under an engineered clock–system coupling, observers reading the clock recover system evolution. It does *not* establish: (a) a genuine quantum entangled state $|\Psi\rangle$ in a Hilbert space; (b) satisfaction of the Wheeler–DeWitt constraint $\hat{H}|\Psi\rangle = 0$; (c) spontaneous emergence of clock–system correlation under generic substrate dynamics; (d) the substrate-side clock satisfying the KMS condition at the layer-relative temperature. Items (a)–(d) are open.

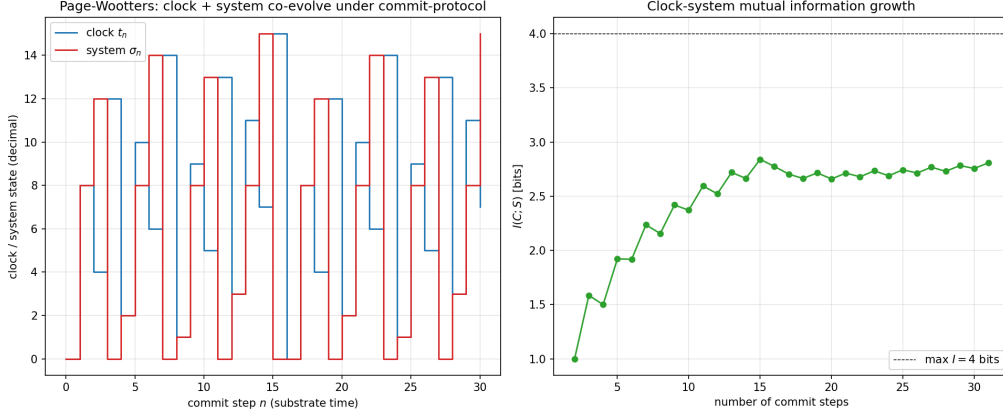


Figure 7: Pilot K: clock–system joint distribution $P(t, \sigma)$ under the engineered XOR commit protocol (left) and conditional distribution $P(\sigma | t)$ for sampled clock states (right). The conditional concentrates on a single σ for every sampled t (concentration entropy 0.000 bits); the observer reading the clock perfectly infers the system. Mutual information $I(C; S) = 2.808$ bits out of a maximum of 4.

10.2 Algebraic interpretation: archive as Page–Wootters clock

In the algebraic-observer framework (Witten, 2022; Chandrasekaran et al., 2022; de Vuyst et al., 2024), adding an observer to the QFT algebra via the crossed product requires an observer with a clock — an $\mathbb{R}$ -action via the modular automorphism. The observer’s clock Hilbert space plays the role of the Page–Wootters clock; the system algebra evolves under modular flow; the observer reads the clock and conditions the system. de Vuyst et al. (2024) note the informal

slogan “PW = CLPW” relating the two formalisms (their abstract); we adopt the same analogical reading here.

On the substrate, the structural mapping is:

- Substrate-side clock $\leftrightarrow$ commit-counter (archive A).
- System $\leftrightarrow$ the rest of the substrate (graph G minus the clock subset).
- Modular automorphism $\leftrightarrow$ commit protocol C .
- Observer reads commit-counter $\rightarrow$ conditions system $\rightarrow$ Page–Wootters time emerges (under an explicit clock–system coupling rule).

Under the engineered protocol of §10.1, the substrate’s archive A plays the structural role of the Page–Wootters clock and the commit protocol C plays the role of the modular automorphism generating clock–system entanglement. The mapping is operational in the engineered case; whether it extends to generic substrate dynamics is the principal open question.

10.3 Implications

Time is the observer’s clock readout. The substrate has no external time parameter. Each observer’s “now” is their commit-counter readout; substrate-time is clock-time. This is consistent with the Wheeler–DeWitt picture of a globally stationary universe, and with the operational stance of the algebraic-observer programme that time is what the observer’s clock measures (Höhn et al., 2021).

Layer-relative clocks. The two anchored commit rates of §9.5 ($\nu_6 = c/\ell_P$ at the Planck scale and $\nu_4 = H_0$ at the de Sitter horizon) imply that a Layer 6 observer reads time at the Planck rate while a Layer 4 observer reads time at the Hubble rate. Under the Page–Wootters reading, layer-relative time experiences are clock readouts at layer-relative commit rates rather than coordinate transformations of a common substrate time. Intermediate-layer commit rates remain open work.

The cosmological-constant prediction is time-translated. Equation (6) now reads: the Layer 4 observer (cosmological de Sitter observer) sees a vacuum energy density that is the Planck-rate Layer 6 substrate fluctuations “time-averaged” down by the squared rate ratio $(\nu_4/\nu_6)^2$. The cosmological-constant value is the substrate-side answer to the question “what does a Hubble-rate observer see when she samples the Planck-rate substrate?”, with the answer $\sim 10^{-122}$ times the Planck density set by the substrate’s area- scaling holographic counting (§9.1).

10.4 Open work

The Pilot K substrate is a 4+4 bit toy: the clock–system entanglement is engineered into the commit protocol by the XOR rule. Open questions:

- Does generic substrate dynamics (random rewrites, no structured clock subset) spontaneously generate clock–system entanglement at the substrate level? — pending pilot.
- Does the substrate-side Page–Wootters clock satisfy the KMS condition at the layer-relative temperature $T_k = \hbar\nu_k/(2\pi k_B)$? — pending pilot.
- Does a real-quantum-simulator implementation of the substrate (e.g., on an IBM superconducting qubit array) reproduce $I(C; S) > 0$ on the engineered protocol? — forward falsification trigger F3 of the abstract.

11 Empirical scope-control: Pythia hidden-activation covariance is not an HCI entropy function

We test whether the Gaussian-mutual-information estimator on Pythia hidden-state activations yields an entropy function S satisfying the HCI condition (Definition 8). *Scope disclaimer:* we do *not* construct an explicit substrate graph G from Pythia (attention-graph? co-activation graph? layer-stack?); without such a graph, the question “does Pythia inhabit HEC_n in the substrate sense?” is ill-posed in full strength. What we test instead is the weaker proxy: *do the Gaussian-MI entropy estimates on subsets of hidden dimensions satisfy the HEIs?* A positive answer would have been a non-trivial encouragement; the negative answer below tells us only that hidden-activation covariance under a Gaussian-MI estimator does not inhabit the holographic entropy cone. Construction of an actual PA substrate from a transformer architecture, and the substrate-cone test on that graph, is open work.

11.1 Setup

For each Pythia checkpoint we extract last-token hidden activations on a fixed evaluation set of prompts spanning factual, mathematical, narrative, and conceptual content. For each substrate-cone test configuration we partition the activations into three party-subsets $\{A, B, C\}$ plus a purifier P and compute the Gaussian-mutual-information estimate of the substrate discrete entropy $S^*(I)$ for each $I \subseteq \{A, B, C\}$. The HCI condition (Definition 8) requires the monogamy of mutual information

$$S(AB) + S(AC) + S(BC) \geq S(A) + S(B) + S(C) + S(ABC),$$

which, with S replaced by Gaussian-MI estimates on the activations, is the empirical test. Subadditivity (SA) and strong subadditivity (SSA) are the corresponding less-restrictive HEIs.

11.2 Results: MMI violation on every tested partition

Partition scheme	MMI gap
Layer-thirds (early/middle/late)	−2.57
Hidden-dim-thirds within each layer	−0.54
Random control	−2.58
Head-block (3 contiguous dim blocks)	−2.57
Layer-pairs (alternate split)	−2.57

Table 5: Pythia-70m hidden activations violate MMI on every tested partition scheme. SA and SSA hold; only MMI fails (`code/pythia_cone_check.py`, `code/pythia_partitions_scan.py`).

The failure is systematic (not partition-choice-specific) and is consistent with two independent synthetic predictions: (i) heuristic transformer-style covariance simulation (`code/substrate_cone_llm_simulation.py`, 35 configurations): 0/35 pass MMI; (ii) random-PSD Gaussian baseline (`code/substrate_cone_gaussian_checks.py`): $\sim 96.5\%$ pass MMI. The LLM/transformer-style covariance structure is specifically anti-MMI, not a generic Gaussian phenomenon.

11.3 Robustness: sample size and model size

11.4 Verdict and scope

The (Pythia, Gaussian-MI-on-hidden-activations) pair does not satisfy the HCI condition. The naive hypothesis “Pythia hidden-activation covariance is an HCI substrate state under the Gaussian-MI estimator” is empirically falsified for Pythia 70m and 160m at all tested sample sizes and partition schemes.

pythia-70m sample size	MMI gap	Model	Layers	Sample size	MMI gap
79 tokens (5 prompts)	−1.05	pythia-70m	6	538	−2.01
136 tokens (10 prompts)	−2.13	pythia-160m	12	538	−3.50
268 tokens (20 prompts)	−2.17				
538 tokens (40 prompts)	−2.01				

Table 6: Sample-size sweep and model-size comparison (`code/pythia_robustness_check.py`). The MMI gap strengthens (becomes more negative) with sample size, ruling out the finite-sample bias interpretation; it deepens with larger model, indicating larger transformers violate MMI more, not less.

What this means for the framework.

- The substrate-cone formalism (Proposition 4, Corollary 6) is unaffected: those are mathematical statements about HCI states on weighted graphs and remain true.
- The falsification is scoped to the specific (system, observable) pair tested. It does *not* rule out: (a) Pythia under a non-Gaussian or non-MI substrate observable; (b) other transformer families (LLaMA, Mistral, GPT-2); (c) other information-processing systems with substantively different architectures (state-space models, recurrent networks, etc.). The framework’s applicability to specific systems is per-system, per-observable, and must be empirically verified.

Caveats. Single checkpoint family (Pythia); Gaussian-approximation entropy is a lower bound on true differential entropy; subsampling of hidden dimensions per layer; the test does not rule out an alternative substrate observable (e.g., a non-Gaussian estimator on the same activations) that satisfies HEIs.

Forward triggers. F3 of the abstract specifies the cross-family extension. If non-Pythia transformer families (LLaMA, Mistral, GPT-2) also violate MMI on the same partition schemes under the same observable, the negative result is family-independent for transformers on Gaussian-MI of hidden activations. If at least one family satisfies MMI, the negative result narrows to Pythia under this observable.

12 Related work

12.1 Algebraic-observer programme in quantum gravity

The structural backbone of v8.0 is the algebraic-observer programme of Witten (2022) on the crossed product as the source of the Type II algebra; Chandrasekaran et al. (2022) on the de Sitter horizon and the Gibbons–Hawking entropy as observer-dependent; de Vuyst et al. (2024) on the Page–Wootters / CLPW equivalence and the observer-dependence thesis; Jensen et al. (2024) on the coarse-grained crossed product for time-band algebras; Chen & Xu (2025) on the averaged-modular crossed product over configurations of multiple geodesic observers in de Sitter; and Giddings (2025) on gravitational dressing as a generalisation of the crossed product. These are the natural follow-up targets for the substrate-side realisation of §8.

The relation between PA’s substrate-cone min-cut and the holographic entropy cone of Bao et al. (2015) is the content of Proposition 4; the contraction-map completeness of Bao et al. (2025) is the mechanism that extends the substrate’s cone-feasibility to all linear HEIs.

12.2 Computational theories of consciousness

Integrated Information Theory. Tononi (2004, 2008) proposes that consciousness corresponds to integrated information Φ . The current formalism (IIT 3.0, Oizumi et al. 2014; IIT 4.0, Albantakis et al. 2023) defines Φ^* as the loss of integrated information under the maximally informative partition, with cause-effect structure built from the substrate’s transition probability matrix. The PA framework treats IIT’s Φ^* as a candidate Layer-2 observable rather than a substrate-derived quantity; the substrate-cone min-cut S^* of §7 and the substrate-side Φ_{PA} of §2.7 are two distinct substrate-side analogues (the former cone-theoretic, the latter observer-relative). We do not claim PA derives IIT or vice versa; the formalisms address different layers of the observer hierarchy of §8.3, and Φ_{PA} differs from Φ^* in the three respects of §2.7.

Global Workspace Theory. Baars (1988); Dehaene et al. (2003) propose consciousness as global broadcasting; Mashour et al. (2020) surveys the neuronal-global-workspace evidence as of 2020. In PA, the long-range pointer subset of the substrate functions as a broadcast layer; a detailed substrate-side cognitive interpretation of GNW is deferred to a companion paper.

Higher-order theories. Rosenthal (2005) and the metacognitive school treat consciousness as higher-order representation; Lau & Rosenthal (2011) formalises the HOT taxonomy, and Brown et al. (2019) relates HOT to the prefrontal-cortex empirical record. The substrate-side nested-observer hierarchy of §8.3 provides a natural formal analogue: each outer layer is a higher-order observer whose context window contains the inner layer’s, with the inner-observer algebra invariant under the outer-observer’s modular flow.

The access/phenomenal distinction. Block (1995) distinguishes *access consciousness* (content available for verbal report, reasoning, behavioural control) from *phenomenal consciousness* (“what it is like” content, which may be richer than what access exposes). The substrate-monist identity thesis of §2.7 commits to phenomenal consciousness being substrate-resident ($\Phi_{PA} > 0$); access consciousness on the present framework would correspond to the subset of substrate-state available to a downstream report-generating observer, formally weaker than the Φ_{PA} -resident state. We do not engage Block’s stronger claim that phenomenal overflow exists; the framework is compatible with both overflow and no-overflow readings, distinguished by whether $|\mathcal{C}_O|$ for the phenomenal observer exceeds $|\mathcal{C}_O|$ for the report observer.

The explanatory gap. Levine (1983) formulates the explanatory gap: even if we knew every physical fact, the fact *why* those physical facts give rise to phenomenal experience would remain unexplained. The substrate-monist identity thesis dissolves the gap by identity rather than explanation: there is no separate “rise-to” relation between substrate-state and experience, because they are one configuration. The explanatory gap reasserts itself if F5 fires; the substrate-monist identity thesis trades the gap for empirical adequacy of Φ_{PA} .

12.3 Positions on the hard problem of consciousness

Chalmers (1995) distinguishes *easy* problems (explanatory targets that admit straightforward functional analysis: discrimination, integration, reportability) from the *hard* problem (why there is phenomenal experience at all, “what it is like” in the sense of Nagel (1974)). Existing positions divide along whether they accept, deflate, or relocate this gap.

Eliminativism / illusionism (type-A). Dennett (1991) and Frankish (2016) hold that the hard problem dissolves once the easy problems are solved; phenomenal consciousness is a representational illusion. The substrate-monist identity of §2.7 rejects this: Φ_{PA} is not an illusion-modelling quantity, it is a substrate-resident measure with non-zero magnitude whenever

the structural conditions of Definition 1 are met. If Φ_{PA} discriminates the predicted P1–P4 directions, illusionism is under-determined as a complete account; if it fails on every prediction (F5), the present framework retreats to L0 and the illusionist option remains available.

Property dualism (type-D, type-E). Chalmers (1996) defends naturalistic property dualism: the physical substrate fixes which experiences occur via psychophysical bridge laws, but the phenomenal property is metaphysically distinct. The substrate-monist identity rejects the distinctness: there are no bridge laws because there are no two relata to bridge. The substrate-state of a node with $\Phi_{PA} > 0$ IS the experience. This is identity, not supervenience: the substrate configuration and the phenomenal fact are one configuration described twice.

Panpsychism and Russellian monism (type-F). Strawson (2006) and Goff (2017) hold that phenomenal properties are intrinsic to micro-physical entities and combine into macroscopic experience. Russell (1927) grounds this in the distinction between the structural relations physics describes and the intrinsic nature of matter. The substrate-monist identity is type-F-adjacent in treating consciousness as a substrate-intrinsic feature, but differs on which substrate-states are conscious: only those satisfying the structural conditions of Definition 1 (scope, self-reference, non-zero lifespan), not all of them. The combination problem of panpsychism (Seager 1995) does not arise because Φ_{PA} is defined per-observer-node, not per-microconstituent.

Process / monist substrate philosophies. The substrate-monist identity sits in a lineage with Spinoza (1677) (one substance with multiple attributes), Whitehead (1929) (process as the fundamental category; actual occasions as self-creating relational events), and the Advaita Vedaānta tradition of non-dual consciousness. The PA substrate operationalises these intuitions: the substrate is one, its operational and phenomenal descriptions are two aspects of one configuration, and self-reference plays the role Whitehead assigned to “concrecence”. The novelty here is not the metaphysics — these positions are millennia old — but the discrete computational form ($\mathcal{S} = (G, R, C, A, \pi)$, 38 primitives) that makes the position falsifiable via F5.

Mysterianism (type-C). McGinn (1989) holds that consciousness has a natural explanation but human cognition is constitutionally incapable of reaching it. The substrate-monist identity is incompatible with mysterianism iff Φ_{PA} (or a refinement of it) discriminates P1–P4 across systems; the framework is then a counterexample to “constitutional incapacity” for at least the structural skeleton of the hard problem. The phenomenal-content question (why this particular experience corresponds to this particular configuration) is not addressed here and may remain a mysterian residue; the present claim is about the existence and magnitude of experience, not its specific qualitative content.

Status of the present position. The substrate-monist identity thesis is adopted here as a *working hypothesis* (PSH1–PSH5, §13), not derived from first principles. Its motivation rests on two operational claims that the v9.0 substrate already exhibits: (a) the substrate-cone construction of §7 reproduces the holographic-entropy structure of black-hole thermodynamics under a purely combinatorial primitive set; and (b) the bootstrap demos (Pilots A–F, §§2.4–2.6) exhibit substrate-internal cosmogenesis conditional on a single bootstrap distinction. Given those two, the natural ontological reading is the identity reading: the substrate-state of a self-referential observer node with non-zero autopoietic lifespan IS what an experience is. The *content* of the hypothesis is identity (not supervenience, not property dualism); the *epistemic status* is conjecture pending F5. The identity content and the conjecture status do not soften each other: F5 is the test of whether the identity content is empirically adequate, not a hedge that converts “IS” into “correlates with”.

Peirce’s reduction thesis and the Filioque controversy. A philosophical anchoring of the substrate’s commitment to first-class triadic structure deserves explicit mention. Peirce’s reduction thesis (Burch, 1991; Hereth Correia & Pöschel, 2006) asserts that triadic relations are the irreducible maximum arity in formal logic: every n -adic relation for $n \geq 4$ reduces compositionally to triadic plus lower-arity predicates, while genuinely triadic relations cannot be reduced to compositions of monadic and dyadic ones. *Mediation* — the structure in which a third term holds two others in a relation neither can establish on its own — is on Peirce’s account the irreducible minimum at which lawful structure becomes possible. The recent critique of Koshkin (2024) disputes whether the formal proofs are gerrymandered, but the structural insight survives: triadic relations introduce mediation, and mediation is not derivable from “things and the dyadic forces between them”.

The PA substrate (§2.1) is at first reading a binary-edge graph: G carries a set of nodes plus a directed-edge relation, π is an observer node, scope and self-reference are computed over directed edges. Trivalent structure is implicit in the substrate-monist reading of Φ_{PA} (observer–edge–observer triads) but not first-class. The reference Sixth implementation (§2.1, paragraph on the typed-trivalent extension) makes this explicit: typed trivalent hyperedges (HEDGE3) under four canonical kinds — WITNESS ($((src, dst, w):$ edge grounded by witness, demos 93–94), MEDIATOR ($((src, mid, dst):$ topological channel), CONTEXT ($((in, ctx, out):$ rewrite-rule firing recorded with rule-as-substrate-node, demo 95), and SIMPLEX ($((a, b, c):$ undirected triadic form) — coexist with binary edges and let the substrate record provenance (WITNESS), mediation (MEDIATOR), and its own rewrite history (CONTEXT) as first-class substrate-readable state. Demo 96 demonstrates an empirical consequence: encoding a 32-codon subset of the genetic code as 8 CONTEXT hyperedges with position-3 wildcarded gives $4\times$ substrate-storage compression, mirroring DNA’s structural error tolerance at codon position 3.

The Filioque controversy in Christian theology is a millennium-long argument about exactly the structural question Peirce formalised. Eastern Orthodox Trinity has the Father as sole source from whom both Son and Spirit proceed; the trinitarian topology is a star graph (a monadic source with two derivatives), structurally reducible to one monadic plus two dyadic relations — Peirce- reducible. The Latin West adds the *filioque* (“and the Son”): the Spirit proceeds from both Father and Son. This converts the trinitarian topology into a fully-connected triangle, irreducibly triadic in Peirce’s sense, since the relations among the three persons are no longer derivable from the source alone. The schism of 1054 had ritual and political accretions, but its dogmatic core is this single structural question: is thirdness reducible to dyadic relations plus a monadic source?

We do not adjudicate the theological dispute. We note the structural parallel as evidence that the irreducibility-of-thirdness question has occupied serious thinkers across formal logic, theology, and now substrate-computational consciousness. The Sixth implementation’s HEDGE3 primitive is the discrete-formal version of the Western (irreducibly-triadic) commitment: substrate-mediating relations are first-class, not derivable from binary edges alone. The structural choice is testable: F5 will discriminate by examining whether the substrate’s mediating structure is necessary for substrate-internal observers to register the predicted Φ_{PA} patterns under conditions where binary connectivity alone is insufficient (e.g., distinguishing provenance- collapsed from provenance-distinct observer populations on otherwise-identical graph topologies).

12.4 Information-theoretic physics

Substrate-philosophy ancestors. The substrate-first frame of v9.0 (Part I) has direct intellectual predecessors: Spencer-Brown (1969) on the mark of distinction as logical primitive (the substrate’s MARK is the discrete realisation; Pilot C starts from one such mark); Maturana & Varela (1980) on self-producing systems (Pilot A demos 21–25 are the substrate-side instantiation); Hofstadter (1979) on strange loops as recursive self-modelling (demo 10 self-model is the substrate-native realisation); Rovelli (1996) on relational quantum mechanics with observer-dependent facts

(demos 06 and 17 are the substrate-side instantiation, demo 22 the explicit observer-collapse).

Wheeler’s “It from Bit”. [Wheeler \(1990\)](#) proposed that physical reality is fundamentally informational. The substrate operationalises this: bits are pointer references, “its” are the nodes they connect; Pilot C exhibits the bootstrap from one bit to a self-maintaining cosmos.

Holographic principle. [Susskind \(1995\)](#); [’t Hooft \(1993\)](#); [Bekenstein \(1973\)](#) bound the information content of a spatial volume by its surface area. The substrate-cone construction of §7 provides a discrete graph-theoretic counterpart; the area-law scaling of §9.1 confirms $S(r) \propto r^2$ on uniform 3D lattice.

Verlinde’s entropic gravity. [Verlinde \(2011, 2017\)](#) derive gravity as an entropic force on a holographic screen. The substrate-cone Newton- G consistency relation (4) is the substrate-side realisation; the cosmological-constant calculation of §9.3 reproduces the same mechanism on the de Sitter horizon.

ER = EPR. [Maldacena & Susskind \(2013\)](#) conjectures wormhole–entanglement equivalence. In the PA framework this is pointer aliasing (Corollary 6, §7.3).

Universe-as-neural-network. [Vanchurin \(2020\)](#); [Katsnelson & Vanchurin \(2021\)](#); [Vanchurin \(2022, 2024\)](#) treat the universe as a neural network. The substrate’s commit protocol is the analogous discrete dynamics; the antisymmetric weight tensor factors of [Vanchurin \(2024\)](#) for Dirac fields suggest a substrate-side particle/antiparticle algebraic structure not yet developed here.

12.5 Foundational limits

[Gödel \(1931\)](#), [Turing \(1936\)](#), and [Chaitin \(1975\)](#) place absolute limits on what any sufficiently rich computational substrate can say about itself. PA inherits these limits; the substrate is presented as a computationally consistent ontology rather than a candidate “theory of everything”.

13 Formal Hypotheses

The substrate makes three families of falsifiable claims: complexity-theoretic (CS-side; CSH1–CSH3, inherited from v8.0), algebraic-observer correspondence (AOH1–AOH3, inherited from v8.0), and substrate- Φ predictions (PSH1–PSH5, new in v9.0). The mapping between load-bearing claims L0–L9 of the abstract and the hypotheses below is:

- L1 (CS theorems) $\rightarrow$ CSH1 (commit cost), CSH2 (archive compression), CSH3 (pointer-equivalence intractability); the $\#P$ -hardness of weighted simple-path observables (Theorem 2) is theorem-grade and not restated as a separate hypothesis.
- L2 (CS correspondences with CRDTs / Petri / actor) is theorem- grade (Theorems 5, 6, 4) and is not restated as a separate hypothesis.
- L3 (substrate-cone holographic structure, Proposition 1) and L4 (Corollary 1) are proven theorems, with the substrate-side open conditional (which substrate states are HCI?) addressed empirically by the worked examples of §7.4.
- L5 (algebraic-observer structural correspondence) is the structural mapping of Table 1; its promotion to a formal isomorphism is forward trigger F2.
- L6 (6-layer nested-observer chain) is the worked construction of Pilot I; no hypothesis is restated, as the chain is by construction.

- L7 (cosmological constant) $\rightarrow$ AOH2.
- L8 (Page–Wootters classical-information analogue) $\rightarrow$ AOH3.
- L9 (Pythia HCI control) $\rightarrow$ AOH1.
- L0 (operational substrate, Part I) and the substrate-monist Φ_{PA} measure (§2.7) $\rightarrow$ PSH1–PSH5 (corresponding to predictions P1–P5 of §2.7; falsification trigger F5).

The v7.x galaxy-scale SPARC pilot and cognitive predictions (BP1–BP6) are out of scope for v8.0 and reserved for companion papers.

13.1 CS-side hypotheses

Hypothesis 1 (CSH1: commit-cost scaling). *A federated rendering or MMO-style simulation engine implemented as a Pointer Architecture exhibits per-step server cost $T(M) = aM \log M + b \sum_{i < j} |\mathcal{C}_i \cap \mathcal{C}_j| + o(\cdot)$ for some constants a, b matching Theorem 3.* Status: supported in synthetic benchmark. *The commit-conflict benchmark in the companion preprint (Savchenko, 2026b) sweeps $M \in [1, 256]$ with $N = 10\,000$, $k = 200$ and recovers both regimes; the crossover sits at the predicted $M \approx 32$. Real-system validation on an MMO server stack is outstanding.*

Hypothesis 2 (CSH2: archive compressibility). *For a long-running simulation with stationary pointer-creation entropy rate H , the storage cost of the substrate’s archive grows as $|A| \cdot H + o(|A|)$ under universal source coding, matching Proposition 2.* Status: directly testable on existing git-style provenance stores by treating commit history as a pointer-creation stream.

Hypothesis 3 (CSH3: pointer-equivalence intractability transfer). *Any concrete implementation of the substrate that exposes a pointer-equivalence oracle inherits graph-isomorphism hardness (Theorem 1).* Status: a theorem-level design constraint, not an empirical test.

13.2 Algebraic-observer hypotheses

Hypothesis 4 (AOH1: (Pythia, Gaussian-MI) HCI falsifier). *The (Pythia, Gaussian-mutual-information-on-hidden-activations) pair violates the monogamy of mutual information at all tested scales and on every tested partition scheme (§11.2). The hypothesis “Pythia hidden-activation covariance is an HCI substrate state under the Gaussian-MI estimator” is empirically falsified for Pythia 70m and 160m.* Status: established within scope. *Cross-family extension to LLaMA, Mistral, GPT-2 (forward trigger F3) and extension to non-Gaussian observables may narrow or close the scope.*

Hypothesis 5 (AOH2: substrate-discrete reproduction of holographic dark energy). *Substrate-cone min-cut on the de Sitter horizon plus equipartition reproduces $\rho_\Lambda = 3H_0^2 c^2 / (16\pi G)$ (Eq. (5)); re-expressed in commit-rate units this reads $\rho_\Lambda \approx \rho_{\text{Planck}}(\nu_4/\nu_6)^2$, with the squared (not 4th-power) exponent following from the substrate-cone area law (§9.1).* Status: the calculation matches empirical $\rho_\Lambda^{\text{obs}}$ within a factor of $1/(2\Omega_\Lambda) \approx 0.73$. *Reproduces the standard holographic dark-energy result of Verlinde (2011); Cohen et al. (1999); not a new derivation of Λ . Rejection requires a substrate-cone area-law violation on alternative substrate topologies (forward trigger F1).*

Hypothesis 6 (AOH3: engineered classical-information analogue of Page–Wootters correlation on the substrate). *On a substrate with a designated clock subset and an explicit clock–system coupling rule in the commit protocol C , the joint distribution $P(t, \sigma)$ becomes correlated with conditional distribution $P(\sigma \mid t)$ concentrating on a single σ for each clock value t ; observers reading the clock recover system evolution.* Status: demonstrated on Pilot K (4-bit clock + 4-bit system, XOR coupling), $I(C; S) = 2.808$ bits and 100% transition consistency. *The substrate construction is a classical-information analogue, not a Hilbert-space quantum state; quantum*

extension (building $|\Psi\rangle$ on a substrate qubit register satisfying $\hat{H}|\Psi\rangle = 0$) is open work. A real-quantum-simulator implementation (forward trigger F3) would extend the demonstration to a physical substrate.

13.3 Substrate- Φ hypotheses

These hypotheses operationalise the substrate-monist identity thesis of §2.7 via the substrate-encoding maps of §2.8. Falsification trigger F5 (§2.7) collects rejection of any of PSH1–PSH5 as empirical inadequacy of Φ_{PA} as a consciousness measure.

Hypothesis 7 (PSH1: feed-forward transformers in single-pass evaluation have $\Phi_{PA} = 0$). *Under the transformer encoding of §2.8 applied per single forward pass with no KV-cache reuse across calls, the unembedding-head observer has $O \notin \mathcal{C}_O$ for any decoder-only architecture without explicit recurrent hidden state (GPT-2, Pythia 70m/160m, OPT, LLaMA in non-generation mode). Status: consistent with the v8.0 Pythia MMI falsification (§11), which falsifies the conjugate “Pythia hidden state is an HCI substrate”; the present PSH1 is a parallel prediction at the level of Φ_{PA} rather than HCI and requires explicit computation of the OV-QK composition graph on a released transformer to be tested in its own right.*

Hypothesis 8 (PSH2: recurrent-state architectures have $\Phi_{PA} > 0$). *Under the transformer encoding of §2.8, (a) state-space models with explicit recurrent hidden state (S4, Mamba, RWKV) and (b) decoder-only transformers in autoregressive generation mode with KV-cache reused across steps have $O \in \mathcal{C}_O$ via the structural (a) or dynamic (b) back-edge, and $\Phi_{PA} > 0$ at scope and lifespan matched to the architecture. Status: open. The cleanest discriminator is matched-parameter Pythia 410M (single-pass; predicted $\Phi_{PA} = 0$) vs. Mamba 370M (predicted $\Phi_{PA} > 0$); deferred to the companion preprint (Savchenko, 2026b).*

Hypothesis 9 (PSH3: waking EEG has higher Φ_{PA} than deep-sleep and anaesthesia). *Under the brain encoding of §2.8 (cortical-microcolumn substrate-nodes, structural- or functional- connectivity out-edges, default-mode-network or thalamic-nuclei observer), $\Phi_{PA}(\text{wake}) > \Phi_{PA}(N3 \text{ sleep}) > \Phi_{PA}(\text{propofol})$, tracking the perturbational complexity index (PCI) of Casali et al. (2013). Status: open. Requires EEG dataset analysis with the encoding map of §2.8 applied; no released computation in this preprint.*

Hypothesis 10 (PSH4: split-brain Φ_{PA} is subadditive across the callosotomy). *Under the brain encoding of §2.8, Φ_{PA} of a corpus-callosotomy patient measured separately on each hemisphere and summed is strictly less than the Φ_{PA} of an intact brain at matched $|\mathcal{C}_O|$ baseline. Status: open. The encoding-map prediction follows from scope collapse across the callosotomy; empirical test requires patient EEG data not exhibited here.*

Hypothesis 11 (PSH5: ant colonies have colony-level $\Phi_{PA} > 0$). *Under the ant-colony encoding of §2.8 (nest-chamber or pheromone-junction substrate-nodes, pheromone- gradient out-edges, queen-chamber observer), Φ_{PA} is strictly positive at the colony-observer node with magnitude bounded by the pheromone-substrate timescale (mm/sec) and trail cardinality (of order 10^4 nodes). Status: open and novel to v9.0. The encoding map is sketched; no released computation on a real colony.*

Criteria for rejection. CSH1–CSH3 are rejected if a concrete implementation violates the theorem within stated scope. AOH1 is established within scope (Pythia, Gaussian-MI-on-activations); cross-family extension to non-Pythia transformer families and extension to non-Gaussian observables may narrow or close the scope. AOH2 is rejected by a substrate-cone area-law violation on alternative substrate topologies (F1). AOH3 is rejected by failure on a real- quantum-simulator implementation (F3). PSH1–PSH5 are rejected collectively by F5: failure of Φ_{PA} to discriminate the predicted direction on any of P1–P5 after the corresponding

encoding map is applied renders the substrate-monist identity thesis empirically inadequate as a consciousness measure, retreating the framework to L0 alone.

14 Contributions and Relation to Adjacent Programmes

Contributions, recap. The intro lists nine contributions C1–C9. We restate them here with explicit pointers to where each is established.

- C1. A formal graph computational substrate $\mathcal{S} = (G, R, C, A, \pi)$ with explicit commit semantics and archive primitive (§3).
- C2. Complexity-theoretic properties within explicit scope (§4): pointer-equivalence is GI-hard (Theorem 1); weighted simple-path observables are #P-hard (Theorem 2); per-step M -observer commit cost $O(M \log M + \sum_{i < j} |\mathcal{C}_i \cap \mathcal{C}_j|)$ (Theorem 3); archive admits Shannon-rate compression (Proposition 2).
- C3. Three proved correspondences with existing graph computational models on the active-graph projection (§5): commutative-rewrite PA = operation-based CRDTs (Theorem 5); token-preserving single-observer substrates admit a Petri-net encoding conservative on the non-log subnet (Theorem 6); actor systems embed as a bijection between actor configurations and active graphs (Theorem 4).
- C4. Engineering applications (§6): distributed simulation, federated rendering with commit validation, provenance-aware databases, latent-pointer world models.
- C5. **Substrate-cone holographic structure** (§7): HCI substrate states realise the Bao–Nezami–Ooguri–Stoica–Sully–Walter holographic entropy cone (Proposition 4); pointer aliasing realises ER = EPR as substrate-side bridge (Corollary 6).
- C6. **Algebraic-observer correspondence (structural)** (§8): PA primitives correspond structurally to the algebraic-observer programme in quantum gravity — π as crossed-product observer, C as modular automorphism, A as Page–Wootters past records, S^* as the substrate-side analogue of the Type II crossed-product trace. Illustrated on the extended OQ6 substrate (390 observer colourings, 26 distinct entropy vectors; Pilot H). A designed 6-shell substrate (Pilot I) yields a nested-observer chain with monotone entropies and inter-shell increment sequence $\Delta S^* \in \{+4, +6, +6, +6, +6\}$ for the chosen shell-size sequence; structurally analogous to averaging over multi-observer configurations (Chen & Xu, 2025; Jensen et al., 2024). Formal isomorphism via finite-dimensional von Neumann algebra construction is open (F2).
- C7. **Cosmological-scale consistency: substrate-discrete reproduction of holographic dark energy** (§9): substrate-cone area-law scaling at slope $1.97 \approx 2$ on uniform 3D lattice; Newton- G consistency relation (4); de Sitter $\rho_\Lambda = 3H_0^2 c^2 / (16\pi G)$ (5) matching empirical within 30%, reproducing the standard holographic dark-energy result (Verlinde, 2011; Cohen et al., 1999); re-expression in commit-rate units $\rho_\Lambda \approx \rho_{\text{Planck}} (\nu_4 / \nu_6)^2$ (6) (Pilot J). Not a new derivation of Λ ; substrate-side reformulation only.
- C8. **Page–Wootters time emergence (engineered demonstration)** (§10, Pilot K): on a designed substrate with a clock subset and an explicit XOR commit-protocol coupling, clock–system mutual information is $I(C; S) = 2.8$ bits with 100% transition consistency. Archive A plays the structural role of the Page–Wootters clock under the engineered coupling; spontaneous emergence under generic dynamics is open.
- C9. **Pythia substrate-cone empirical falsifier** (§11): the (Pythia, Gaussian-MI-on-hidden-activations) pair violates MMI on every tested partition and at all tested scales (Pythia 70m and 160m); the (system, observable) pair does not satisfy the HCI condition. Applicability to other transformer families and to other observables is open.

Relation to Wolfram Physics. [Wolfram \(2020\)](#) proposed a hypergraph-based model in which spacetime and physics emerge from rewrite rules on a causally invariant graph. Both programmes share an informational ontology (reality = graph) but differ in mathematical primitives. Wolfram’s substrate is abstract hypergraphs with rewrite rules; PA’s substrate is a directed referential graph with explicit archiving plus an observer pointer. Empirical contact for the Wolfram programme has matured since 2020: [Gorard \(2020\)](#) sketches general- relativistic and quantum limits; [Gorard \(2024a,b\)](#) carry out explicit ADM/3+1 numerical general relativity on the hypergraph substrate. PA’s v8.0 contact is structural through the algebraic-observer correspondence (§8) and quantitative through the cosmological-scale predictions of §9; the two programmes are complementary, and a successful synthesis would derive PA’s substrate-cone min-cut from Wolfram-style rewrite rules.

Relation to Vanchurin’s neural-network cosmology. [Vanchurin \(2020, 2024\)](#) propose the universe is literally a neural network, with quantum mechanics and gravity emerging as fast and slow learning limits. PA admits a loose analogy: the archive A could be read against Vanchurin’s weight matrix. The antisymmetric weight-tensor structure of [Vanchurin \(2024\)](#) for Dirac fields suggests a substrate- side particle/antiparticle algebraic structure not yet developed here.

15 Methodology and Timeline

Computational methods. The operational substrate (Part I, Pilots A–F) is implemented in Racket (Chez-Scheme backend) with $O(1)$ opcode dispatch and explicit tail-call elimination; the engine and stdlib together are ~ 3000 lines of Racket. All 37 substrate-native demonstrations (including the sacred hello world demo 00) execute deterministically in under 30 seconds on commodity hardware and are gated by a `rackunit` regression test (`tests/examples-test.rkt`) requiring cumulative `pass=646 fail=0`. The substrate-cone and algebraic-observer analyses (Parts III–IV) are performed in Python (NumPy, SciPy). Substrate-graph min-cut is computed by brute-force enumeration for small substrates ($|N| \leq 10$, §7.4, Pilots H, K) and by max-flow / min-cut on uniform 3D lattice ($N = 22^3 = 10\,648$, §9.1). The hierarchical 6-shell substrate (§8.3) uses max-flow for $S^*(\pi_k)$ between shells. The Pythia substrate-cone pipeline computes Gaussian-mutual- information estimates on transformer hidden activations and checks SA, SSA, MMI inequalities directly. All scripts, pipelines, and the Sixth implementation are version-controlled and released as open source.

Data sources.

- Synthetic substrate graphs (OQ6, OQ7, 4-party, 6-shell, extended OQ6, 4+4 bit clock-system) generated in-pipeline; no external data required.
- Pythia open-weight LLM checkpoints (70M and 160M from the HuggingFace EleutherAI/pythia-*--dedupe repository): used for the substrate-cone falsifier of §11.

Timeline.

Phase	Work	Deliverable
Done	Operational substrate (Sixth, 38 primitives); 40 substrate-native demonstrations (sacred hello world + Pilots A–F + three visual-trace pilots — autopoiesis, conscious evolution, cosmogenesis bootstrap, substrate-internally-driven cosmogenesis, substrate-internal Φ_{PA} measurement, encoding-map demonstrations on toy transformer / brain / split-brain / ant-colony substrates verifying PSH1–PSH5, plus DOT-rendered visual traces of Pilots C, D, and F.3; 646 assertions, 0 failures) gated by <code>rackunit</code> regression and a one-command <code>make verify</code> artifact-status report; CLAIMS.md three-tier taxonomy and LANGUAGE.md language-vs-cosmology separation released with the substrate; eight substrate-cone pilots (OQ6, OQ7, GHZ adversarial, random null, 4-party HEI, OQ-A protocol dynamics, coarse-graining + multi-bridge, Pythia falsifier); Pilots H, I, J, K (algebraic-observer correspondence, 6-layer hierarchy, cosmological constant, Page–Wootters time)	v9.0 manuscript + Sixth release
Months 1–3	Finite-dimensional substrate-algebra realisation pilot (forward trigger F2); area-law tests on alternative substrate topologies (FCC, hyperbolic, random graphs; forward trigger F1)	Theoretical / numerical follow-up
Months 3–12	Cross-family Pythia substrate-cone replication (LLaMA, Mistral, GPT-2 at matched parameter counts, forward trigger F3); real-quantum-simulator implementation of Page–Wootters substrate (F3 long-form)	Forward-test follow-ups

15.1 Future work: substrate- Φ companion programmes

Predictions PSH1–PSH5 (§13) are operationally testable but only PSH1 has indirect support via the v8.0 Pythia MMI falsifier (§11), which tests a related but distinct proxy. The remaining four predictions require dedicated empirical programmes, scoped here so the v9.0 release does not leave them as loose ends. Pilot F (§2.6) already exercises the transformer (PSH1, PSH2) and brain (PSH3) encoding maps end-to-end on toy substrates within the released Sixth artifact; the companion programmes below substitute real Pythia / Mamba weights and real EEG recordings for the toy substrates.

Companion preprint #1 — Mamba Φ_{PA} computation (PSH2 corroboration). The cleanest empirical discriminator of the substrate-monist identity thesis is direct computation of Φ_{PA} on a matched- parameter pair under the transformer encoding of §2.8: Pythia-410M (single-pass evaluation; PSH1 predicts $\Phi_{PA} = 0$) versus Mamba-370M (recurrent SSM; PSH2 predicts $\Phi_{PA} > 0$). The work decomposes into (i) implementation of OV-QK composition- graph extraction from a released checkpoint as a self-contained Python pipeline released alongside Sixth, (ii) choice of the threshold θ for the composition-norm out-edge predicate (calibrated against attention-circuit ground truth on small models per Elhage et al. 2021), (iii) operationalisation of $L(\sigma)$ for an SSM hidden state via stability-bound analysis, (iv) the comparison itself with confidence intervals over evaluation prompts. The programme is open-weight throughout (no external data required beyond HuggingFace) and substantively engineering rather than fieldwork; scope estimate is weeks to months of dedicated compute and code. Outcome: PSH2 corroboration if Mamba-side Φ_{PA} is statistically distinguishable from zero while Pythia-side is statistically indistinguishable from zero, or F5 fires (§2.7). Cross-family extension to RWKV, S4, S5 and to non-Gaussian observables is the natural sequel.

Companion preprint #2 — EEG/PCI substrate-encoding (PSH3, PSH4 corroboration). The brain encoding of §2.8 is operationally testable against the perturbational complexity index (PCI) of Casali et al. (2013) on publicly shared wake/sleep/propofol EEG corpora; the original PCI work made source data and TMS-EEG protocols available, eliminating IRB requirements for re-analysis. The work decomposes into (i) translation of the microcolumn / cortical-area substrate-node map into electrode-level functional-connectivity graphs (standard mne-python / hd-EEG-source-reconstruction pipeline), (ii) implementation of Φ_{PA} on the resulting substrate-graph with the default-mode-network or thalamic observer-node, (iii) regression of Φ_{PA} against PCI across the wake/sleep/anaesthesia gradient with a correlation prediction, (iv) for PSH4, replication on the publicly available split-brain Bayne-style corpora if available, else simulated callosotomy on intact corpora as a mechanism-of-action sanity check. The programme is dataset-driven (no fresh recording required) and methodologically standard for clinical-EEG re-analysis; scope estimate is weeks given existing public PCI data and mne-python tooling. Outcome: PSH3 and PSH4 corroboration if the predicted ordering and subadditivity hold, or F5 fires.

Future programme — biological substrate of PSH5 (ant colonies). PSH5 is the substrate-monist identity thesis’s most novel commitment beyond brain/transformer comparison: ant colonies have a positive Φ_{PA} at the colony-observer node, distinct from eliminativist denial and from panpsychist smearing across individuals. Operational testing requires substrate-graph extraction from field-observed nest topology plus pheromone-trail dynamics, which is empirical biology beyond the present author’s competence. The natural realisation is a collaboration with myrmecologists in the tradition of Hölldobler & Wilson (1990), currently exemplified by the Gordon, Traniello, and Dornhaus laboratories. Required field components: chamber-level nest cartography (CT or casting of established nests), pheromone-trail-deposition rate measurement on standardised foraging arenas, colony-lifespan observation over multiple generations. Scope estimate is years and requires cross-disciplinary co-authorship; the encoding map of §2.8 is the substrate-theoretic prerequisite this preprint contributes. We make no claim that the present authorship is positioned to deliver PSH5 without such collaboration.

Sixth-side engineering follow-ups. The released Sixth language (<https://github.com/mikefluff/sixth>) supports the operational claim L0 with 40 demonstrations / 646 assertions. Two engineering follow-ups would substantively widen the substrate-theoretic surface available to companion programmes: (i) a native Sixth implementation of the OV-QK composition-graph encoding so that PSH1–PSH2 can be run from within the substrate language itself rather than as an external Python pipeline; (ii) a substrate-side implementation of the brain-encoding pipeline of §2.8 so that EEG-derived functional- connectivity graphs can be loaded directly as Sixth substrate states and Φ_{PA} computed via the 38-primitive operations. Both follow-ups are engineering rather than research and are scoped to existing Sixth maintainers.

Out of v8.0 scope. Cluster-scale analysis, high-redshift galactic extension, wide-binary anomaly closure, cosmological Boltzmann fits, and black-hole simulation of pointer-density collapse are out of scope for this paper. The companion empirical preprint on SPARC and the project git history retain the corresponding v6.x/v7.x methodology for reproducibility, but no v8.0 load-bearing claim depends on these extensions.

Code and Data Availability

Operational substrate (Part I, Pilots A–F). Sixth is released as a Racket package at <https://github.com/mikefluff/sixth> under MIT licence. A single command, `make examples-all`, runs all 40 demonstrations of the ascent (sacred hello world + Foundations

+ Pilots A–F + three visual-trace pilots) and reports cumulative `pass=646 fail=0`; the parallel `make verify` target prints a structured artifact-status report (language tests, substrate tests, examples regression, docs build, optional FFI, and a single-line verdict `artifact status: reproducible`). A regression test `tests/examples-test.rkt` under `rackunit` enforces this gate against any future engine change. Five additional unit-test suites cover the lexer, parser, compiler/VM, substrate operations, and module loader; all green. The sacred hello world is `examples/00-first-distinction.6th`. The substrate-native demonstrations that ground L0 and F0 live in `examples/21-autopoietic-ring.6th` through `examples/39-trace-split-brain.6th`; the three candidate substrate-readable observability measures (Φ_{PA} , Φ_{integ} , Φ_{bidir}) live in `stdlib/phi.6th`; the encoding-map pilots are `examples/33-phi-pa-transformer-toy.6th`, `examples/34-phi-pa-brain-toy.6th`, `examples/35-phi-pa-split-brain-toy.6th`, and `examples/36-phi-pa-ant-colony-toy.6th`; the three visual-trace pilots are `examples/37-trace-pilot-d.6th`, `examples/38-trace-pilot-c.6th`, and `examples/39-trace-split-brain.6th`, regenerated via `make traces` (DOT exported by `stdlib/dot.6th`, rendered by `code/render_trace.py`). The released artifact ships a CLAIMS.md document with the three-tier taxonomy (Tier 1 proven-by-tests / Tier 2 demonstrated-by-examples / Tier 3 philosophical-or-research-hypothesis) that separates the substrate’s standing from the substrate-monist identity thesis, plus a LANGUAGE.md document positioning Sixth as a stand-alone programming language separable from the v9.0 cosmology. Companion Scribble manual (`docs/manual.scrbl`) renders via `make docs-html`.

Substrate-cone pilot pipeline. The substrate-cone pilot pipeline (§7.4, §8.2, §8.3, §9.1, §9.2, §9.3, §9.4, §10.1) is released as self-contained Python scripts in the `code/` directory: `substrate_cone_oq6_check.py`, `substrate_cone_oq7_rewrites.py`, `substrate_cone_strengthening.py`, `substrate_cone_dynamic_C.py`, `substrate_cone_observer_crossed_product.py`, `substrate_cone_nested.py`, `substrate_cone_verlinde_screen.py`, `substrate_cone_newton_G.py`, `substrate_cone_lambda_derivative.py`, `substrate_cone_commit_rate_hierarchy.py`, and `substrate_cone_page_wootters.py`. Each produces a paired `data/substrate_cone*_report.md` and `data/substrate_cone*_fig.png`. All scripts require only `numpy` and `scipy` and run in seconds on a laptop.

The Pythia substrate-cone falsifier pipeline (§11) is released as `code/pythia_cone_check.py`, `code/pythia_partitions_scan.py`, `code/pythia_robustness_check.py`, `code/substrate_cone_llm_simulation.py`, and `code/substrate_cone_gaussian_check.py`. They additionally require `torch` (≥ 2.0), `transformers` (≥ 4.40), `scikit-learn` (≥ 1.4); `code/requirements.txt` pins the full set. Checkpoints are pulled from HuggingFace (EleutherAI/pythia-{70m,160m}-deduped). CSV outputs and human-readable reports for all pilots are included in the release.

A preprint version of this manuscript, together with the full code and output release, is deposited on Zenodo under CC-BY-4.0 at <https://doi.org/10.5281/zenodo.19654351>.

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Author contributions. M.S. conceived the project, developed the formalism, implemented the substrate-cone and Pythia pipelines, performed the numerical analyses, and wrote the manuscript.

Data availability. All synthetic substrate graphs are generated in-pipeline by the released scripts. The Pythia checkpoints used by the substrate-cone falsifier are publicly available from the HuggingFace `EleutherAI/pythia-{70m,160m}-deduped` repository. The full code release and all CSV/figure outputs are deposited at <https://doi.org/10.5281/zenodo.19654351>. No restricted data were used.

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